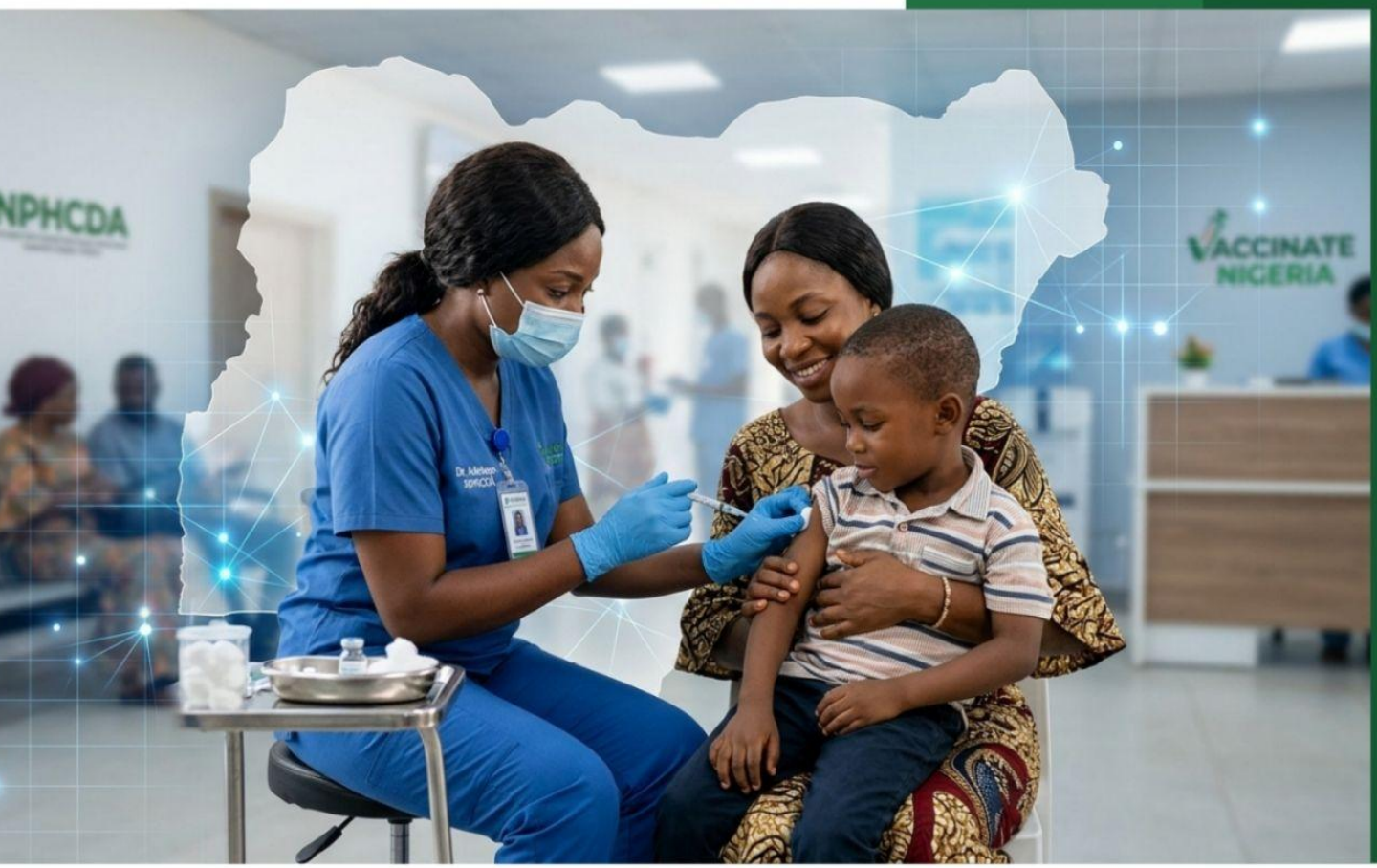


Integrating Predictive Modelling into Existing Immunization Programmes to Enhance Decision-Making and Reduce the Zero-Dose Burden in Nigeria

Consolidated Final Technical Report on Zero-Dose Forecasting, Immunization Risk Modelling, and Baseline–Endline Institutional Readiness

Prepared For
National Primary Health Care Development Agency (NPHCDA)

Prepared By
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List of Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
ACSM	Advocacy, Communication and Social Mobilisation	AFP	Acute Flaccid Paralysis
AI	Artificial Intelligence	AICc	Corrected Akaike Information Criterion
AMAC	Abuja Municipal Area Council	ANC	Antenatal Care
BCG	Bacille Calmette-Guérin vaccine	BI	Business Intelligence
CHI	Composite Hesitancy Index	CHIPS	Community Health Influencers, Promoters and Services
CI	Confidence Interval or Credible Interval, as applicable	COVID-19	Coronavirus Disease 2019
DHIS2	District Health Information Software 2	DIH	Digital Innovation Hub
DTP	Diphtheria-Tetanus-Pertussis	DTP1	First dose of a DTP-containing vaccine
DTP3	Third dose of a DTP-containing vaccine	ECHIS/eCHIS	Electronic Community Health Information System
EMR	Electronic Medical Record	EPI	Expanded Programme on Immunization
EVM	Effective Vaccine Management	FCT	Federal Capital Territory
FDR	False Discovery Rate	Gavi	Gavi, the Vaccine Alliance
GBD	Global Burden of Disease	Gi*	Getis-Ord Gi* hotspot statistic
GLM	Generalized Linear Model	GRID3	Geo-Referenced Infrastructure and Demographic Data for Development
GWR	Geographically Weighted Regression	HMIS	Health Management Information System
HTR	Hard-to-Reach	ICT	Information and Communication Technology
IDSR	Integrated Disease Surveillance and Response	IEV	Identify-Enumerate-Vaccinate
IHME	Institute for Health Metrics and Evaluation	LASSO	Least Absolute Shrinkage and Selection Operator
LGA	Local Government Area	LMIC	Low- and Middle-Income Country
LQAS	Lot Quality Assurance Sampling	M&E	Monitoring and Evaluation
MCV1	First dose of Measles-Containing Vaccine	MCV2	Second dose of Measles-Containing Vaccine
MGWR	Multiscale Geographically Weighted Regression	NCDC	Nigeria Centre for Disease Control and Prevention
NDHS	Nigeria Demographic and Health Survey	NITDA	National Information Technology Development Agency
NPC	National Population Commission	NPHCDA	National Primary Health Care Development Agency
ODK	Open Data Kit	OLS	Ordinary Least Squares
Penta1	First dose of Pentavalent vaccine	Penta3	Third dose of Pentavalent vaccine
PHC	Primary Health Care	PRS	Planning, Research and Statistics
QGIS	Open-source Geographic Information System software	R-hat	Potential Scale Reduction or model-convergence statistic
RI	Routine Immunization	RTS,S	Malaria vaccine targeting *Plasmodium falciparum*
SIA	Supplementary Immunization Activity	SMS	Short Message Service
SPHCDA	State Primary Health Care Development Agency	SPSS	Statistical Package for the Social Sciences
TBA	Traditional Birth Attendant	UNICEF	United Nations Children's Fund
VPD	Vaccine-Preventable Disease	WHO	World Health Organization
WUENIC	WHO/UNICEF Estimates of National Immunization Coverage	ZARA	Zero-dose Analytics and Risk Assistant

Abbreviation	Meaning	Abbreviation	Meaning
ZD	Zero-Dose		
ACLED	Armed Conflict Location and Event Data		
BHCPF	Basic Health Care Provision Fund		
LBD	Local Burden of Disease (IHME)		
NmDHS	Nigeria mini-Demographic and Health Survey		
PCV	Pneumococcal conjugate vaccine		
REC	Reaching Every Community		
Rota	Rotavirus vaccine		
WorldPop	Gridded population estimate programme		
SAE	Small-area estimation	BYM2	Besag-York-Mollie spatial model, version 2
ARI	Adjusted Rand index	CRPS	Continuous ranked probability score
MAE	Mean absolute error		

Executive Summary

Introduction and Background

Nigeria continues to carry a substantial and highly unequal burden of zero-dose and under-immunized children. Although national immunization indicators suggest gradual recovery and broadly stable routine delivery, national averages conceal severe subnational disparities, particularly across the North-West and parts of the North-East. Persistent challenges include unreliable population denominators, incomplete private-sector reporting, fragmented data systems, limited advanced analytics capacity, insecurity, geographical barriers, and uneven use of available data for proactive decision-making. The NPHCDA, with GAVI and UNICEF support, therefore implemented this programme to integrate predictive modelling into immunization planning to strengthen national and subnational capacity for data-driven prevention, preparedness, and response to vaccine-preventable diseases.

Objectives

The programme sought to build the capacity of the NPHCDA and State Primary Health Care Development Agencies to use surveillance and routine immunization data for trend analysis, predictive modelling, and decision support. Specific objectives included integrating modelling into routine planning and resource allocation, developing sustainable government capacity, strengthening staff proficiency in data analysis and predictive methods, and promoting the use of modelling for programme adaptation, vaccine introduction, and implementation strategy review.

Preparatory Engagement and Participant Selection

The implementation process included formal preparatory engagement with participating states and relevant institutional stakeholders. Letters were sent to states to introduce the assessment, clarify the purpose of the work, request cooperation, and support access to state, LGA and facility-level respondents. This was followed by state entry and community entry processes conducted in collaboration with state and local immunization structures. These activities supported respondent identification, scheduling of interviews, field team introduction, and smooth administration of the readiness checklist and key informant interviews.

Participant selection for the modelling fellowship was structured to support national and subnational ownership. A total of 20 national-level participants were selected, comprising 18 NPHCDA participants and 2 NCDC participants. In addition, 37 subnational participants were selected to represent state-level routine immunization and data-use structures. This ensured that capacity-building activities were not limited to the national level but extended to officers expected to interpret and apply modelling outputs in routine planning, supervision and corrective action.

Modelling Domains and Questions

Seven priority domains were completed: antigen-coverage forecasting; vaccination dropout and completion; equity and zero-dose risk stratification; vaccine hesitancy; zero-dose forecasting, hotspots and community archetypes; polio risk and interruption; and vaccine-preventable-disease outbreak risk. A planned eighth domain covering campaign optimization, counterfactual scenarios, and cost-effectiveness remains dependent on programme expenditure data.

Data Sources and Analytical Methods

The analyses triangulated routine DHIS2 immunization data, NDHS survey estimates, a 37-state zero-dose master dataset, ACSM activity records, equity indicators, GRID3 spatial boundaries, the national AFP line list, population estimates, and Institute for Health Metrics and Evaluation (IHME) Global Burden of Disease (GBD) 2023 mortality estimates. Methods included Prophet forecasting, LASSO feature selection, geographically weighted regression, composite indices, Bayesian hierarchical beta regression, Bayesian small-area estimation with a spatial (BYM2) model, Getis-Ord hotspot analysis, hierarchical clustering, spatial scan statistics, negative-binomial modelling, and false-discovery-rate-corrected correlation tests. The results were operationalized through a live no-code predictive-modelling application and the ZARA analytical assistant.

Findings by Modelling Domain

National forecasts for BCG, Penta1, Penta3, and Measles1 remained above 80% of their 2024 level, but 1,302 of 3,092 LGA-antigen forecasts, in 513 of 773 LGAs, fell below that mark, 195 of them with the decline already visible in 2025, demonstrating that risk is local rather than national. A wider panel of nine additional antigens (OPV3, IPV1, PCV3, Yellow Fever and Meningitis A) showed no decline warnings; IPV2 continues to rise, while rotavirus doses dipped in 2025 and need watching. Measles1-to-Measles2 remained the most important completion gap, while dropout drivers differed by antigen transition and therefore require differentiated interventions. Equity analysis showed that locally varying models substantially outperformed a single national model and identified severe structural deprivation in the North-West. Hesitancy was not significantly associated with zero-dose burden, indicating that access and supply constraints must be addressed separately from demand-generation concerns. Bayesian forecasts identified Sokoto, Kebbi, and Zamfara as critical-rate states, while Kano, Katsina, Kaduna, and other populous states carried large absolute numbers of zero-dose children. The North-West accounted for approximately 1.08 million forecast zero-dose children in 2026. Two Bayesian methods estimated about 2.1 million zero-dose children in 2026: 2.10 million under Method 1, a Bayesian hierarchical model with DHIS2-calibrated LGA allocation (773 LGAs), and 2.15 million under Method 2, Bayesian small-area estimation (all 774 LGAs). About 150 LGAs hold half of the burden and 319 to 342 LGAs are needed to reach 80%; 107 LGAs rank in the top 155 under both methods. Five community archetypes, defined at local government level across all 774 LGAs, demonstrated that intervention packages must be adapted to combinations of distance, fertility, poverty, education, facility delivery, and demand barriers. Polio risk was spatially concentrated, with residual vaccine-derived risk requiring sustained surveillance and campaign quality. Seven of eight coverage-to-mortality hypotheses were supported, and the zero-dose forecast correlated strongly with vaccine-targeted mortality, confirming convergence of immunization, equity and child-survival risks.

The forecasts were validated out-of-sample against the NmDHS 2025-26, released after this work concluded, which ranked the 37 states in almost the same order ($\rho = 0.89$ for Method 1 and 0.87 for Method 2) and matched the North-West level to within one percentage point, and against an independently constructed IHME LGA coverage surface ($\rho = 0.65$ and 0.89).

Readiness Assessment

Baseline readiness ranged from 59.5% to 97.6%, while endline scores ranged from 78.6% to 97.6%. Bayelsa recorded the largest improvement, increasing from 59.5% to 90.5%. Routine data availability and use were generally strong. Qualitative findings showed progress from general awareness to confirmed operational use of modelling in identifying and reviewing priority zero-dose LGAs, although national capacity remained concentrated in a small number of staff and continued to depend on external partners.

Stakeholder Engagement, Digital Remediation and Close-out Activities

The programme included structured stakeholder engagement and close-out activities to support interpretation, ownership and uptake of the modelling outputs. Stakeholders engaged included NPHCDA leadership, the Executive Director NPHCDA, directors and key NPHCDA programme leads, the Digital Innovation Hub team, the Routine Immunization team, the Routine Immunization Technical Working Group, Gavi in-country and technical stakeholders, UNICEF, NCDC, national participants, subnational participants and the CIDRE–Quantium Insights Consortium.

Following the meeting with the GAVI in-country team, the consortium held a further presentation with the Deputy Director NPHCDA, other directors and key NPHCDA staff, including the Head of DIH and Head of RI. The team presented to the Executive Director, who recommended inclusion of additional antigens such as OPV and IPV, ranking of LGAs from worst to best for prioritization, and development of LGA-specific problem statements with corresponding proposed interventions and solutions. These inputs were intended to strengthen outbreak preparedness, intervention planning and operational use of the priority list. The team also presented to the Routine Immunization Technical Working Group.

A remedy plan was developed to address the digital and DIH gaps identified during implementation. The plan prioritizes DIH operationalization, routine DHIS2 data refresh, model-output visualization, state-level analytics participation, data-quality automation, systems integration, power and connectivity improvements, role-based access, approval workflows, documentation, and continued capacity-building. The gap analysis showed that DHIS2/HMIS and routine review structures provide a foundation, but predictive modelling requires a stronger digital ecosystem capable of linking data sources, assuring quality, running reproducible models, refreshing dashboards, protecting data, and embedding interpretation into routine decision forums.

Conclusions and Recommendations

Nigeria’s zero-dose burden is concentrated, persistent, and actionable. Immediate investment should prioritize Sokoto, Kebbi, and Zamfara, followed by other convergent high-risk states and the ranked high-burden LGAs. Programmes should protect Penta1-3 coverage, close the Measles2 completion gap, differentiate interventions by each LGA's measured barriers, strengthen polio surveillance, institutionalize the modelling platform within the Digital Innovation Hub, establish a small, dedicated analytics core, provide sustained mentorship, and assign explicit modelling responsibilities with leadership oversight. Systems and infrastructure support should be concentrated in weaker units, alongside decisive action to address denominator instability and private- and tertiary-facility reporting gaps.

Limitations and Next Steps

Outbreak-risk estimates currently use GBD mortality as a proxy pending IDSR/NCDC line-list data. Routine administrative data remain affected by reporting and denominator errors; hesitancy findings reflect recorded ACSM concerns rather than population prevalence; and readiness findings cover thirteen administrative units rather than every state. The next phase should update the models with surveillance and expenditure data, complete campaign optimization and cost-effectiveness modelling, validate LGA prioritization, include additional antigens such as OPV and IPV, develop LGA-specific problem-and-intervention profiles, and embed routine model refresh, review and decision-use processes within the DIH.

1. Introduction and Background

Immunization is one of the world's most cost-effective health interventions and is responsible for saving millions of lives every year from vaccine-preventable diseases. Although there has been great progress around the world, there are still significant inequalities in vaccine coverage, especially in LMICs. There are still far too many zero-dose and under-immunized children in Sub-Saharan Africa, undermining progress toward universal health coverage and the global immunization goals. Nigeria is the most populous country in Africa, with a significant contribution to the burden of unvaccinated children globally. In 2018, 8.9 million Nigerian children were unvaccinated for the diphtheria-tetanus-pertussis vaccine (DTP1), particularly in northern regions. These differences are attributed to poor health systems, insecurity, population mobility, socioeconomic inequalities, and data quality and use in routine immunization programmes.

Nigeria has adopted several measures, such as supplemental immunization activities (SIAs) and health system strengthening at the primary health care (PHC) level, but gaps in health coverage suggest a need for rethinking planning and monitoring to address these inequities. Lack of integration of data systems, lack of integration of surveillance and program data, and lack of analytical capacity at the national and subnational levels are challenges to evidence-based decision-making. Recent developments in predictive modelling and geospatial analytics can be used to boost routine immunization systems. Data can be used to identify zero-dose children, predict outbreak risks, and prioritize resources. Institutionalization will require structured capacity building, data governance, and long-term partnerships, however, within government systems.

In this context, the National Primary Health Care Development Agency (NPHCDA), supported by GAVI and UNICEF, aims to mainstream predictive analytics into Nigeria's existing immunization system to improve decision-making, ensure equity, and drive progress towards reaching global immunization goals. Routine immunization is a public health intervention that is one of the most resource-efficient and saves an estimated 2-3 million lives every year around the world. High and sustained vaccination coverage in many low- and middle-income countries (LMICs) still represents a major challenge, especially in sub-Saharan Africa. Nigeria, Africa's most populous country, has significant systemic challenges in immunization, due to which the burden of vaccine-preventable diseases is disproportionately high in the country.

With the assistance of GAVI, UNICEF, and other partners, the National Primary Health Care Development Agency (NPHCDA) has carried out several activities to enhance routine immunization services. But coverage gaps, geographic disparities, and data quality problems persist and hamper program effectiveness. The traditional way of planning and monitoring immunization has been reactive rather than proactive, responding to immunization coverage gaps and outbreaks after they have occurred.

Objectives

To build the capacity of the NPHCDA and SPHCDAs to use surveillance and immunization data to identify ways to improve immunization programme effectiveness in preventing disease

through improved data-driven decision support for prevention, detection, preparedness, and response to vaccine-preventable diseases through trend analytics and mathematical modelling.

Specific Objectives

- To integrate predictive modelling into existing routine immunization programs based on the surveillance and routine immunization data to enhance decision-making and improve outcomes.
- To integrate modelling into immunization programs and routine decision-making processes from surveillance and routine immunization data to enhance planning, resource allocation, and overall program effectiveness.
- To build and institutionalize government capacity to adopt, sustain, and scale modelling practices for long-term impact and resilience in primary health care (PHC) initiatives.
- To enhance staff capacity in data modelling and predictive analysis by leveraging surveillance and immunization data from multiple platforms to inform evidence- based decision-making.
- To promote staff adoption of predictive modelling techniques for strategic program planning, including vaccine introductions and shifts in implementation strategies

1.1 The Zero-Dose Child

Zero-dose children are those who have received no routine vaccinations, operationally identified as children who have not received the first dose of a DTP-containing vaccine (DTP1, also Penta1). DTP1 is used worldwide as a proxy for access to the immunization system, because a child who never receives it has typically never been reached by routine services. In 2024, an estimated 14.3 million children globally were zero-dose, and just nine countries account for more than half of them, with Nigeria among the largest contributors. Geographic remoteness, poverty, low maternal education, insecurity, and weak health-system reach are the recurring drivers of this exclusion.

1.2 Where Nigeria Stands, NDHS 2024

The 2023 to 2024 Nigeria Demographic and Health Survey (NDHS 2024) is the most recent nationally representative measurement of routine coverage. Anchoring this study's state estimates to the NDHS series (2008 to 2024) and weighting by the 12-to-23-month cohort, the national zero-dose rate has been essentially flat for a decade: 48.7% in 2008, 33.1% in 2018, and 33.5% in 2024. National averages, however, conceal a profound subnational divide: independent ecological analysis of NDHS data finds North-West DTP1 coverage declining from 37.3% (2018) to 33.4% (2024) even as the South-East holds above 90%. This inequity, not the national mean, is the problem this modelling programme is built to localize and act on.

1.3 Why Zero-Dose Matters: The Link to Vaccine-Preventable Child Deaths

Nigeria carries one of the world's largest absolute burdens of under-five mortality, and the leading causes are dominated by conditions that vaccination prevents or mitigates. Global and Nigeria-specific estimates consistently rank malaria, diarrhoeal disease, and pneumonia (lower-respiratory infection) as the foremost causes of death among children aged 1 to 59 months, alongside neonatal causes (prematurity, birth asphyxia, sepsis). Measles, pertussis,

diphtheria, tetanus, tuberculosis, and yellow fever, all directly preventable by EPI antigens, add a further avoidable layer of mortality. Because DTP1 non-receipt marks the children the system never reached, zero-dose concentration is, in effect, an early-warning map of where vaccine-preventable deaths are most likely to occur. Domain 7 tests that link formally and find it strongly supported.

1.4 How the Programme Was Delivered

The programme ran as one integrated six-month cycle from January to June 2026, moving from assessment through capacity-building and applied modelling to institutional handover.

- **Month 1: Inception and governance.** Kickoff, workplan, data inventory and stakeholder alignment, alongside a baseline assessment documenting existing status, infrastructure, staff capacity and governance.
- **Months 1 to 2: Baseline and modelling questions.** Readiness review and definition of decision-use questions, refined into the eight domains aligned to NPHCDA decision needs set out in Section 3.
- **Month 3: Capacity building.** Structured self-learning, a virtual class, an in-person fellowship and competency testing for the selected national and subnational participants.
- **Months 4 to 5: Applied modelling.** Group mentorship, abstract development, model validation and production of the domain outputs, with abstract presentations converting model outputs into policy-facing products.
- **Month 6: Endline, integration and handover.** Delivery of the no-code application and ZARA, standard operating procedures and user guides, the endline assessment, and close-out.

The intended shift is from one-off training to a modelling workflow the agency runs in-house. Whether that shift holds is the question the readiness assessment in Section 6 is designed to answer.

2. Literature Review

This review draws on evidence published up to mid-2026, focusing on the issues most directly relevant to the programme: the state of immunization systems globally and in Nigeria, developments in predictive modelling methods, and the factors that determine whether analytics capacity becomes sustainable institutional practice rather than a short-term project output.

2.1 Global and Nigerian Immunization Challenges

Since the COVID-19 pandemic, global immunization has seen both recovery and continued difficulty. DTP3 coverage has returned to near pre-pandemic levels in many low- and middle-income countries, but gains are uneven, with fragile and conflict-affected settings lagging. Nigeria reflects these trends: national DTP3 has improved modestly, yet several northern states still record coverage below 40%. Measles, pertussis, and rotavirus gastroenteritis remain significant contributors to under-five mortality where services are weakest.

Zero-dose children have become a central indicator of health-system reach and equity. Geographic analysis alone is insufficient; gender, ethnicity, socioeconomic status, and intersecting deprivation must also be considered. In Nigeria, although coverage has improved

since 2018, inequities have widened, with unvaccinated children clustering in specific communities, wards, and disadvantaged groups increasingly in peri-urban and informal settlements as well as remote rural areas. In 2024, Nigeria was among the ten countries with the largest numbers of zero-dose children, recording an estimated 2.1 million DTP-unvaccinated and 3.1 million MCV-unvaccinated children.

The Immunization Agenda 2030 set targets of at least 90% national and 80% district-level coverage and a sharp reduction in zero-dose children. Nigeria, like many countries, remains off track. Closing the gap requires strategies capable of identifying and prioritising consistently missed children, which is precisely where predictive modelling has become relevant. Research in Nigeria and comparable settings shows machine-learning methods (random forests, gradient boosting, ridge classifiers) reliably identify high-risk facilities and communities from routine DHIS2 data, outperforming standard coverage-threshold approaches without requiring costly additional surveys.

2.2 Data Quality, Access, and Logistics in Nigeria

The scale-up of DHIS2 has materially improved the timeliness and completeness of routine immunization data, with reporting completeness above 80% in many states by 2019, though variation persists at the LGA and ward levels. Denominator accuracy remains a core challenge: discrepancies between 2006-census projections and field headcounts produce over- or under-estimates of coverage. Electronic registries, supervisory verification, community-based surveillance, and data-triangulation methods (including LQAS) have been introduced in priority states to validate figures.

On the supply side, expanded fixed sites, outreach and mobile teams, and emergency posts in conflict-affected areas have improved physical access. Demand-side barriers persist, including mistrust and misinformation in parts of the north; social mobilization, engagement with religious and traditional leaders, and community health influencers have shifted perceptions, while caregiver autonomy and the availability of female health workers shape uptake in conservative communities. Cold-chain investments, such as solar refrigerators, electronic temperature monitoring, and EVM assessments, have reduced failures, though maintenance gaps and last-mile transport remain constraints, especially in Borno, Zamfara, and Kebbi.

2.3 Predictive Modelling in Sub-Saharan Africa

Across the region, forecasting models for demand, outbreak-risk stratification, and coverage estimation increasingly shape strategic planning and resource allocation. Countries that have invested systematically tend to show superior, more consistent coverage. Rwanda is the most comprehensive regional example: forecasting integrated into its supply chain and district information system has helped sustain DTP3 and measles coverage above 90% and reduce zero-dose in hard-to-reach areas. Kenya and Senegal are developing outbreak-response risk-stratification models linked to surveillance. By contrast, settings with less consolidated data systems show volatile coverage and high urban-rural disparity.

Key transferable lessons recur: data-quality improvement is a prerequisite for reliable modelling; supply-chain managers and health workers must be engaged in model design so outputs are actionable; phased rather than system-wide rollout works better; and clear data-

governance frameworks are essential. Nigeria's adoption of the GRID3 platform underscores the value of subnational calibration in heterogeneous contexts and of public-private data partnerships. The clear implication is that predictive capability must be built into institutional structures, namely routine planning processes, dedicated analytics units, and information-system procurement, rather than added as an afterthought.

2.4 Outcomes and Challenges of Modelling Application

In Nigeria, UNICEF and partners have piloted data-driven models for zero-dose identification, vaccine distribution, and outbreak risk, drawing on administrative data, surveys, satellite-derived environmental measures, and mobile-network data. Uptake has been uneven, concentrated where data infrastructure and technical capacity are strongest. Geospatial risk maps have helped target outreach to the highest-risk wards. Implementation studies show modelling identifies zero-dose children more accurately and earlier than administrative data alone and supports outbreak preparedness for measles and meningitis by guiding the timing and extent of campaigns. Cost-effectiveness work in LMICs suggests data-driven allocation can achieve materially more coverage per unit of resource.

The barriers are equally well documented. Data quality and availability are the most fundamental: model accuracy depends directly on completeness, consistency, and granularity, and gaps in reporting, inconsistent denominators, and limited survey data reduce reliability and erode manager confidence. Technical-capacity constraints limit the interpretation of complex outputs at subnational levels. Institutional barriers, including limited buy-in, weak integration into formal planning, and competing priorities, constrain the translation of insight into action. Addressing these requires sustained capacity-building, data-system strengthening, and institutional change management alongside the technical work itself. This is the precise gap the readiness assessment in Section 6 is designed to measure.

3. Modelling Domains and Research Questions

Seven of the eight framework domains are reported here. Domain 8 (RI campaign design, counterfactual scenario modelling, and budget-constrained optimization) is the planned next phase and awaits the programme and expenditure data required for cost-effectiveness analysis.

Table 1: Modeling Domains and Research Questions

#	***Domain***	***Research question (status)***
1	Antigen coverage forecasting	Which antigens are projected to fall below targets in the next 6–12 months at LGA/ward level? (Completed)
2	Coverage continuity / dropout	What are predicted dropout rates between key antigen pairs, and what drives incomplete vaccination? (Completed)
3	Equity & zero-dose stratification	How does coverage and zero-dose risk vary by location, poverty, maternal education and HTR status? (Completed)
4	Hesitancy & behaviour	To what extent is hesitancy contributing to shortfalls, and how do patterns shift? (Completed)
5	Zero-dose modelling & hotspots	Where do zero-dose children concentrate, and what archetypes explain it? (Completed)
6	Polio mapping & interruption	Which areas are highest-risk for polio, and what is the interruption pathway? (Completed)
7	Surveillance & outbreak risk	Which areas are highest-risk for VPD outbreaks given coverage gaps? (Completed, GBD 2023 proxy)
8	Campaign design & effectiveness	Which states/LGAs to prioritize; access, demand or supply? (Planned next phase)

4. Methodology including Data Sources

The methodology including data analysis integrates NPHCDA-provided datasets with additional sources that the consortium curated. All models are reproducible from the consortium's analytical notebooks, and the analyses are operationalized in a live, no-code web application that the NPHCDA Digital Innovation Hub can run on uploaded data.

4.1 Principal Datasets

Table 2. Principal Datasets Mapped to the Domains They Feed

Dataset	***Description***	***Domains***
DHIS2 routine RI	Monthly administrative dose counts, all states, 2021–2025	1, 2, 5
NDHS 2008–2024	Survey DTP1 coverage and zero-dose, 12–23-month cohort	1, 3, 4, 5
Zero-dose master dataset	37-state file: NDHS coverage plus 30+ predictors	2, 3, 4, 5
NPHCDA ACSM register	2,593 social-mobilization activities, 36 states	4
Equity indicators	Wealth, education, HTR-distance, urbanicity by state	3, 5
GRID3 boundaries	admin1/admin2 spatial layers	3, 5, 6
AFP national line list	~103,000 surveillance records, 2015–2025	6
IHME GBD 2023	State under-5 cause-of-death fractions (proxy)	7
LGA population (NPC 2022)	Distributes state cohort across 774 LGAs	5
NmDHS 2025-26	National mini survey released after this work concluded; out-of-sample check	Validation
IHME Local Burden of Disease	DTP1 admin-2 coverage surface, 2000–2018; independent LGA comparator	Validation
WorldPop	LGA population surfaces used with NPC 2022 for absolute-burden conversion	1, 4, 5, 7
Meta Relative Wealth Index	LGA-level wealth covariate for deprivation index and archetypes	3, 5
Weiss travel-time surface	Motorised travel time to nearest facility; remoteness covariate	3, 5
ACLED	Political-violence event and fatality counts; conflict covariate	3, 5

Pending from NPHCDA: IDSR/NCDC surveillance line-lists (Domain 7) and programme/expenditure data (Domain 8).

4.2 Analytical Methods

Table 3. Analytical method by Domain

Domain	*Method*	*Output*
1	Prophet time-series forecasting, 80%/95% intervals, per antigen and per-LGA	Coverage trajectories to 2027; threshold-breach flags
2	Prophet dropout forecasting + LASSO feature selection	Dropout heatmaps; ranked driver importance
3	Multiscale geographically weighted regression (MGWR); composite equity index	Spatially varying coefficients; four-tier map
4	Field-coded concern taxonomy → Composite Hesitancy Index; Gi*; correlation test	Concern inventory; zonal CHI; decoupling test
5	Method 1: Bayesian hierarchical Beta regression with DHIS2-calibrated LGA allocation; Method 2: Bayesian small-area estimation (survey-anchored, BYM2 spatial model); Getis-Ord Gi*; Ward clustering (k=5)	State and LGA estimates with credible intervals (773 and 774 LGAs); priority probabilities; hotspots; archetypes
6	Bernoulli case-control spatial scan; negative-binomial GLM + Prophet	Significant AFP cluster; interruption-year projection
7	Eight FDR-corrected coverage-to-mortality tests; composite risk surface	Hypothesis grid; ranked risk surface; triangulation

Bayesian convergence for Domain 5 is satisfactory for both methods: Method 1 maximum R-hat = 1.006 and Method 2 maximum R-hat = 1.003, with no divergent transitions. Full step-by-step methods are held in the consortium's technical appendix and the Domain 5 notebooks.

4.3 Analytical Coverage of Local Government Areas

The Domain 5 estimates cover all of Nigeria's 774 LGAs. Method 1 produces estimates for 773 LGAs; Guzamala (Borno) reported no Penta1 doses to DHIS2 in 2021-2024 (partial reporting began in 2025), and Method 1, which distributes each state's estimate using LGA shares of reported Penta1 doses, cannot place it. Method 2 estimates all 774 LGAs, including Guzamala, from survey anchors, local conditions and neighbouring LGAs. Domain 1 forecasts also cover 773 LGAs; Guzamala has no 2024 doses and therefore no Domain 1 baseline (Section 5.1).

Table 3b. Local government areas covered by the Domain 5 estimates, by zone.

Zone	*LGAs estimated, Method 1*	*LGAs estimated, Method 2*
North West	186 of 186	186 of 186
North East	111 of 112	112 of 112
North Central	121 of 121	121 of 121
South West	137 of 137	137 of 137
South South	123 of 123	123 of 123
South East	95 of 95	95 of 95
Nigeria	773 of 774	774 of 774

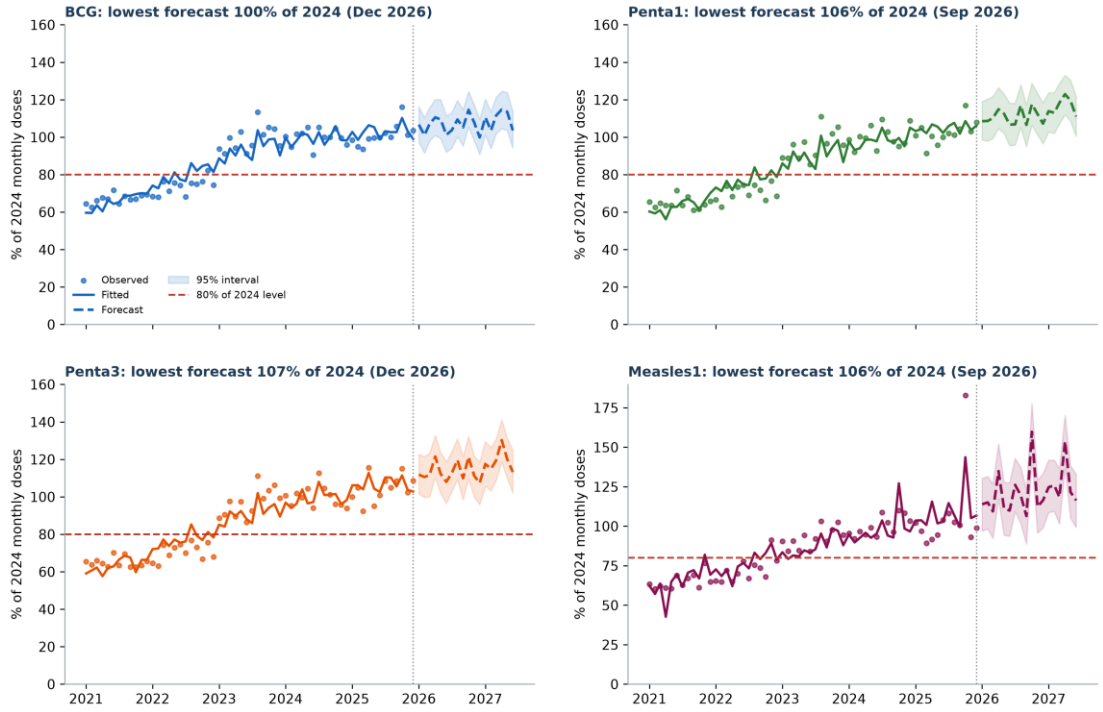
Guzamala (Borno) is the only LGA without a Method 1 estimate. Source: DHIS2 routine antigen export by LGA and month, 2021-2025; GRID3 boundaries (774 LGAs).

Routine reporting quality still matters. Because Method 1 distributes each state's estimate using LGA shares of reported Penta1 doses, LGAs whose reported doses are far above or below their estimated child population carry the largest allocation error; these are also the LGAs where the two methods differ most (Section 5.5.5). The Digital Innovation Hub should include them in its monthly data-quality checks, beginning with restoring Penta1 reporting in Guzamala.

5. Findings by Modelling Domain

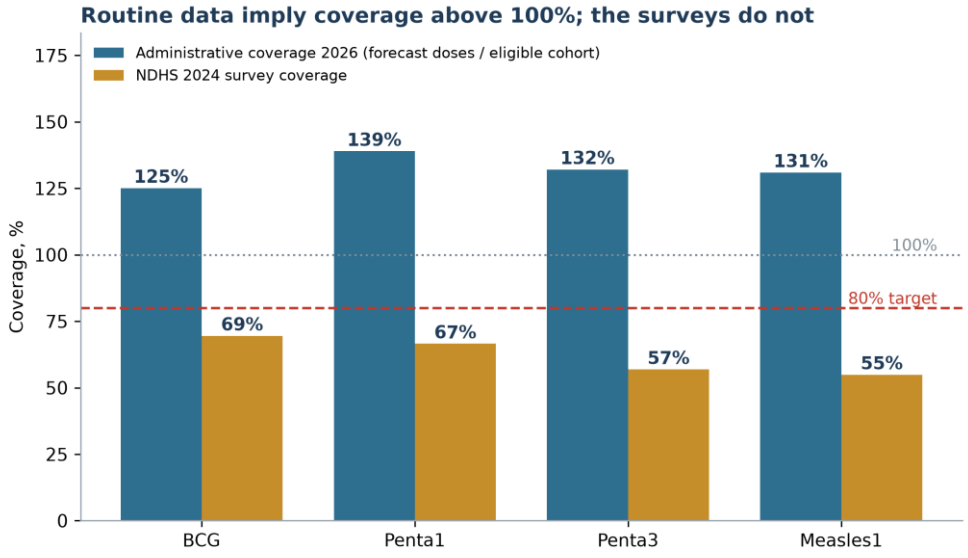
5.1 Domain 1 — Antigen Coverage Forecasting

Domain 1 is read two ways. An early-warning view compares each antigen with its own 2024 baseline, treating the 80% line as a decline tripwire. At the national level all four tracer antigens (BCG, Penta1, Penta3, Measles1) are forecast to stay above their 2024 level throughout the next 6-12 months; the lowest national points are 100% for BCG (Dec 2026), 106% for Penta1, 107% for Penta3 and 106% for Measles1. The national aggregate conceals the operational risk. Each of the 773 LGAs with 2024 data carries its own forecast for each of the four tracer antigens, giving 3,092 LGA-and-antigen forecasts. Of these, 1,302 dip below the 80% mark within the next 6 to 12 months, across 513 LGAs. In 195 of them the decline is already visible in 2025 data (2025 doses below 80% of 2024), and these should be followed up first; the remaining 1,107 are projected dips in series still close to their 2024 level. The risk is real but local, and must be managed LGA by LGA rather than read off the national average.



DHIS2 2021-2025, all 774 LGAs. Katsina, Kogi, Kwara, Lagos and Niger have no rows before January 2023. Model estimate.

Figure 1a. National tracer-antigen doses as a percent of their 2024 level, with Prophet forecasts and 95% prediction intervals.



Eligible cohort = 2024 under-five projection / 5 (7.02 million). NDHS 2024 national values weighted by state under-five population. Model estimate.

Figure 1b. 2026 administrative coverage of the eligible cohort against NDHS 2024 survey coverage.

Administrative coverage of the eligible cohort tells a different story from the survey. Dividing forecast 2026 doses by the 12-23-month cohort gives 125-139% for the four tracer antigens, against 55-69% in NDHS 2024. Routine doses exceed the projected cohort, which points to

denominators that are too small and to doses counted where children are vaccinated rather than where they live. This is why the early-warning flag uses each area's own 2024 level, and why Domain 5 anchors local estimates to the surveys.

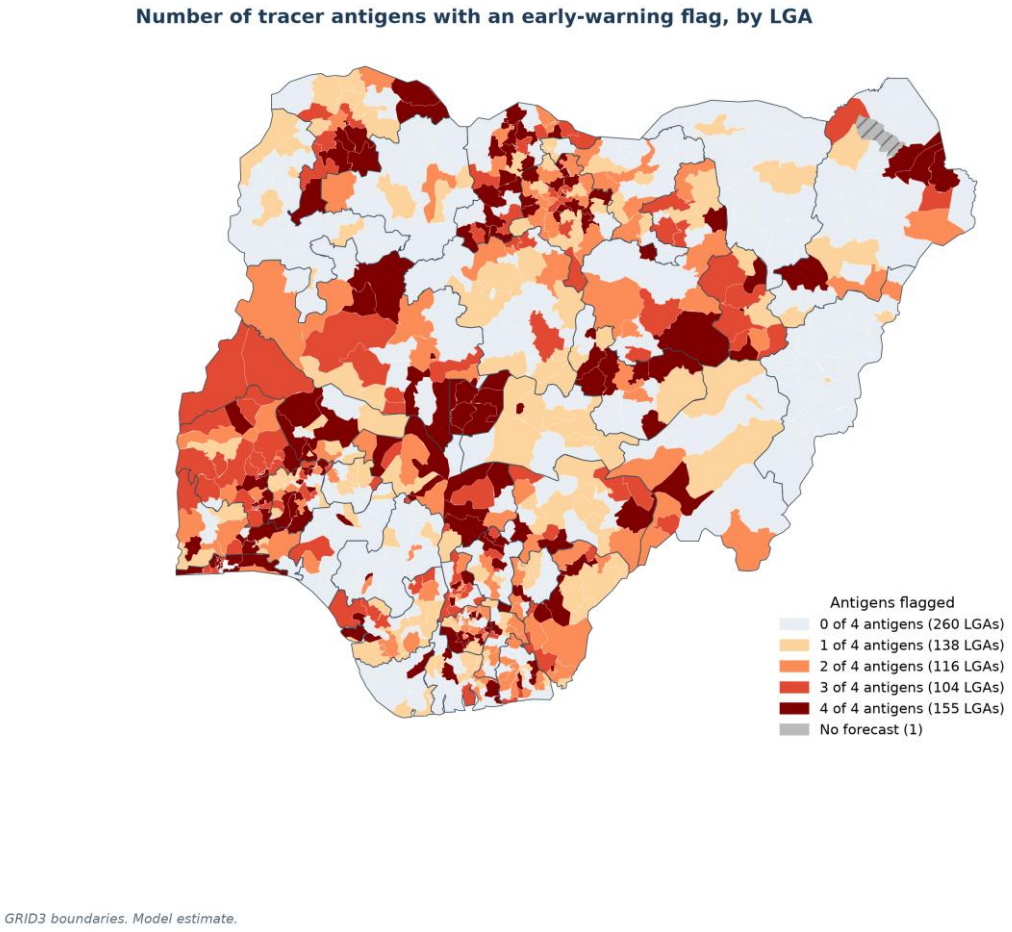


Figure 1c. Number of the four tracer antigens with an early-warning flag, by LGA.

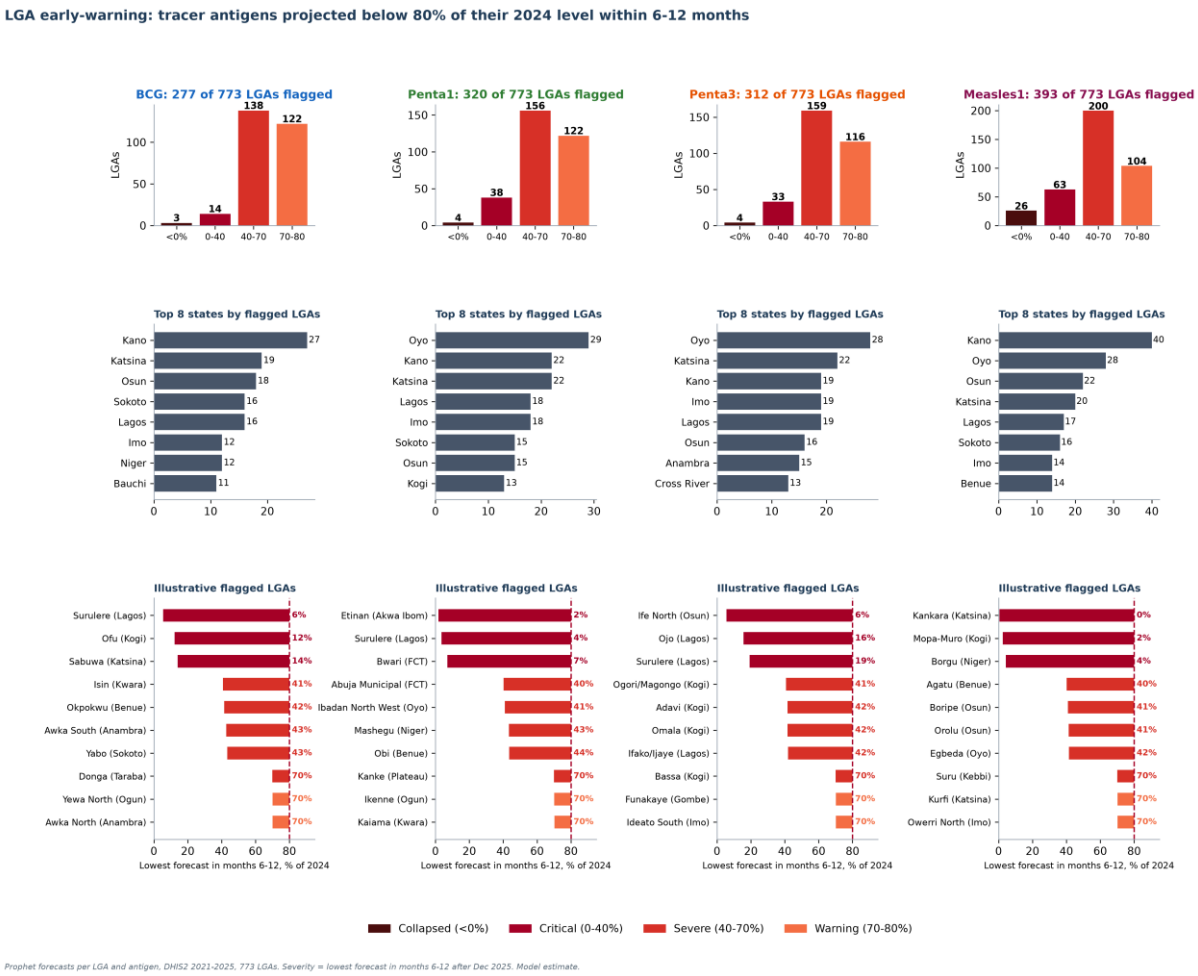


Figure 1d. LGA early-warning flags by antigen: severity, states with the most flagged LGAs, and illustrative LGAs.

Model estimate. Forecasts below 0% (37 of 3,092) indicate an erratic LGA series, for example a sudden change in reporting, and should be verified in DHIS2 before action. The full LGA list is in the Domain 1 government workbook.

5.1.1 The Wider Antigen Panel

The antigen panel was extended beyond the four tracers to cover nine further antigens, using the same Prophet specification. This applies decline screening across the routine schedule rather than to the tracer set alone, and covers the antigens that account for the majority of delivery volume.

Five established antigens (OPV3, IPV1, PCV3, Yellow Fever and Meningitis A) are projected to hold at or above 80% of their own 2024 delivery level through the forecast window, and none raises a decline warning. Four recently introduced series, IPV2 and Rotavirus doses 1 to 3, follow different paths: IPV2 is still climbing (2025 doses 109% of 2024), while rotavirus doses fell to 89-91% of their 2024 level in 2025 and should be watched; these are monitored for uptake rather than screened for decline, since a rising series cannot meaningfully be tested against a decline tripwire.

The programme consequence is that the decline signal identified in Section 5.1 is specific to the tracer antigens and to particular LGAs, and is not a general deterioration across the antigen schedule. Supervision can therefore be routed narrowly. The polio antigens in particular, OPV3, IPV1 and IPV2, show no national decline signal, which is consistent with the interruption pathway described in Section 5.6.

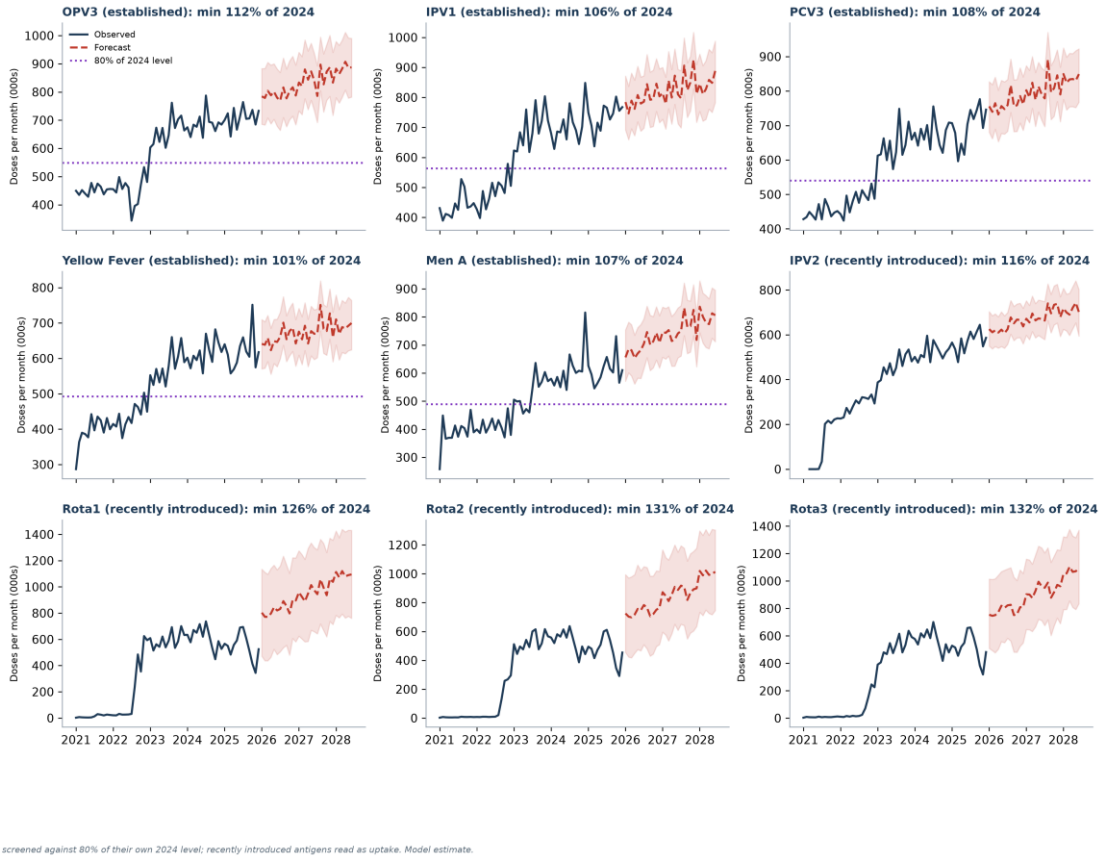


Figure 1e. Forecast trajectories for the nine additional antigens against the 80% at-risk-of-decline mark (Prophet, DHIS2 2021-2025, projected to 2028).

5.1.2 Reading the At-Risk-of-Decline Flag Correctly

The 80% figure used throughout Domain 1 is an early-warning threshold rather than a coverage statistic. The distinction is material to how the results are applied, and is set out in four points below.

- **Definition.** Projected monthly doses for an antigen in an LGA, expressed as a percentage of that same antigen's 2024 delivery level in that same LGA.
- **Interpretation.** A reading below 80% means doses are slipping against a settled benchmark. It does not mean that only 80% of children were reached.
- **Denominator independence.** No child population enters the calculation, so the flag is unaffected by the denominator instability documented in Section 9.3. That is precisely why it is useful in a setting where denominators are contested.
- **Programme use.** Catch the decline early and route supervision to the exact antigen, month and LGA.

The benchmark is 2024 rather than 2025 because 2024 is the last complete, fully reported year and therefore holds still. The 2025 series is partial and continues to shift as late reports arrive; benchmarking against it would generate alerts that are in truth reporting lag.

True coverage answers a different question. It divides doses by the eligible 12-to-23-month cohort and is reported separately, both in Figure 1b and in the platform under "Estimated coverage (eligible cohort)". The early-warning flag tells the programme when delivery is falling; coverage tells it how many children were reached. Different questions, different denominators, and the two should not be quoted interchangeably.

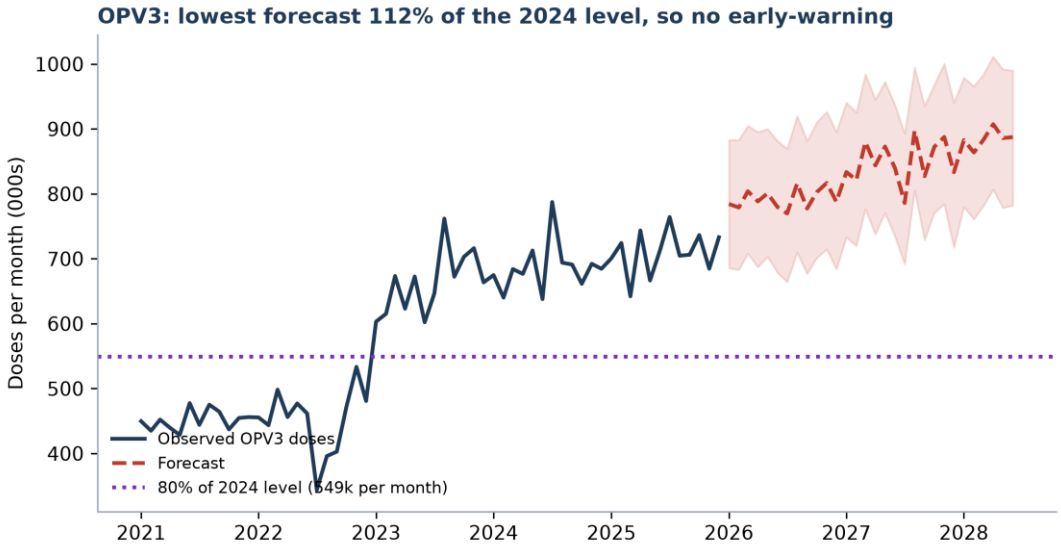


Figure 1f. Worked example: the dotted line is 80% of OPV3's own 2024 doses. The forecast remains above it, so OPV3 is not flagged.

5.2 Domain 2 — Dropout and Incomplete Vaccination

The three antigen transitions tell different stories. Penta1-to-Penta3 dropout is low and stable: 6 to 7% a year nationally in 2021-2024 and 4.8% in 2025, with a forecast of about 6.6% for December 2026 (95% prediction interval 2.6 to 10.8%), so most children who start the Penta series complete it. Penta1-to-Measles1 dropout is moderate, about 11 to 16% a year (10.8% in 2025), with wide month-to-month swings. The decisive transition is Measles1-to-Measles2: national dropout fell from 60% in 2021 to 34% in 2025 and is forecast at about 26% by December 2026 (20 to 32%), much improved but still by far the largest of the three gaps. In 2025, 34 of 37 states lost more than 10% of children between the two measles doses, with the highest rates in Anambra (60%), Akwa Ibom (58%), Imo (54%), Abia (53%) and Oyo (48%).

Cross-validated LASSO driver selection points to different mechanisms for each transition, so a single retention strategy will not fix all three. Across states (2024), no equity covariate consistently predicts Penta1-to-Penta3 dropout; across LGAs (2021-2024), it is higher where conflict fatalities are higher and lower where oral rehydration salts (ORS) coverage and facility delivery are higher. Penta1-to-Measles1 dropout is higher where distance to a facility is a problem and fertility is higher, and lower where four or more antenatal visits are common; across LGAs it rises with child wasting and stunting. Measles1-to-Measles2 dropout is lower where vaccination cards are seen and births are registered; its positive association with

antenatal care reflects the high second-dose dropout of southern states, where antenatal coverage is also high, rather than a causal effect. These are ecological associations (37 states; 743 LGAs, R-squared 0.10 to 0.12) that indicate direction rather than size. Retention efforts therefore need to be tailored by transition: defaulter tracing and security-sensitive outreach for the Penta series in conflict-affected LGAs; linking the nine-month visit to outreach and nutrition contacts where distance and undernutrition are barriers; and home-based records, card retention and second-year reminders for the measles second dose, concentrated on the high-dropout states of the South-East and South-South and on Oyo.

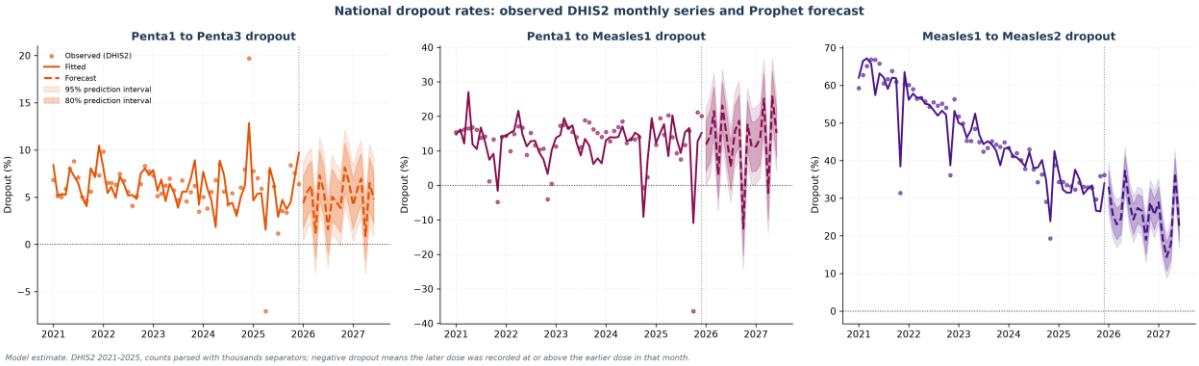


Figure 2. National dropout rates for the three antigen transitions: observed DHIS2 monthly series and Prophet forecast with 80% and 95% prediction intervals.

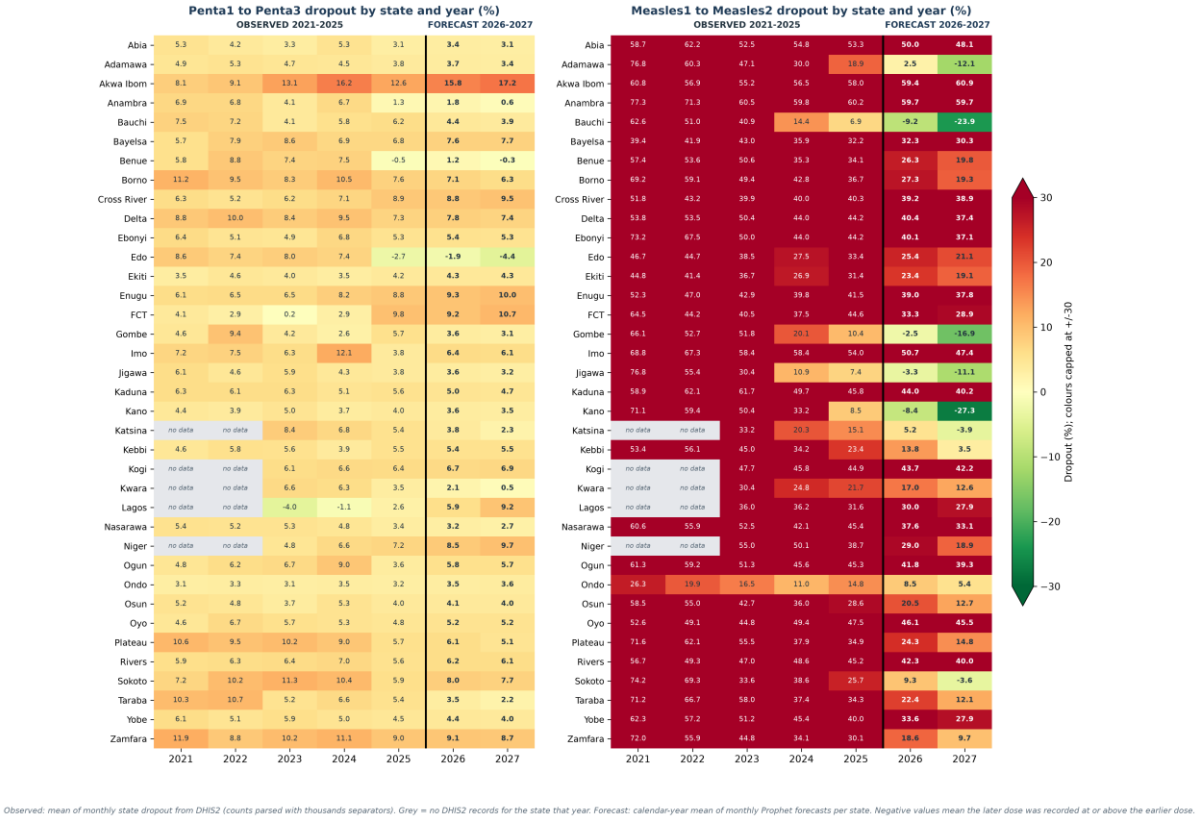


Figure 3. Dropout by state and year, observed 2021-2025 and Prophet forecast 2026-2027 (Penta1-to-Penta3 and Measles1-to-Measles2).

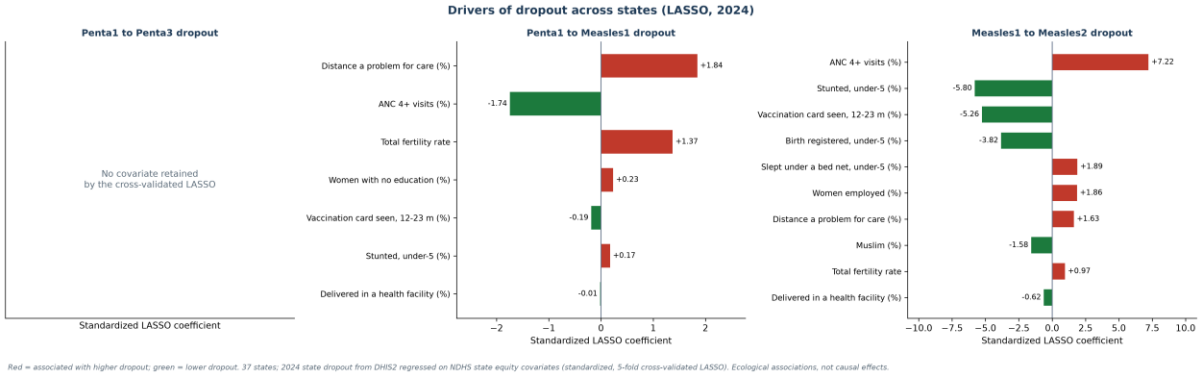


Figure 4. Drivers of dropout across states: signed LASSO coefficients, 2024 (red = associated with higher dropout; green = lower).

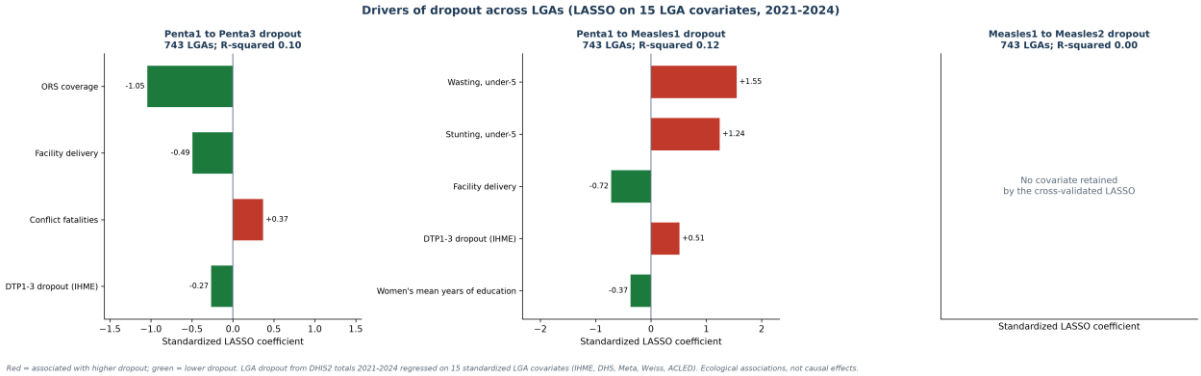


Figure 4b. Drivers of dropout across LGAs: signed LASSO coefficients on 15 LGA covariates, 2021-2024.

Model estimate. DHIS2 monthly counts 2021-2025, read with thousands separators removed (DHIS2 exports counts of 1,000 or more as text, for example "1,234"). Negative dropout means the later dose was recorded at or above the earlier dose in that period. The same calculations run in the Dropout & Completion page of the web application.

5.3 Domain 3 — Equity and Zero-Dose Risk Stratification

Zero-dose risk does not respond to the same drivers everywhere. A multiscale geographically weighted regression fits substantially better than a single national model (the fit statistic rises from 0.743 to 0.868), and that 13-point gain is the share of the pattern that is purely local. It is the statistical justification for targeting states individually rather than applying one national playbook.

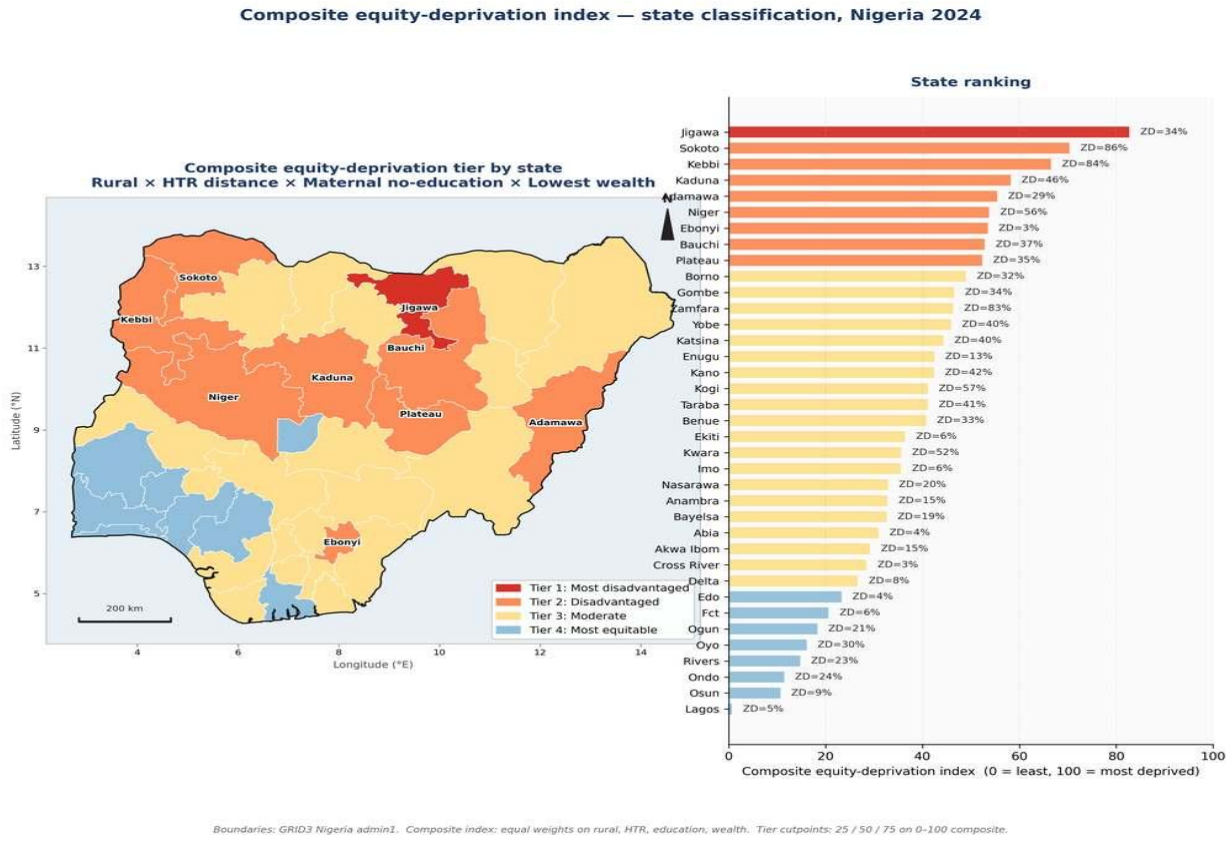
Table 4. Model comparison for equity stratification.

Model	*R^2*	*Adjusted R^2*	*AICc*
Global OLS	0.743	n/a	n/a
GWR (bandwidth 35)	0.821	0.755	303.21
MGWR (per-predictor)	0.868	0.803	301.65

Higher R^2 and lower AICc indicate the better fit.

A composite equity-deprivation index combines rural residence, distance to a facility, mothers with no formal education, and lowest-wealth-quintile status into a single 0–100 score, sorting states into four tiers. The most-deprived tier is dominated by the North-West, with Jigawa recording the highest deprivation score. The ranking reflects structural disadvantage, which is

distinct from the current observed zero-dose rate: Jigawa ranks most disadvantaged on structure even though its observed rate is mid-range, while Sokoto and Kebbi combine very high deprivation with very high observed zero-dose. Tier 1 and Tier 2 states are where equity-focused investment should be concentrated.



Figures 5–6. Observed zero-dose by state (NDHS 2024) and the composite equity-deprivation tiers.

5.3.1 Equity Analysis at LGA Level

The state-level analysis in Table 4 was repeated across all 774 LGAs to give the resolution that microplanning actually requires. The result is consistent in direction and larger in magnitude: allowing each driver to act differently in different places raises the share of variation explained from 39% to 55%. More than a third of the variation a single national model cannot account for is recovered by letting the drivers vary geographically.

The composite equity-deprivation index, built from travel time, maternal education, poverty and the Meta Relative Wealth Index, scored 0 to 100 and grouped into four equal bands, was likewise recalculated at LGA level on GRID3 boundaries. Tier 1, the most deprived band, is concentrated in the North-East and North-West, consistent with the state-level tiering but far more actionable, since it identifies specific LGAs within otherwise mixed states.

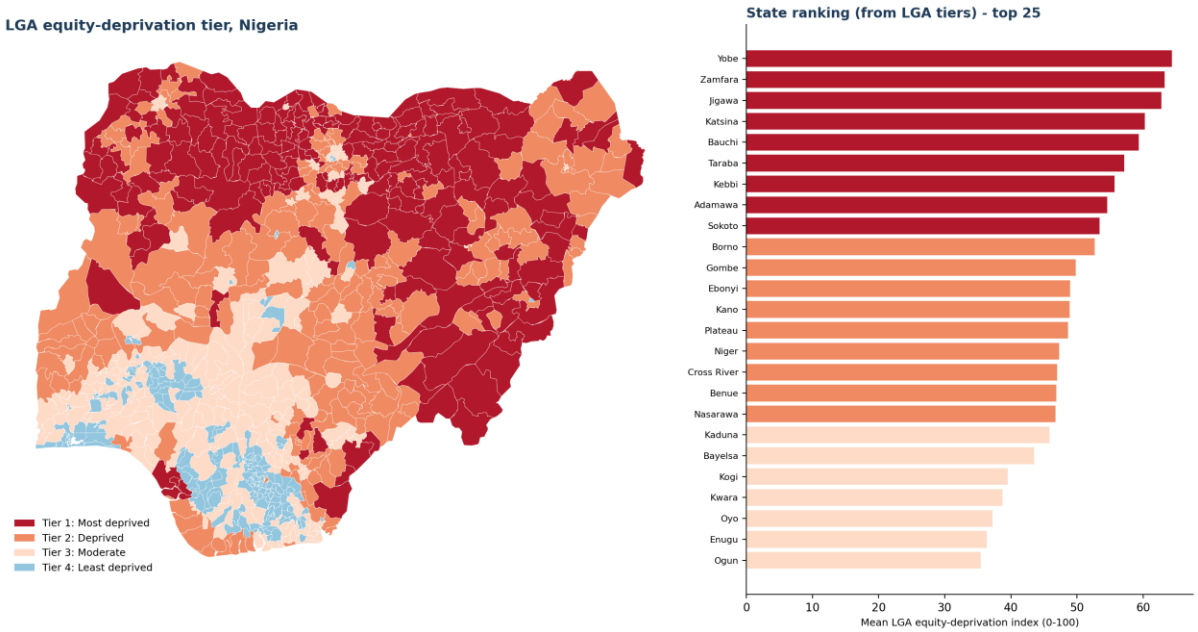


Figure 6b. LGA-level composite equity-deprivation tiers, 774 LGAs (GRID3 boundaries).

The two outputs answer different questions and should be used together. The burden ranking in Section 5.5.2 shows where the most children are; the deprivation tier shows where the barriers are deepest. LGAs that are high on both should take precedence over LGAs that are high on either alone.

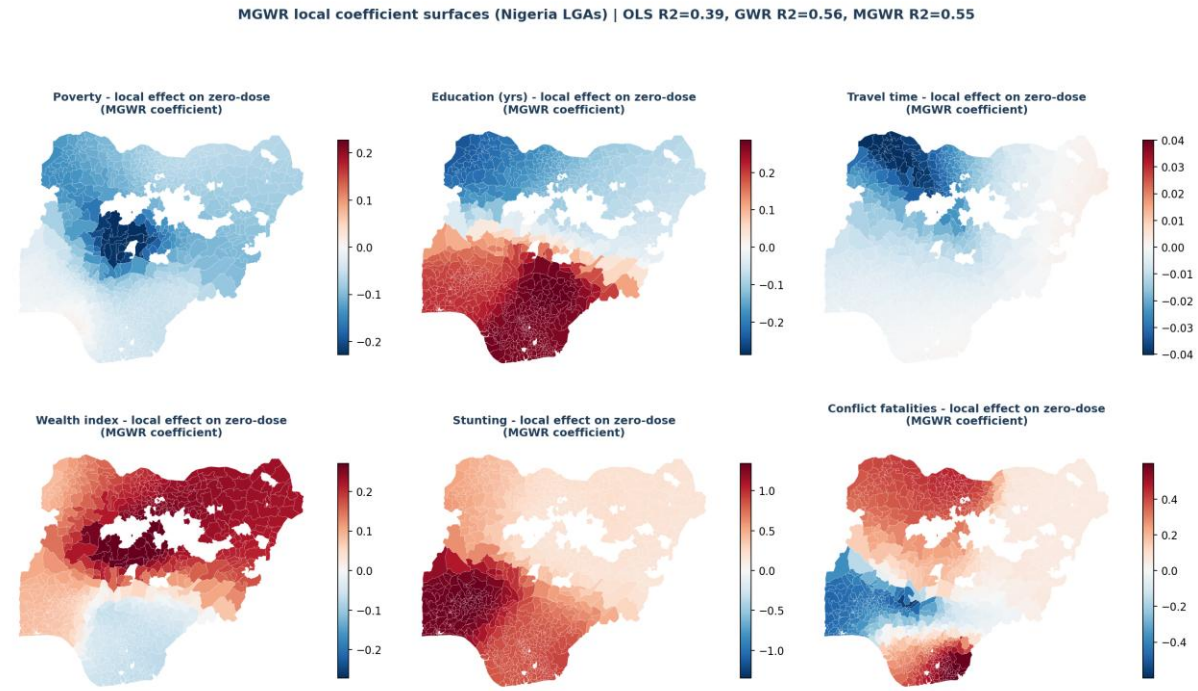


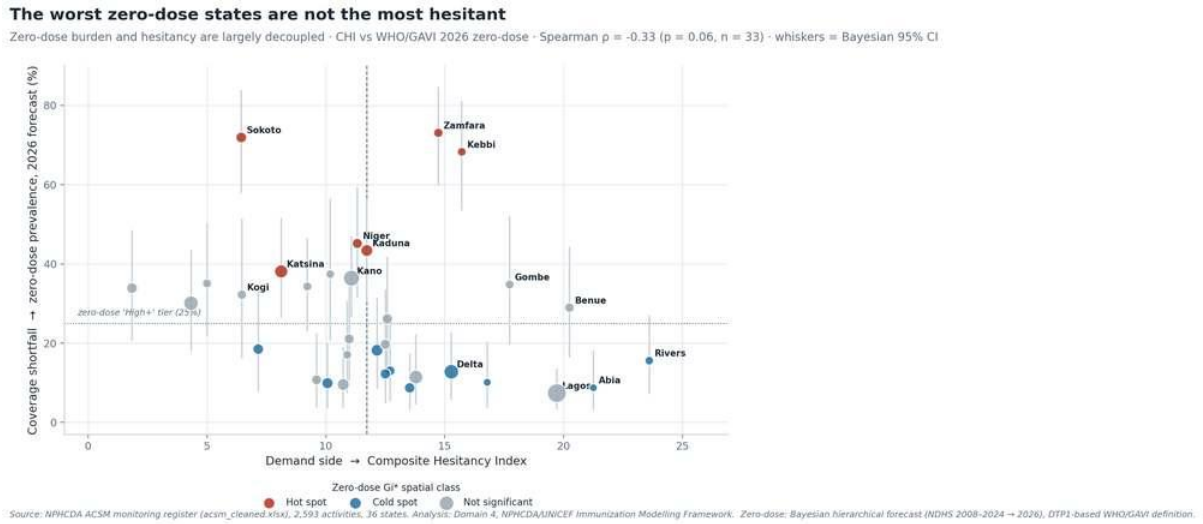
Figure 6c. MGWR local coefficient surfaces: which barrier is strongest where. Red indicates association with higher zero-dose, blue the reverse; unmodelled LGAs are blank.

Two cautions govern interpretation. The LGA coefficients are local, standardised, partial associations and are not causal effects. LGAs without a DTP1 series in the Domain 3 analysis

carry no estimate at all rather than a zero and appear blank on the maps; they should not be read as low-risk.

5.4 Domain 4 — Vaccine Hesitancy

Across 2,593 ACSM field activities in 36 states, vaccine-safety fear is the dominant caregiver concern (710 safety questions and 108 rumours), far ahead of infertility, conspiracy and political concerns. To compare states fairly, a Composite Hesitancy Index normalizes coded concerns by each state's activity volume. Plotting it against the forecast zero-dose rate shows the two are largely decoupled (Spearman rho = -0.33, p = 0.06): the most hesitant states are not the worst zero-dose states. Sokoto, Kebbi, and Zamfara sit high on burden but only mid-range on hesitancy. The practical consequence is that supply-and-access interventions and demand-generation messaging address different maps; treating zero-dose purely as a demand problem would misdirect resources.



Figures 7–8. Coded caregiver concerns and the Composite Hesitancy Index versus zero-dose burden (decoupling).

5.5 Domain 5 — Where Zero-Dose Children Concentrate, and the Archetypes Behind Them

A Bayesian hierarchical Beta regression on the NDHS 2008-2024 series with a DHIS2 Penta1 trend term (Method 1; maximum R-hat = 1.006, no divergent transitions) produces state-level posterior forecasts with full credible intervals for 2026-2028. The three Tier-1 (Critical) states, Sokoto, Kebbi and Zamfara, carry the highest predicted rates, each with a 2026 posterior mean near or above 69%. The North-West sits in a band of its own throughout; most other states drift slowly downward, and credible intervals widen after 2024, where recent survey coverage is thinnest, reflecting honest forecast uncertainty rather than noise.

Table 5. The nine highest-burden states by estimated zero-dose children in 2026 (extract of the full 37-state Bayesian forecast).

State	*Obs 2024 (%)*	*2026 mean (95% CI)*	*ZD children 2026*
Kano	42.4	36.8 (27-47)	197,005
Sokoto	86.5	72.7 (59-84)	170,749
Katsina	39.8	39.0 (27-52)	169,686
Kebbi	84.0	69.0 (54-82)	155,754
Zamfara	82.6	70.9 (58-83)	151,485
Kaduna	45.6	42.0 (29-55)	133,970
Bauchi	36.9	33.3 (22-46)	109,358
Jigawa	34.3	34.8 (22-49)	106,265
Niger	56.0	45.0 (31-59)	104,030

Burden is the posterior-mean rate applied to each state's 12-to-23-month birth cohort (state total 2.07 million). LGA totals in Section 5.5.2 sum to 2.10 million because LGA rates are bounded between 1% and 99% during allocation.

Aggregated to zones, the North-West alone holds about 1.08 million zero-dose children in 2026, more than the other five zones combined, with the North-East and North-Central following at roughly 340,000 each. The Getis-Ord Gi* hotspot analysis confirms a single contiguous high-risk block across the North-West and into the North-East that persists across 2026–2028, while the South remains a coherent cold-spot. The burden is not only high in the North-West, it is entrenched and spatially stable.

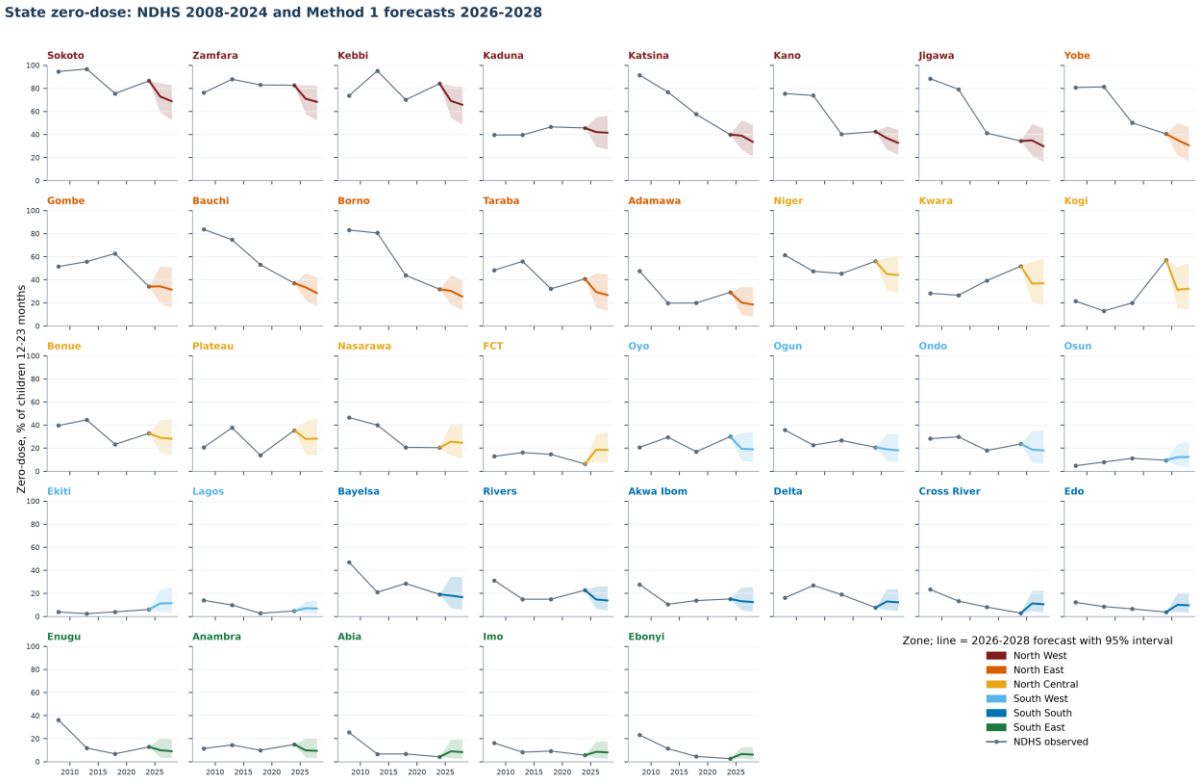


Figure 9. State-level zero-dose: NDHS 2008-2024 observations and Method 1 forecasts for 2026-2028 (shaded bands are 95% credible intervals).

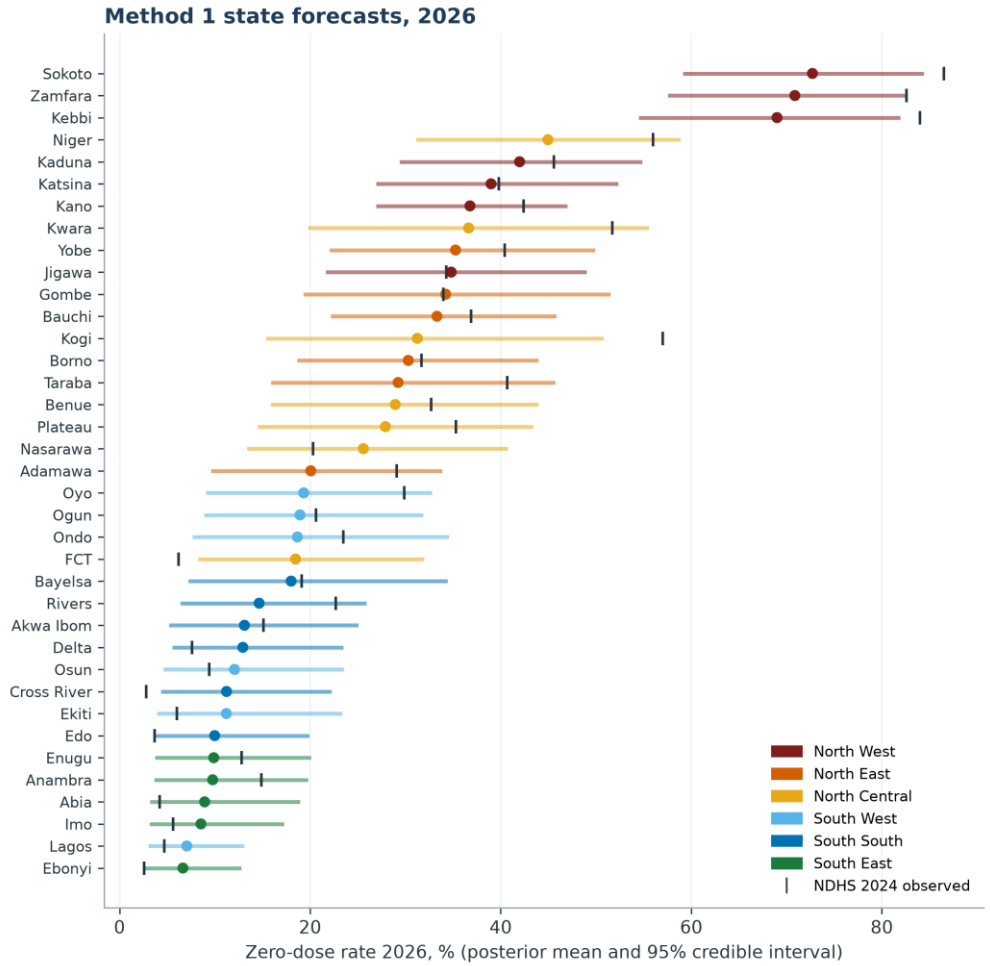
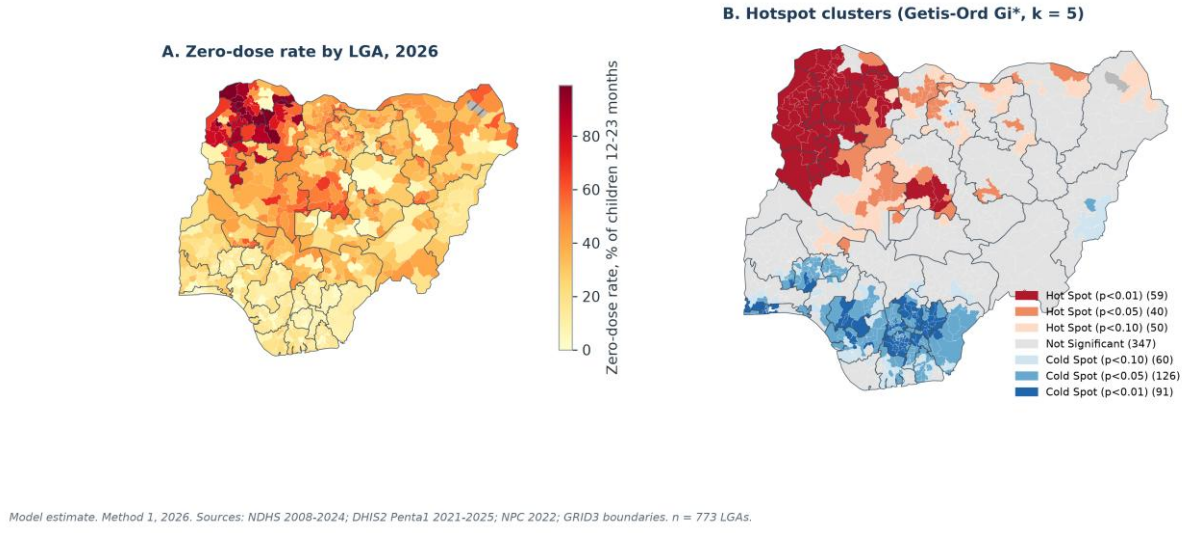


Figure 10. Predicted state zero-dose rate (2026) with 95% credible intervals, by zone, against NDHS 2024 observed values.



Model estimate. Method 1, 2026. Sources: NDHS 2008-2024; DHIS2 Penta1 2021-2025; NPC 2022; GRID3 boundaries. n = 773 LGAs.

Figures 11-12. Method 1 LGA zero-dose rate (2026) and Getis-Ord Gi* hotspot clusters, 773 LGAs.

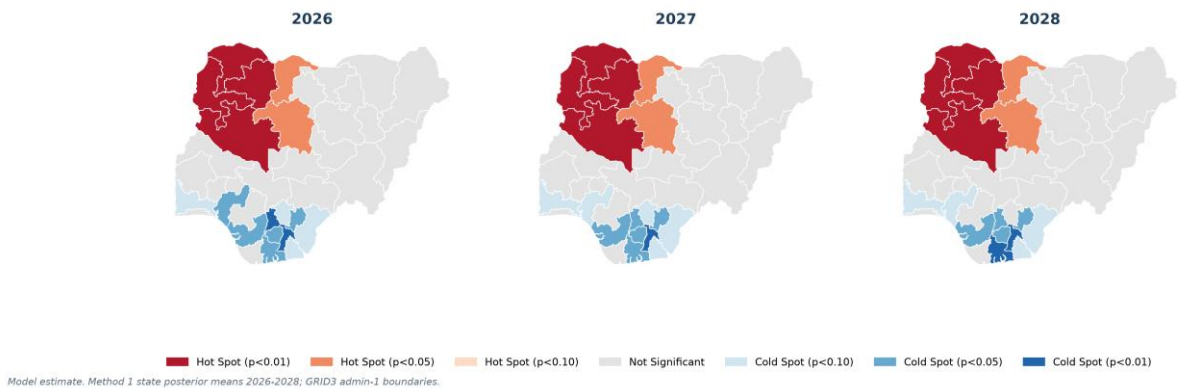


Figure 13. Forecast state zero-dose hotspots, 2026-2028 (Getis-Ord Gi*, Queen contiguity; Method 1).

A ranked list of all 773 modelled LGAs orders the burden for microplanning. The burden is concentrated, but less sharply than an 80/20 distribution would imply. The highest-burden 155 LGAs, the top 20% of LGAs, contain 51% of estimated zero-dose children. Half of the burden is reached with 151 LGAs, 60% with 199 and 80% with 342, about 44% of those modelled.

The practical consequence is a two-wave effort. A first wave targeting about 151 LGAs would reach half of unreached children and is fundable as a single package. Coverage of 80% cannot be achieved from that first wave alone, so the second, extending toward 342 LGAs, should be planned alongside it rather than treated as contingent.

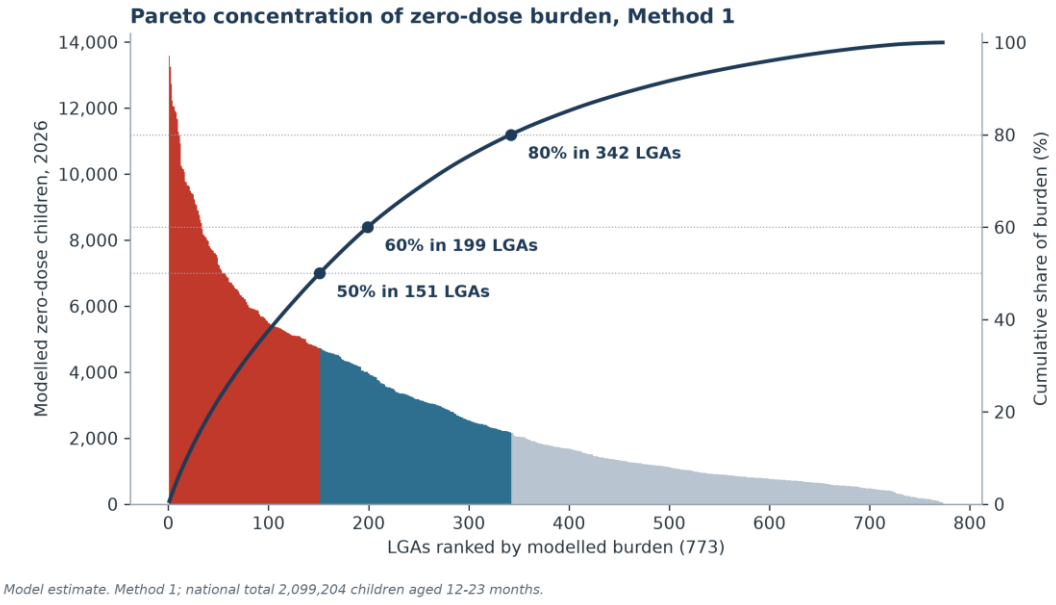


Figure 14. Pareto concentration of zero-dose burden across 773 LGAs (Method 1; total 2.10 million children), with the LGAs needed to reach 50%, 60% and 80%.

5.5.1 Community Archetypes at LGA Level

Community archetypes are the structural profiles behind zero-dose burden: groups of areas that share a barrier mix and therefore need the same response. An earlier version of this clustering was built at state level as a proof of concept, before LGA-level predictors were available. It is not carried forward here. Stakeholder review confirmed that a state-level archetype is too

coarse for operational use, since microplanning, supervision and outreach are organised at local government level and a single state commonly contains areas with materially different barriers. Once LGA-level predictors were identified, the clustering was rebuilt at that resolution, and all archetype results reported here are LGA-level.

The clustering uses agglomerative (Ward) linkage on fifteen indicators in six domains (socioeconomic, maternal care, nutrition, access, insecurity and programme performance) drawn from IHME, DHS, the Meta Relative Wealth Index, Weiss travel-time surfaces and ACLED conflict data, across all 774 LGAs. Five archetypes, also called contextual profiles, are retained, each with a distinct barrier signature. Their labels describe only what the indicators measure; Section 5.5.7 reports how stable the profiles are and how candidate packages are assigned.

The clustering also provides a consistency check on the zero-dose surface. The archetypes were built from contextual indicators: education, nutrition, poverty, wealth, travel time, water, maternal care, conflict and the IHME 2016 DTP1-3 dropout surface. No zero-dose or coverage estimate entered the clustering. The archetypes nevertheless reproduce the geography of missed children: Archetypes 1 and 2 together hold approximately 1.52 million of the 2.10 million modelled zero-dose children (Method 1), about 72% of the national burden (77% under Method 2), across 328 of the 774 LGAs. Because several indicators are also inputs to the estimation models, this is a consistency check rather than independent validation; the independent state-level check against the NmDHS 2025-26 is reported in Section 5.5.7.

Table 6. The five LGA-level community archetypes, their signatures and matched intervention packages.

#	*Profile (example states)*	*LGAs*	*Zero-dose children*	*Mean rate*	*Defining signature*	*Priority intervention*
1	Deprived northern rural (Kano, Jigawa, Katsina)	136	695,337 (33%)	44.2%	Highest child undernutrition; lowest maternal education (~2.5 yrs) and facility delivery (~10%)	Reaching-Every-Community microplanning, BHCPF outreach and community health workers; co-deliver with nutrition and antenatal care
2	Low maternal care, high conflict exposure (Borno, Niger, Adamawa)	192	820,871 (39%)	39.0%	Highest political-violence fatalities; high antigen dropout; weak antenatal care and delivery	Security-integrated microplanning, negotiated access, mobile outreach teams and defaulter tracing
3	Geographically isolated (Cross River, Bayelsa, Delta)	20	23,514 (1%)	17.7%	Extreme travel time to care (~108 minutes); low improved-water access	Outreach and mobile sessions (boat-based where waterways apply), community health worker networks, and multi-antigen bundling per visit
4	Near-average (Osun, Akwa Ibom, Oyo)	280	427,791 (20%)	20.6%	Near the national average on every indicator; fair care-seeking (ANC 72%, delivery 61%)	Core fixed-site and outreach services, ward enumeration, and Periodic Intensification of Routine Immunization
5	Relatively advantaged (Imo, Anambra, Lagos)	146	131,691 (6%)	9.9%	Highest wealth and education (~9.6 yrs) and facility delivery (78%); moderate conflict events	Targeted social mobilisation, a fixed-plus-outreach hybrid, and private-sector last-mile reach

Model estimate. Agglomerative (Ward) clustering of 15 covariates across all 774 LGAs. Example states are illustrative rather than exhaustive, since archetypes are assigned per LGA, so a single state may contain several. Zero-dose children and mean LGA rates are Method 1 estimates; percentages are of the 2.10 million national total. Method 2 values are given in Figure 15 and the government workbook. BHCPF = Basic Health Care Provision Fund. Labels describe measured indicators only; population mobility, migration, riverine settlement and informal settlements are not measured by any input.

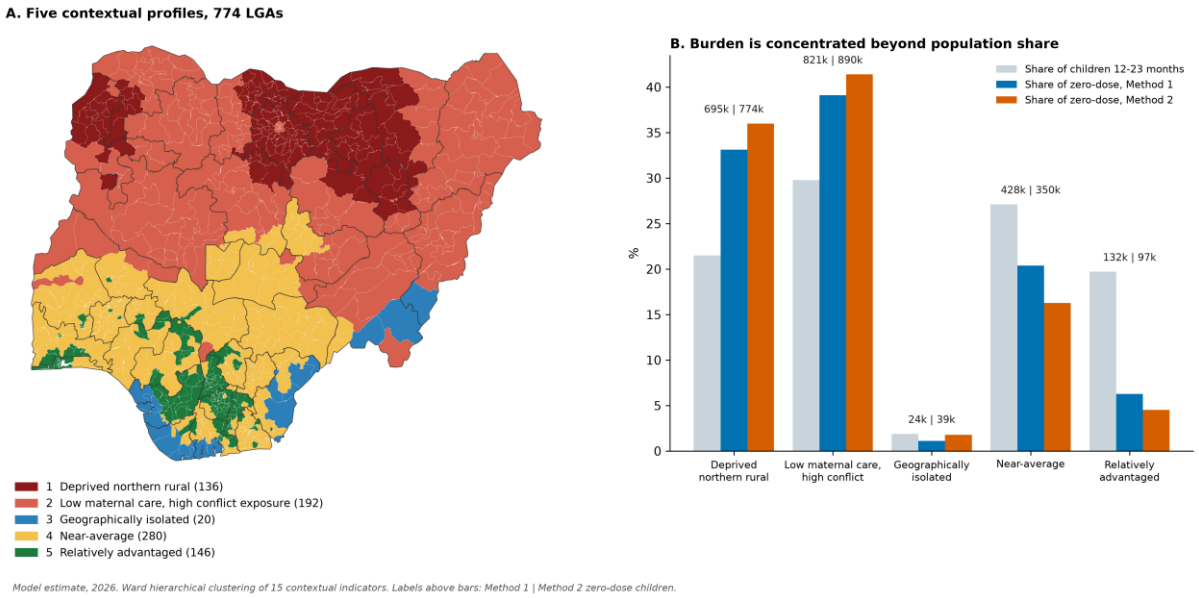


Figure 15. Contextual profiles (community archetypes) of the 774 LGAs and each profile's share of children and of modelled zero-dose children under both methods.

Archetype boundaries do not coincide with state boundaries. A single state may contain both Archetype 1 and Archetype 5 LGAs, which is why intervention packages should be assigned at local government level rather than delivered through whole-state campaigns.

5.5.2 The Ranked LGA Priority List

Every modelled LGA is ranked worst-to-best by estimated zero-dose burden, providing the prioritisation ordering required for microplanning. As Figure 14 shows, the 151 highest-burden LGAs reach half of all zero-dose children, 199 reach 60% and 342 reach 80%. The ten highest-burden LGAs under Method 1 are shown below; the full ranked list for both methods, filterable by state and priority tier, is held in the platform and in the government workbook.

Table 6b. The ten highest-burden LGAs by modelled zero-dose children, 2026 (Method 1).

Rank	*State*	*LGA*	*Zone*	*Zero-dose children*	*Rate (%)*
1	Zamfara	Bukkuyum	North West	13,589	87.8
2	Zamfara	Bakura	North West	13,253	99.0
3	Sokoto	Tambuwal	North West	12,738	83.2
4	Zamfara	Birnin Magaji	North West	12,223	92.8
5	Zamfara	Maradun	North West	12,057	81.2
6	Zamfara	Maru	North West	12,052	57.5
7	Zamfara	Zurmi	North West	11,913	56.6
8	Sokoto	Sokoto North	North West	11,869	75.2
9	Zamfara	Talata Mafara	North West	11,666	75.6
10	Zamfara	Kaura Namoda	North West	11,282	55.3

Model estimate. Source: Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation, 2026; GRID3 boundaries. Hotspot classification: Getis-Ord Gi, k = 5 nearest neighbours (Figures 11–12).*

Two features of this ranking have direct programme implications. First, the top of the list is concentrated in Zamfara and Sokoto, which hold 10 of the ten highest-burden LGAs, while the large birth cohorts of Kano, Katsina and Kaduna place many of their LGAs further down the list; rate-based and burden-based prioritisation therefore produce different lists, and

microplanning needs both. Second, 10 LGAs sit at the 99% upper bound of the Method 1 allocation, which should trigger verification of the local denominator and of reported doses before campaign resources are committed.

This ranking is the operational bridge between the modelling and the annual RI workplan. It should be paired with the archetype assignment in Table 6 so that each priority LGA carries both a position in the queue and a matched response package.

5.5.3 Maternal Care and Immunization Gaps Occupy the Same Geography

The LGAs with the weakest antenatal and delivery care are the same LGAs missing the most zero-dose children. Across the 766 LGAs with maternal-care data and a Method 1 estimate, antenatal care with four or more visits correlates with the modelled zero-dose rate at r = -0.63, and facility delivery at r = -0.59. Both associations are strong and consistent, and they indicate an operational entry point that immunization delivery does not currently use.

The two archetypes with the weakest maternal care, deprived northern rural, and low maternal care with high conflict exposure, hold approximately 1.52 million of the 2.10 million modelled zero-dose children, on antenatal coverage of 36 to 40% and facility delivery of 10 to 19%. Where a maternal contact does occur in these places, it is an existing, funded, already-scheduled point of contact with precisely the households the immunization programme is failing to reach.

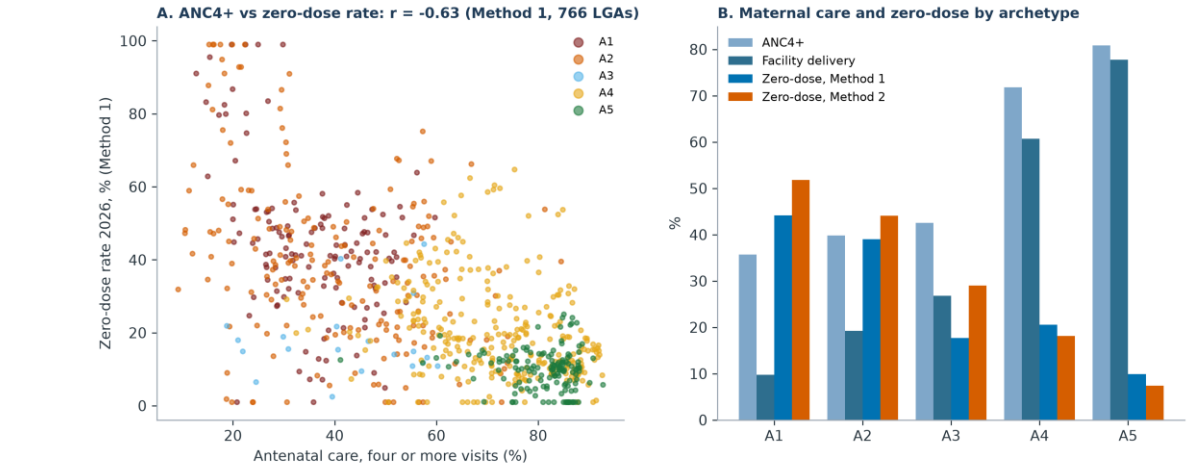


Figure 16. Antenatal care (four or more visits) against the Method 1 zero-dose rate (766 LGAs), and maternal-care indicators and zero-dose under both methods, by archetype.

The implication is that in the highest-burden archetypes, immunization should be co-delivered with antenatal and delivery contacts, so that one visit provides several services, rather than mounted as standalone immunization outreach. This lowers the cost per child reached and uses a contact the household has already accepted.

Table 6c. The RMNCH–EPI operational list: maternal-care indicators and the matched package, by archetype.

Profile	*LGAs*	*ANC4+ (%)*	*Facility delivery (%)*	*Zero-dose rate (%)*	*Zero-dose children*	*RMNCH–EPI intervention*
Deprived northern rural	136	35.7	9.8	44.2	695,337	Reaching-Every-Community microplanning, BHCPF outreach and community health workers; co-deliver immunization with antenatal and nutrition contacts
Low maternal care, high conflict exposure	192	39.8	19.2	39.0	820,871	Security-integrated microplanning, negotiated access, and mobile outreach teams delivering immunization and antenatal care together
Geographically isolated	20	42.6	26.8	17.7	23,514	Outreach and mobile sessions (boat-based where waterways apply), community health worker networks, and multi-antigen and antenatal bundling per visit
Near-average	280	71.8	60.7	20.6	427,791	Ward enumeration and Periodic Intensification of Routine Immunization linked to antenatal clinics
Relatively advantaged	146	80.9	77.8	9.9	131,691	Targeted social mobilization, a fixed-plus-outreach hybrid, and private-sector last-mile reach for immunization and antenatal care

Model estimate. Source: LGA archetype master and the RMNCH–EPI operational workbook (IHME and DHS antenatal-care and facility-delivery covariates; Method 1 modelled zero-dose). All 774 LGAs are ranked worst-to-best in the workbook, each carrying its archetype, antenatal and facility-delivery figures and matched package, ready for microplanning. Facility-level primary health care lists and facility density are not included; those require a GRID3 or National Health Facility Registry dataset.

5.5.4 Method 2: Bayesian Small-Area Estimation for All 774 LGAs

Method 1 estimates zero-dose at state level and distributes each state's estimate to its LGAs using routine DHIS2 Penta1 reporting. A second, independent method was built to estimate LGA zero-dose directly, without routine coverage data, so that LGA priorities could be checked against an approach that does not depend on reported doses or administrative denominators.

Method 2 is a Bayesian small-area estimation (SAE) model. Each LGA has its own zero-dose rate on the logit scale. The population-weighted average of a state's LGA rates is linked to that state's NDHS zero-dose estimate (2008, 2013, 2018 and 2023-24) through a measurement-error likelihood that allows for survey sampling error (design effect 2), so state totals stay consistent with the surveys. Differences between LGAs within a state are explained by six LGA covariates (a maternal-care index combining antenatal care with four or more visits and facility delivery, improved water, the Meta Relative Wealth Index, travel time to a health facility, conflict events and poverty), by a BYM2 spatial random effect on the GRID3 LGA adjacency graph (2,165 neighbour pairs), and by zone effects. The 2026 estimate carries forward each state's survey-anchored level with added process uncertainty. The model was fitted in PyMC with four chains of 8,000 draws after 3,000 tuning steps; every LGA receives a posterior mean, a 95% credible interval, and a probability of being among the 155 highest-burden LGAs.

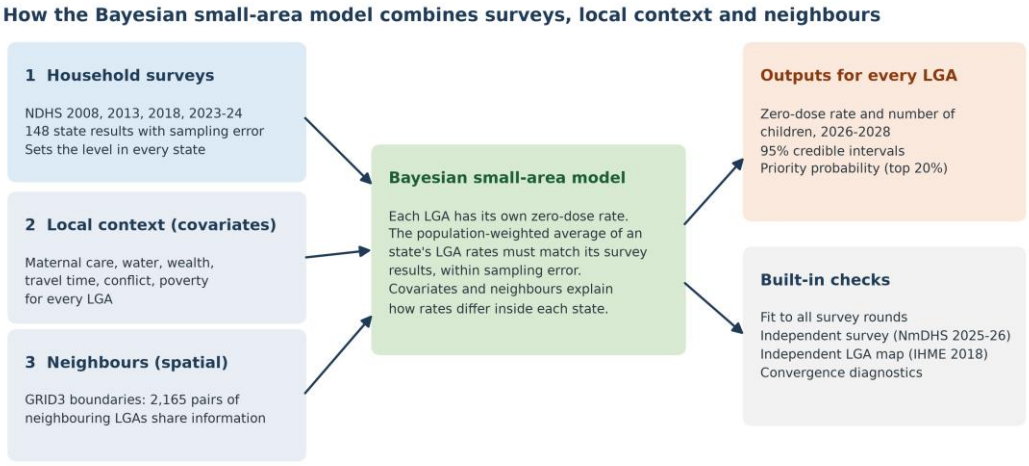


Figure 16b. Structure of the Bayesian small-area model (Method 2).

Method 2 estimates 2.15 million zero-dose children in 2026 (95% credible interval 1.86 to 2.46 million), a national rate of 30.6%. The North-West holds 53% of the total at a zone rate of 50.6%. The 155 highest-burden LGAs hold 54% of zero-dose children, and 40 LGAs are among the top 155 in at least 90% of posterior simulations. Credible intervals are widest where survey anchors and covariates leave most room for error, which identifies the LGAs where local verification is most valuable before large investments.

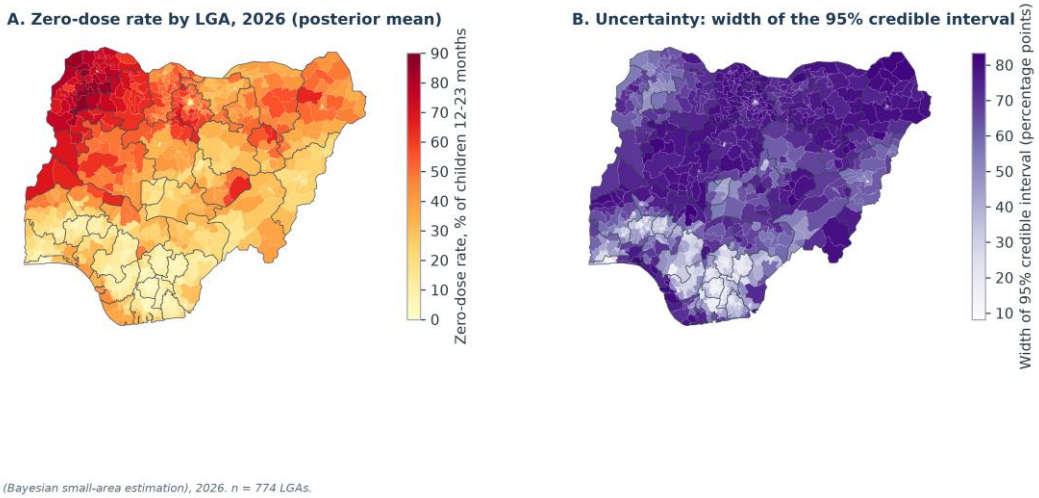


Figure 16c. Method 2 LGA zero-dose rate for 2026 and the width of its 95% credible interval, 774 LGAs.

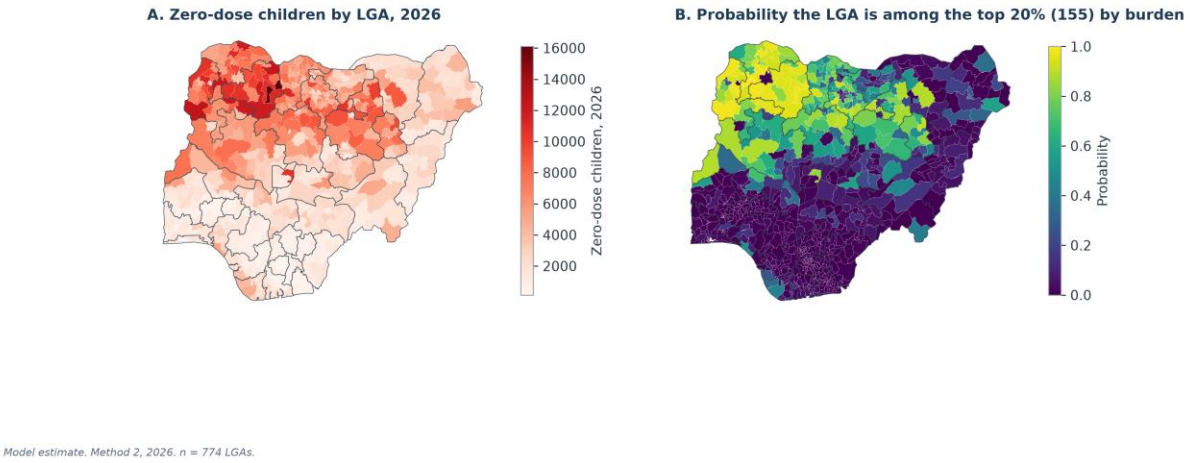


Figure 16d. Method 2 modelled zero-dose children per LGA and the probability that each LGA is among the 155 highest-burden LGAs.

Table 6d. The ten highest-burden LGAs under Method 2, 2026.

Rank	*State*	*LGA*	*Zero-dose children (95% CrI)*	*Rate % (95% CrI)*	*P(top 155)*
1	Zamfara	Gusau	16,100 (5,733-23,565)	64.4 (23-94)	0.98
2	Kebbi	Bagudo	12,567 (5,740-16,083)	76.2 (35-97)	0.98
3	Zamfara	Zurmi	12,406 (4,266-18,113)	64.7 (22-95)	0.96
4	Sokoto	Tambuwal	12,170 (7,507-14,125)	85.0 (52-99)	0.99
5	Kebbi	Danko/Wasagu	12,041 (3,977-17,413)	65.5 (22-95)	0.96
6	Zamfara	Maru	11,904 (3,755-17,925)	62.3 (20-94)	0.95
7	Sokoto	Gada	11,871 (5,327-15,319)	75.2 (34-97)	0.98
8	Kebbi	Birnin Kebbi	11,293 (3,063-17,295)	60.7 (16-93)	0.94
9	Katsina	Kankara	11,171 (3,058-17,195)	61.2 (17-94)	0.94
10	Zamfara	Tsafe	11,115 (3,690-16,396)	63.9 (21-94)	0.95

Model estimate. Source: Method 2: Bayesian small-area estimation (SAE), 2026. CrI = credible interval; P(top 155) = share of 4,000 posterior draws in which the LGA ranks among the 155 highest-burden LGAs.

Maternal care is the strongest and most certain predictor of differences between LGAs within a state: one standard deviation higher on the maternal-care index is associated with a log-odds change of -0.83 (95% credible interval -1.29 to -0.37), an odds ratio of 0.44. Improved water is also associated with lower zero-dose; the remaining effects are smaller and their intervals include zero. These are partial associations used to distribute survey levels within states, not causal effects.

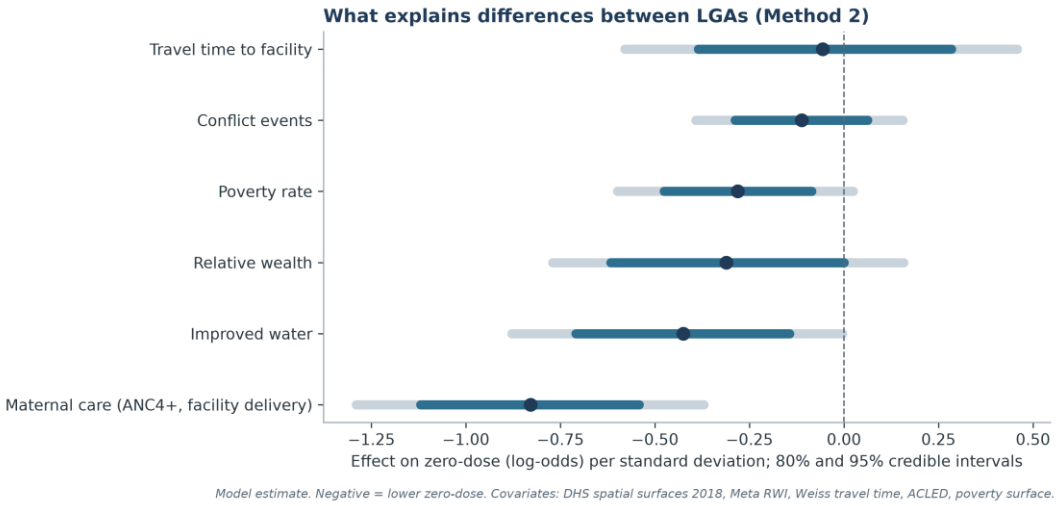


Figure 16e. Posterior covariate effects in the Bayesian small-area model (log-odds of zero-dose per standard deviation; 80% and 95% credible intervals).

The model converged cleanly: maximum R-hat 1.003, minimum bulk effective sample size 2,549, minimum tail effective sample size 2,524 and no divergent transitions across 1,649 monitored parameters. Posterior predictive intervals contain 98% of the 148 NDHS state-round observations, and the 2026 state estimates rank the 37 states in close agreement with the NmDHS 2025-26, which was not used in fitting (ρ = 0.87; mean absolute error 5.9 percentage points; bias +0.1).

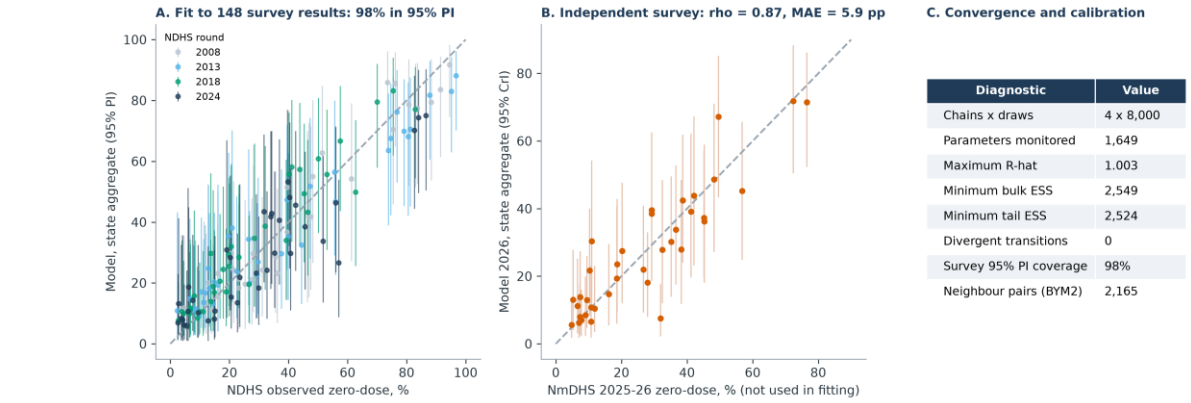


Figure 16f. Method 2 model performance: fit to all NDHS rounds, independent NmDHS 2025-26 check, and Markov chain Monte Carlo diagnostics.

Method 2 does not use routine administrative coverage. Dividing DHIS2 Penta1 doses by the estimated LGA cohort gives coverage above 100% in 81% of LGAs, and state zero-dose derived from routine data alone agrees poorly with the NmDHS 2025-26 (ρ = 0.18; mean absolute difference 27 percentage points). Routine data remain essential for tracking trends and data quality, but survey-anchored estimates are more reliable for the level of zero-dose in each LGA.

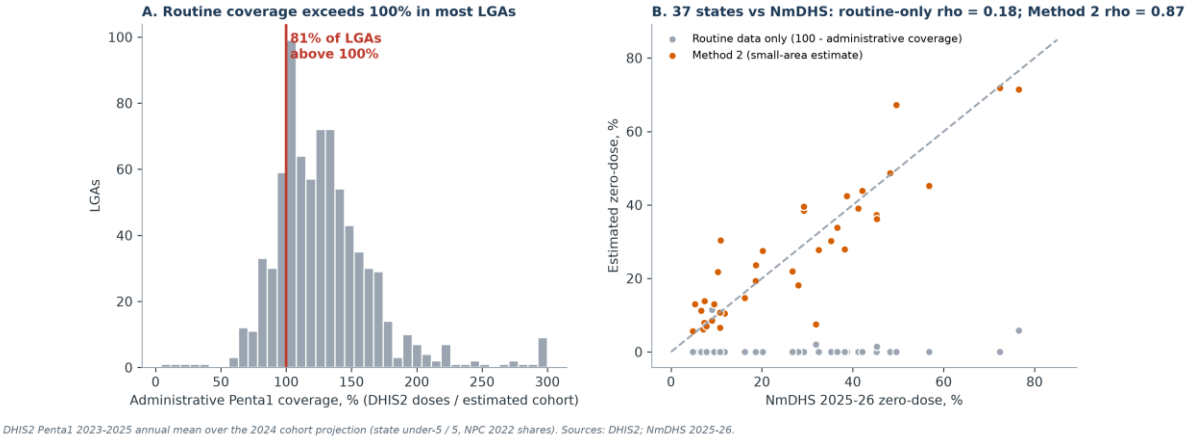


Figure 16g. Routine administrative Penta1 coverage compared with survey-anchored small-area estimates.

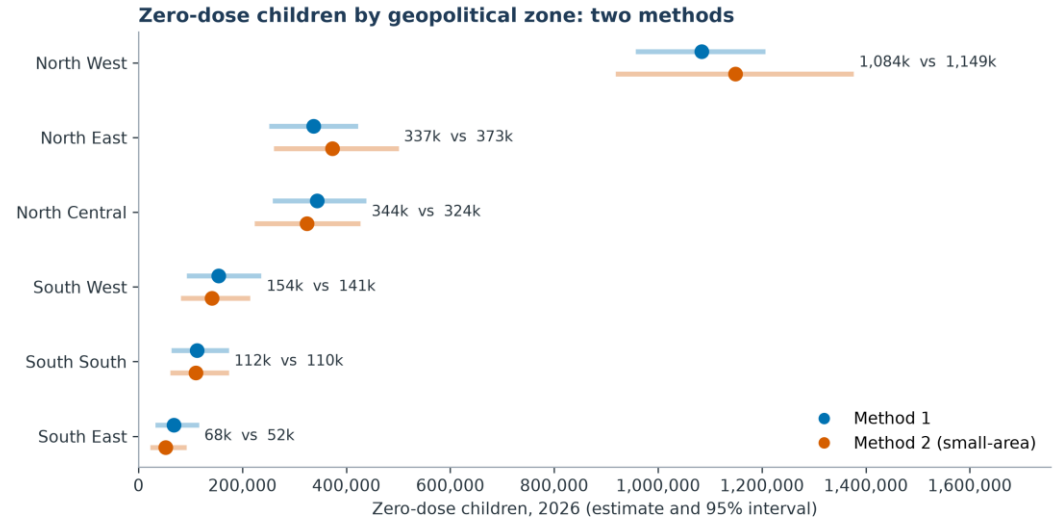
5.5.5 Comparison of Method 1 and Method 2

The two methods agree closely on national, zone and state totals. Nationally they differ by 2.4% (2.10 and 2.15 million), the 95% intervals overlap in every zone, and both place about half of all zero-dose children in the North-West. Across the 37 states, the rank correlation of modelled zero-dose children between methods is ρ = 0.96.

Table 6e. Modelled zero-dose children in 2026 by zone, Method 1 and Method 2.

Zone	*LGAs (M1/M2)*	*Method 1 zero-dose children (95% interval)*	*M1 rate %*	*Method 2 zero-dose children (95% CrI)*	*M2 rate %*	*M2 vs M1*	*NmDHS 2025-26 %*
North West	186/186	1,084,155 (961,608-1,201,516)	47.8	1,149,017 (923,311-1,371,914)	50.6	+6.0%	48.7
North East	111/112	336,687 (255,921-417,822)	31.4	372,892 (265,189-495,882)	34.7	+10.8%	38.7
North Central	121/121	343,731 (262,802-433,604)	33.1	324,138 (228,447-421,964)	31.2	-5.7%	34.2
South West	137/137	154,451 (97,752-231,362)	14.5	141,155 (86,336-210,104)	13.2	-8.6%	14.4
South South	123/123	112,309 (67,941-169,284)	13.7	110,354 (65,945-169,111)	13.4	-1.7%	8.8
South East	95/95	67,871 (36,662-111,588)	9.1	52,005 (26,981-87,430)	7.0	-23.4%	11.2
Nigeria	773/774	2,099,204 (1,898,902-2,308,432)	29.9	2,149,561 (1,856,459-2,455,411)	30.6	+2.4%	-

Model estimate, 2026. Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation; Method 2: Bayesian small-area estimation (SAE). NmDHS 2025-26 zone rates for reference (not used in fitting).



Model estimate. Method 1 intervals reflect state-model uncertainty; Method 2 intervals are full posterior credible intervals.

Figure 16h. Modelled zero-dose children by zone, 2026, under both methods, with 95% intervals.

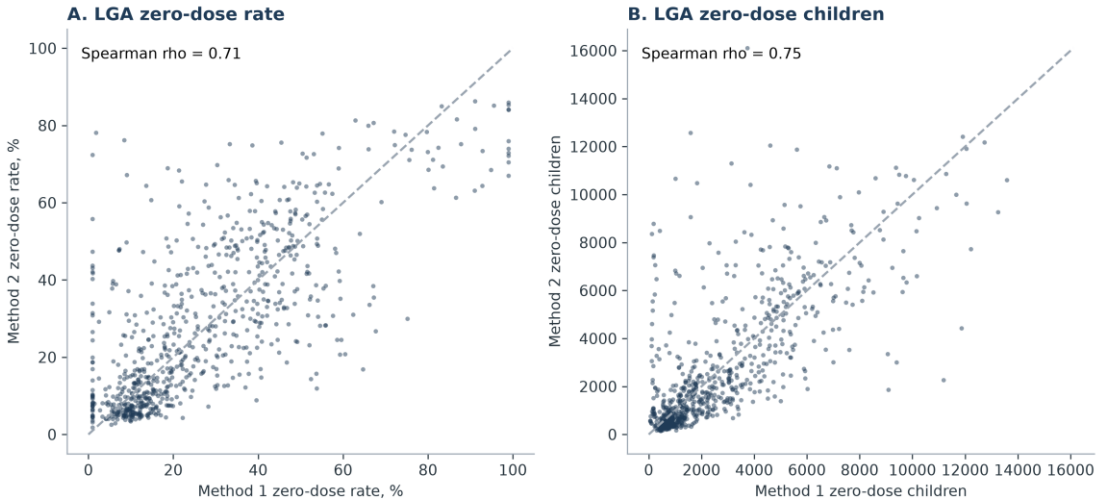
Table 6f. Modelled zero-dose children in 2026 by state, Method 1 and Method 2.

State	*Zone*	*Method 1 children*	*M1 rate %*	*Method 2 children*	*M2 rate %*	*NmDHS 2025-26 %*
Kano	North West	198,101	36.8	227,157	42.4	38.7
Katsina	North West	169,685	39.0	211,744	48.6	48.2
Sokoto	North West	166,648	72.7	167,720	71.4	76.5
Kebbi	North West	153,659	69.0	162,123	71.8	72.4
Zamfara	North West	150,220	70.9	143,697	67.2	49.6
Kaduna	North West	139,578	42.0	119,060	37.3	45.2
Jigawa	North West	106,264	34.8	117,518	38.5	29.2
Bauchi	North East	111,948	33.3	118,781	36.2	45.3
Borno	North East	60,604	30.3	70,746	39.1	41.2
Gombe	North East	50,128	34.2	57,947	39.6	29.2
Yobe	North East	43,282	35.3	53,059	43.8	42.1
Taraba	North East	35,731	29.2	34,085	27.9	38.2
Adamawa	North East	34,994	20.1	38,275	22.0	26.7
Niger	North Central	104,030	45.0	104,633	45.2	56.8
Benue	North Central	52,312	28.9	42,277	23.6	18.7
Kogi	North Central	48,476	31.3	41,779	27.5	20.2
Kwara	North Central	47,518	36.6	42,528	33.8	36.6
Plateau	North Central	42,309	27.9	44,941	30.2	35.2
Nasarawa	North Central	28,735	25.6	29,256	27.8	32.5
FCT	North Central	20,351	18.5	18,725	19.3	18.6
Oyo	South West	45,265	19.3	42,253	18.1	28.0
Ogun	South West	32,123	18.9	22,819	14.7	16.2
Ondo	South West	28,122	18.7	32,119	21.7	10.3
Lagos	South West	22,095	7.1	18,709	6.2	7.1
Osun	South West	14,337	12.1	12,763	10.7	10.8
Ekiti	South West	12,509	11.2	12,492	11.2	6.6
Rivers	South South	30,795	14.7	26,122	13.0	9.5
Delta	South South	21,940	12.9	23,510	13.9	7.4
Akwa Ibom	South South	19,144	13.1	13,242	10.5	11.8
Edo	South South	15,905	10.0	12,516	8.5	9.0
Bayelsa	South South	12,434	18.0	20,938	30.3	10.9
Cross River	South South	12,091	11.2	14,026	13.0	5.3
Anambra	South East	18,507	9.8	14,459	7.9	7.3
Enugu	South East	14,715	9.9	10,442	7.0	7.8
Imo	South East	14,271	8.5	9,100	5.6	4.8
Abia	South East	12,741	8.9	10,508	7.5	31.9
Ebonyi	South East	7,637	6.7	7,496	6.6	10.8

Model estimate, 2026. Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation; Method 2: Bayesian small-area estimation (SAE). States ordered by zone and Method 1 burden.

At LGA level the methods agree on which LGAs carry the burden (rank correlation ρ = 0.75 across 773 LGAs; 107 of the top 155 LGAs are shared), but individual LGAs can differ

substantially. Method 1 follows each LGA's share of reported Penta1 doses: very high reported doses push its rate toward the 1% floor (for example Rabah and Goronyo in Sokoto) and low reported doses push it up (for example Sokoto North). Method 2 is not affected by reporting volumes. LGAs with large differences between the methods are therefore priorities for data-quality review as well as for programme attention.



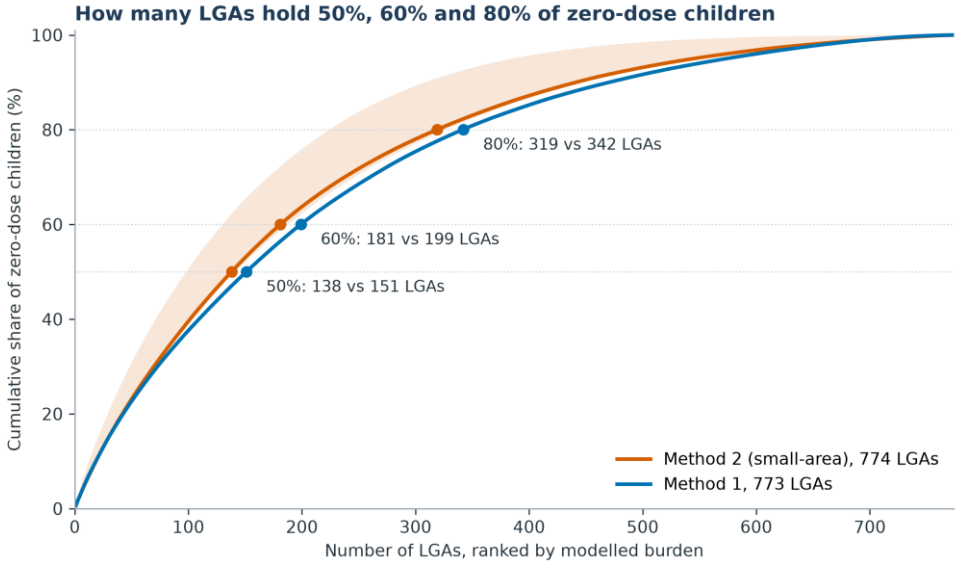
Model estimate, 2026.

Figure 16i. LGA-level agreement between Method 1 and Method 2, 773 LGAs.

Table 6g. LGAs needed to reach 50%, 60% and 80% of zero-dose children, by method.

Share of zero-dose children	*Method 1: LGAs (% of LGAs)*	*Method 2: LGAs (% of LGAs)*
50%	151 (19.5%)	138 (17.8%)
60%	199 (25.7%)	181 (23.4%)
80%	342 (44.2%)	319 (41.2%)
Share held by the top 155 LGAs (20%)	50.9% (46.9-54.7)	54.2% (49.6-58.6)

Model estimate, 2026. LGAs ranked from highest to lowest modelled burden; intervals are 95% credible intervals.



Model estimate, 2026.

Figure 16j. Cumulative share of modelled zero-dose children by number of LGAs targeted, both methods.

Table 6h. Which estimate to use for which decision.

Decision	*Recommended estimate*	*Reason*
National, zone and state totals	Either; report Method 1 with Method 2 as the sensitivity range	Totals agree within 2.4% nationally; state ranks agree (ρ = 0.96); both match NmDHS 2025-26
Ranking LGAs within a state	Method 2	Covers all 774 LGAs with full uncertainty; within-state agreement with IHME ρ = 0.54 vs 0.08
Selecting priority LGAs for funding	The 107 LGAs in the top 155 under both methods first	Robust to the choice of method; then add LGAs with high Method 2 priority probability
Tracking change and early warning	Method 1	Uses the DHIS2 trend and updates with monthly routine data
Data-quality review	LGAs where the methods differ most	Large gaps flag reported doses out of line with population

5.5.6 Validation, Calibration and Uncertainty-Aware Prioritization

Method 2 was tested against the NmDHS 2025-26 state estimates, which were not used to fit any model, alongside four alternative approaches. It had the lowest mean absolute error (MAE) of the approaches tested: 5.9 percentage points, with near-zero bias (+0.1) and a rank correlation of 0.87 (95% CI 0.73 to 0.95). Method 1 was close (MAE 6.3; difference 0.4 points, 95% CI -0.8 to 1.6), so the two methods are statistically equivalent at state level. Gradient boosting (7.5) was less accurate, and a Fay-Herriot area-level model (7.9) and the latest survey carried forward (8.2) were less accurate with intervals that just include zero. Zero-dose derived from routine administrative data alone was far less accurate (MAE 26.5).

Table 6i. External validation against the NmDHS 2025-26, 37 states.

Approach	*MAE*	*Bias*	*Spearman ρ (95% CI)*	*CRPS*	*MAE difference from Method 2 (95% CI)*
Method 2: Bayesian small-area estimation (SAE)	5.9	+0.1	0.87 (0.73-0.95)	4.4	Reference
Method 1: Bayesian hierarchical model with DHIS2-calibrated LGA allocation	6.3	-0.6	0.89 (0.75-0.96)	4.5	+0.4 (-0.8 to 1.6)
Gradient boosting	7.5	+1.4	0.89 (0.75-0.95)	-	+1.6 (0.3 to 2.8)
Fay-Herriot area-level model	7.9	+4.2	0.84 (0.68-0.93)	5.7	+2.1 (-0.0 to 4.2)
Latest survey carried forward	8.2	+3.6	0.84 (0.69-0.92)	6.1	+2.3 (-0.2 to 4.9)
Routine data only	26.5	-26.4	0.18	-	+20.6 (14.9 to 26.3)

Model estimate. MAE and bias in percentage points of state zero-dose prevalence; CRPS = continuous ranked probability score (lower is better); CI = bootstrap confidence interval over states. The NmDHS 2025-26 was not used in fitting.

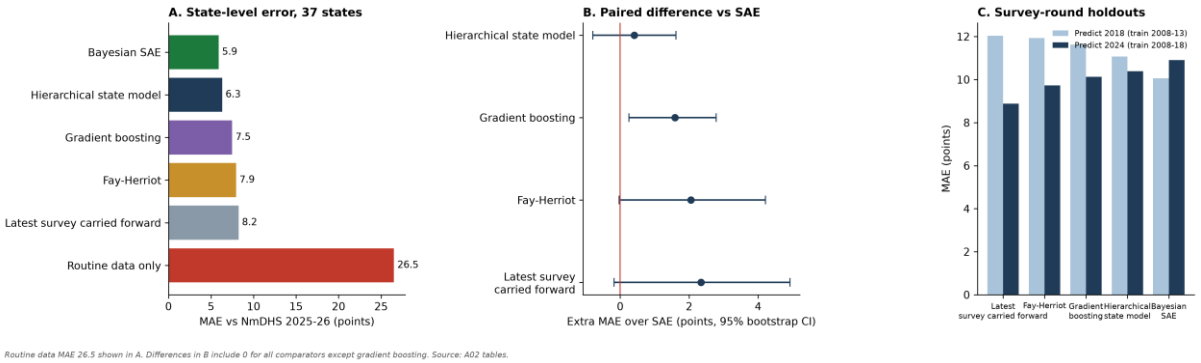
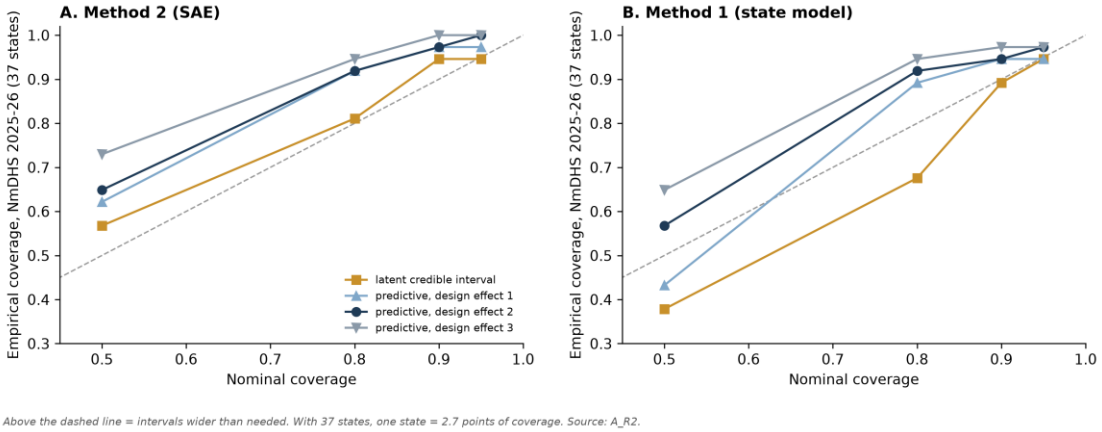


Figure 16k. Comparative performance: state-level error against the NmDHS 2025-26, paired differences from Method 2, and survey-round holdouts.

The uncertainty intervals are well calibrated. Allowing for survey sampling error with a design effect of 2, the 95% predictive interval contained the NmDHS value in 37 of 37 states and the 50% interval in 24 of 37, consistent with nominal coverage (p = 0.10 and 0.26). The narrower credible interval for the underlying rate, which excludes sampling error, covered 57% and 95%. The continuous ranked probability score (CRPS), which rewards both accuracy and sharpness, was 4.4 for Method 2 and 4.5 for Method 1. Within states, Method 2 LGA estimates agree with

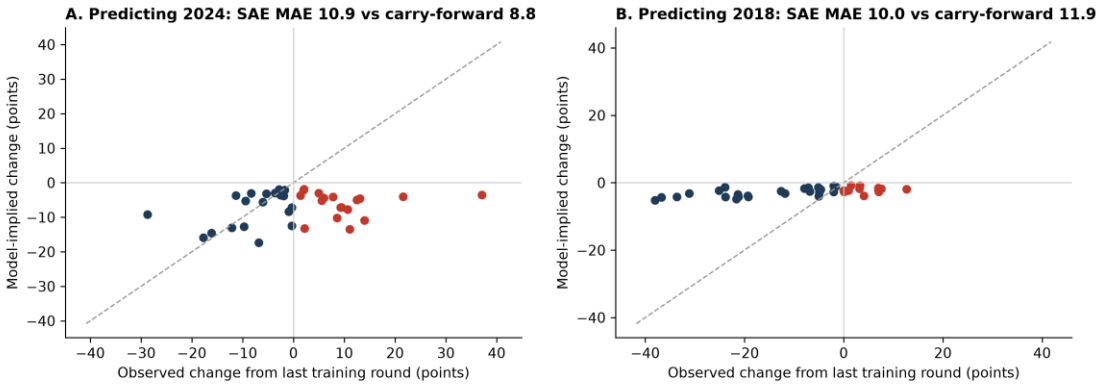
the independently modelled IHME 2018 LGA estimates (within-state rank correlation 0.54, positive in 36 of 37 states), compared with 0.08 for allocation by routine-data shares and 0.04 for a model without covariates.



Above the dashed line = intervals wider than needed. With 37 states, one state = 2.7 points of coverage. Source: A_R2.

Figure 16l. Calibration: empirical against nominal coverage of credible and predictive intervals, by design effect, for both methods.

Projecting ahead is harder than describing the present. When survey rounds were held out, Method 2 predicted 2018 better than carrying the 2013 survey forward (MAE 10.1 vs 12.0) but predicted 2024 less well than carrying 2018 forward (10.9 vs 8.9). In 18 states the model projected the wrong direction of change between 2018 and 2024, and errors there were nearly twice as large (MAE 14.3 vs 7.7). The 2026 estimates therefore carry real trend uncertainty and should be refreshed whenever a new survey round becomes available.



Red = model projected the wrong direction of change. 2018-2024 stagnation broke the 2008-2018 decline. Source: A_R4.

Figure 16m. Temporal extrapolation: observed against model-implied change from the last training round; red marks a wrong projected direction.

Uncertainty matters most for targeting. Across 4,000 posterior draws, 40 LGAs are near-certain members of the 155 highest-burden LGAs (probability 0.9 or more), 517 are unlikely members (0.1 or less) and 217 are uncertain, of which 102 fall outside the ranked list of 155. Ranking by Method 2 expected burden captures 54% of modelled zero-dose children with 155 LGAs, compared with 48% when ranking by rate alone and 30 to 34% when ranking by routine administrative coverage. Each model scores its own list highest, so these capture figures are model-based rather than observed outcomes.

Table 6j. Share of modelled zero-dose children (%) captured by priority lists of different sizes.

Priority rule	*50 LGAs*	*100 LGAs*	*155 LGAs*	*155 LGAs, scored under Method 1*	*200 LGAs*
Routine administrative coverage	11.8	20.3	30.2	33.9	33.4
Method 2 rate only	18.1	34.5	47.7	41.8	57.6
Method 1 expected burden	18.2	32.2	45.3	50.8	54.4
Method 2 expected burden	23.2	39.6	54.2	42.7	63.8
Method 2 probability of top-155 membership	22.6	39.1	54.2	43.0	63.7
Upper bound (true top-K in each draw)	26.4	44.2	59.0	51.7	68.3

Model estimate. Shares are scored under the Method 2 posterior unless stated. The upper bound selects the true highest-burden LGAs in each posterior draw. Model-based, not observed outcomes.

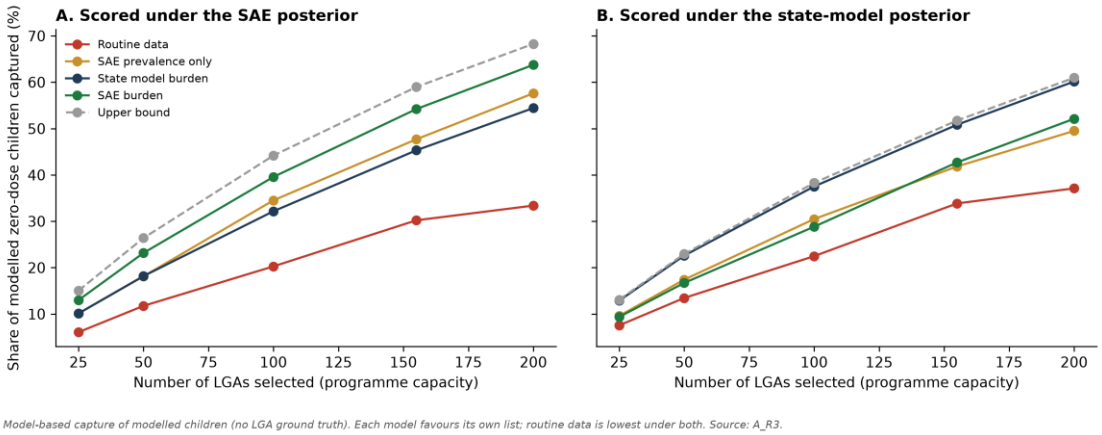


Figure 16n. Share of modelled zero-dose children captured by priority lists of 25 to 200 LGAs, scored under each model's posterior.

For planning, the 40 near-certain LGAs should be funded first, and uncertain LGAs should be verified through microplanning or rapid assessment before full investment. The results are robust: across design-effect, prior and spatial specifications the state MAE ranges from 5.86 to 6.02. Updating the denominator to the 2025 under-five projection raises the national total by 12.7% but changes only 4 of the 155 priority LGAs.

5.5.7 Contextual Profiles and Candidate Intervention Packages

The five community archetypes in Section 5.5.1 were re-examined as contextual profiles: descriptive groupings of LGAs by fifteen indicators in six domains (socioeconomic, maternal care, nutrition, access, insecurity, and programme performance measured as DTP1-3 dropout). The labels now describe only what the indicators measure: deprived northern rural; low maternal care with high conflict exposure; geographically isolated; near-average; and relatively advantaged. Earlier labels referring to nomadism, migration, riverine settlement and urban slums were retired because no input measures those attributes. Population mobility in particular must be established locally before transit-point strategies are considered.

Five profiles are an operational choice rather than a statistical optimum. Four profiles separate the data more cleanly (silhouette 0.31 vs 0.18) by merging the near-average and relatively advantaged groups, but those two groups differ in burden (16.3% vs 4.5% of modelled zero-dose children) and in dominant barrier. Assignments are moderately stable (bootstrap adjusted Rand index (ARI) 0.54, 95% interval 0.42 to 0.72), and the profiles are geographically coherent: 79% of neighbouring LGAs share a profile, against 26% expected by chance.

Membership is clear for 46% of LGAs (60% of modelled zero-dose children), intermediate for 31% and mixed for 23%.

Table 6k. Contextual profiles: size, share of children and of modelled zero-dose children, and main barriers.

Profile	*LGAs*	*Share of children (%)*	*Share of zero-dose, Method 2 (%)*	*Mean rate (%)*	*Domains at or above +0.5 SD*	*Candidate components (centroid rule)*
Deprived northern rural	136	21.5	36.0	51.3	Socioeconomic, maternal care, nutrition, DTP1-3 dropout	Defaulter tracing, demand and cost, mch integration, nutrition integration
Low maternal care, high conflict exposure	192	29.8	41.4	42.6	Socioeconomic, maternal care, nutrition, insecurity, DTP1-3 dropout	Defaulter tracing, demand and cost, mch integration, nutrition integration, outreach, security-sensitive access
Geographically isolated	20	1.9	1.8	29.1	Access	Mch integration, outreach
Near-average	280	27.1	16.3	18.4	None	Core only
Relatively advantaged	146	19.7	4.5	7.0	None	Security-sensitive access

Model estimate. Ward hierarchical clustering of 15 indicators, 774 LGAs. Domain scores are standardized means (higher = more disadvantaged); SD = standard deviation. Shares use the Method 2 zero-dose estimates. A common core of fixed-site and outreach routine immunization with microplanning applies to every LGA.

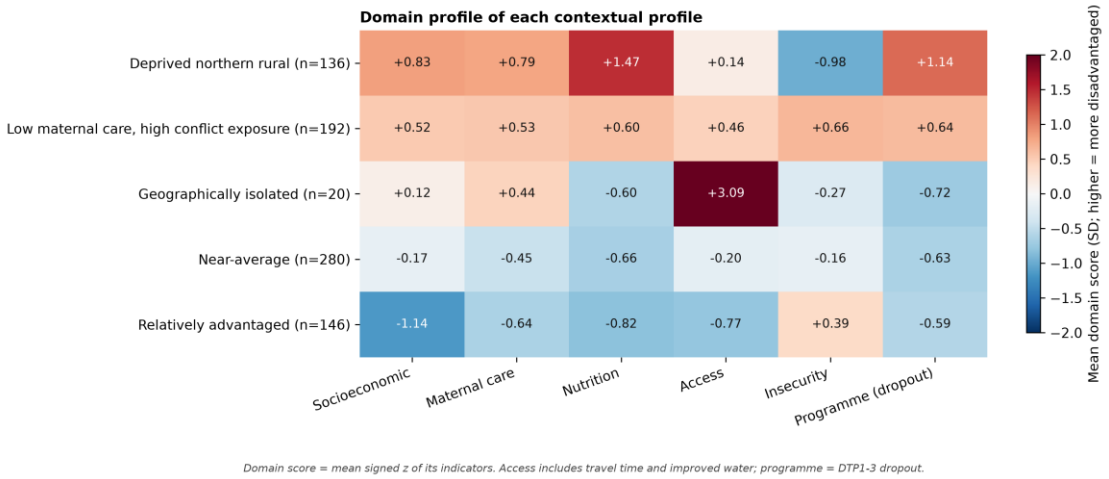


Figure 16o. Mean domain score of each contextual profile across six domains (higher = more disadvantaged).

Contextual barriers vary within states as much as between them for access and insecurity: 56% of the variance in access and 59% in insecurity lies within states, compared with 22% for maternal care and 5% for nutrition. Thirty-five of the 37 states contain LGAs with different dominant barriers. Among LGA pairs with nearly identical estimated zero-dose rates (within one point), 49.6% belong to different profiles and 67% have a different dominant barrier, although such pairs are contextually closer than random pairs (median distance 1.8 vs 2.8). The same zero-dose rate can therefore reflect different barriers in a substantial minority of LGAs, which a rate-only ranking cannot show.

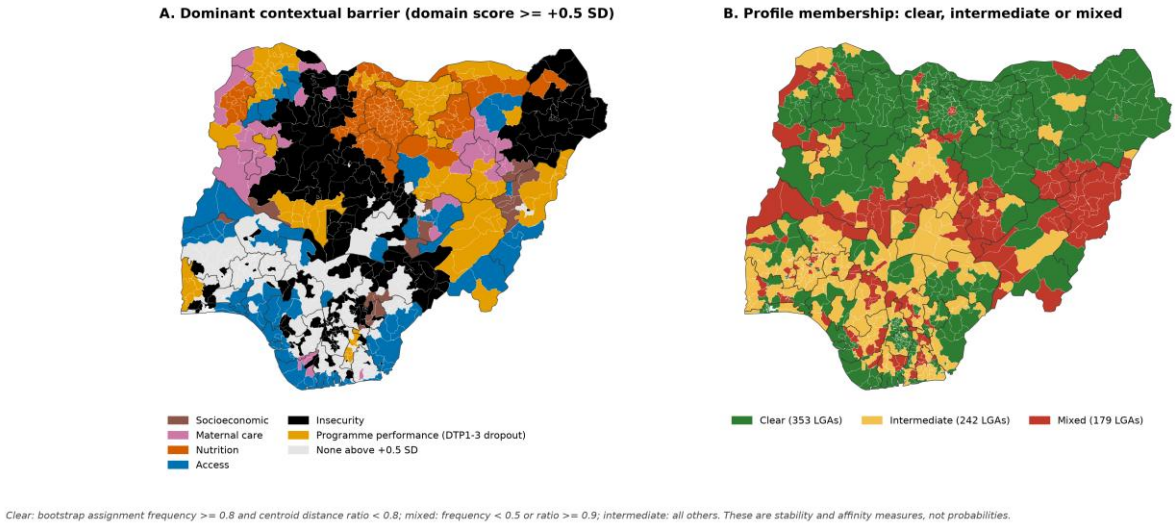


Figure 16p. Dominant contextual barrier by LGA, and profile membership classified as clear, intermediate or mixed.

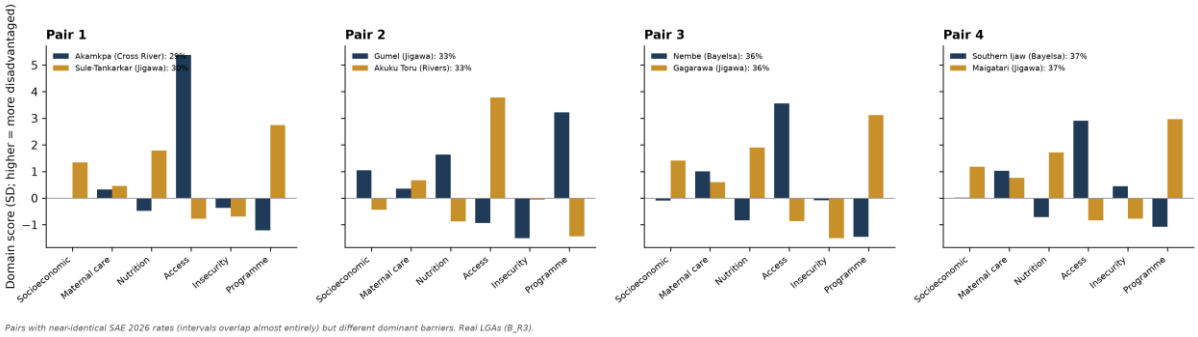


Figure 16q. LGA pairs with near-identical estimated zero-dose rates but different dominant barriers.

The two highest-burden profiles hold 77.4% of Method 2 modelled zero-dose children but 51.3% of the birth cohort. At state level, the share of children living in these two profiles improves prediction of the NmDHS 2025-26 beyond the NDHS 2024 (leave-one-out R-squared 0.71 to 0.78); the NmDHS was not used to build any indicator. Out of sample at LGA level, profiles add information beyond geopolitical zone (R-squared gain +0.08) but little beyond zone and a deprivation quintile (+0.01), partly because they share inputs with the estimation models. Profiles are therefore a summary for planning, not an independent predictor of zero-dose.

Candidate packages are assigned from each LGA's own flagged barriers using documented rules (Table 6l): a barrier is flagged when its domain score is in the national top quartile, or travel time is in the top decile. Of the 774 LGAs, 36%, holding 67% of modelled zero-dose children, have two or more flagged barriers, so combined packages are the norm where burden is highest.

Table 6l. Barrier-to-intervention mapping: flagging rules, candidate components and local-validation requirements.

Barrier	*Flagging rule*	*Candidate components*	*Evidence or guideline*	*Local validation required*
Access	Access domain score in national top quartile, or travel time in top decile	Outreach and mobile sessions; transport support and last-mile logistics; boat-based delivery where waterways apply	WHO Reaching Every District / Reaching Every Community (RED/REC); integrated outreach guidance	Microplan catchment mapping; confirm terrain and settlement pattern
Maternal care	Maternal care domain score in national top quartile	Integrate immunization with antenatal, delivery, postnatal and child-welfare contacts	WHO/UNICEF integrated MNCH service delivery; Gavi IRMMA 'Reach'	Facility functionality and service-integration review
Nutrition	Nutrition domain score in national top quartile	Co-deliver with nutrition screening, growth monitoring and child-health days	WHO integrated child health days; nutrition-immunization co-delivery guidance	Confirm nutrition partner coverage
Socioeconomic	Socioeconomic domain score in national top quartile	Community engagement and demand generation; reduce indirect costs (transport, time); targeted social support	WHO Behavioural and Social Drivers of vaccination (BeSD); Big Catch-Up guidance	BeSD survey or rapid assessment
Insecurity	Insecurity domain score in national top quartile	Security-sensitive flexible scheduling; negotiated or community-supported access; coordination with humanitarian services	WHO/UNICEF immunization in humanitarian and conflict settings; polio negotiated-access experience	Security assessment with state and humanitarian partners
Programme performance (DTP1-3 dropout)	Programme domain score in national top quartile	Defaulter tracing, reminder-recall, home-based records	WHO RED/REC defaulter tracking; reminder-recall evidence (Cochrane reviews)	Check register quality and DHIS2 dropout
Population mobility (not measured)	Cannot be flagged from current data	Transit-point vaccination; portable records; cross-catchment coordination	WHO permanent transit-point strategy	Required: mobility must be established locally before this component is considered

Rules are applied to each LGA's domain scores. Components are candidates for local validation, not measured intervention effects. Population mobility has no input indicator and is never assigned from these data.

How packages are assigned matters. Measured against each LGA's barrier-based package (burden-weighted Jaccard similarity, where 1 means identical), assigning by state agrees best (0.65), followed by contextual profile (0.58), profile plus secondary profile (0.53), geopolitical zone (0.42), deprivation quintile (0.41), zero-dose rate quintile (0.40) and one national package (0.13). State packages refined with each LGA's flagged barriers are therefore the most consistent with the data, and a single national package the least. These are agreements with documented rules, not measured effects of interventions.

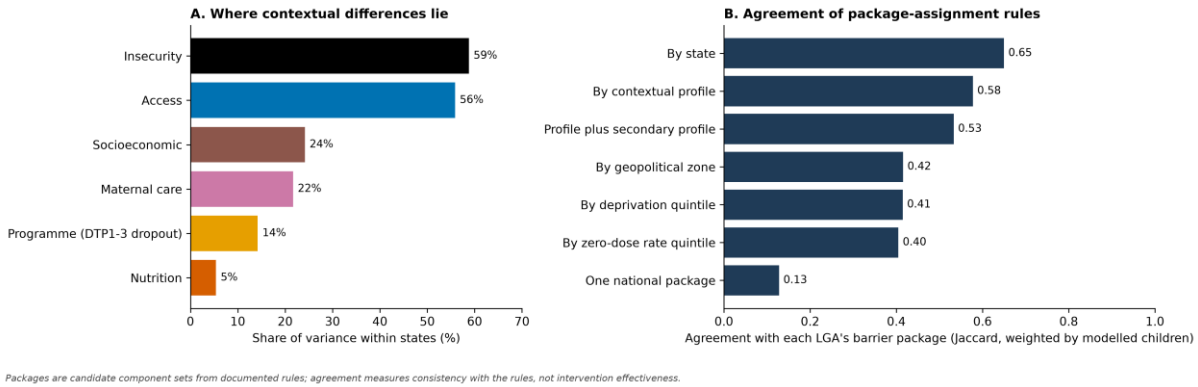
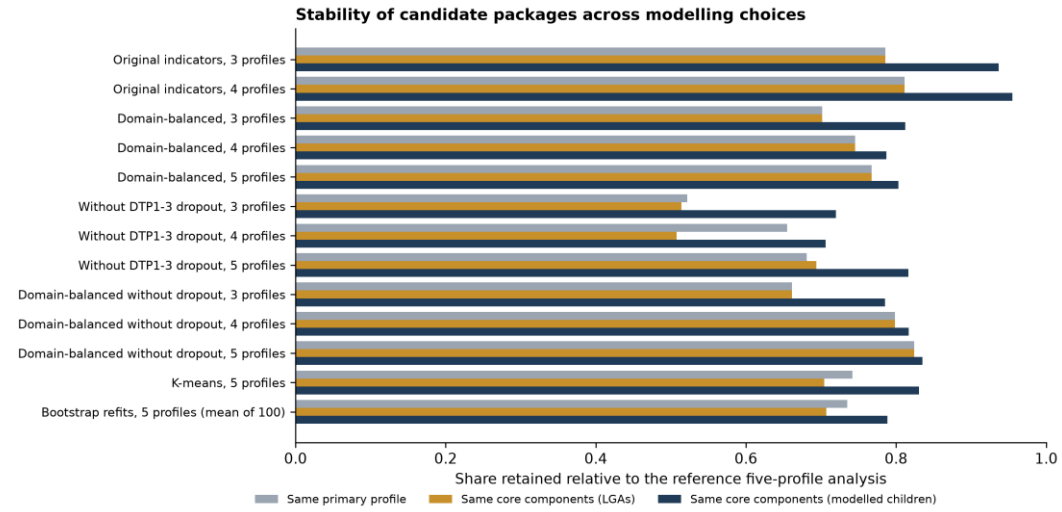


Figure 16r. Share of each domain's variance lying within states, and agreement of package-assignment rules with each LGA's barrier package.

Core package components are retained for 51 to 82% of LGAs (71 to 96% of modelled zero-dose children) across alternative indicator sets, numbers of profiles, clustering algorithms and

bootstrap refits, and for 73 to 82% of LGAs when the barrier-flagging threshold is changed. The LGA barrier flags do not depend on the clustering at all. Packages should be validated locally through microplanning before adoption.



Core components = Jaccard similarity >= 0.67. LGA-level barrier flags do not depend on clustering; changing the flagging threshold retains 73-82%.

Figure 16s. Stability of candidate packages across indicator sets, numbers of profiles, algorithms and bootstrap refits.

5.6 Domain 6 — Polio Risk and the Interruption Pathway

A Bernoulli case-control spatial scan (71 confirmed polio cases against 78,862 non-polio AFP controls) detects one statistically significant residual cluster in the far north-east near the international border (relative risk about 10.5). Wild poliovirus has been absent since 2017; the remaining signal is vaccine-derived. Because the scan uses non-polio AFP from the same surveillance system as controls, it adjusts for surveillance intensity, so the cluster is a true excess rather than an artefact. Everywhere else, confirmed cases are statistically indistinguishable from background; there is no active reservoir elsewhere.

Modelling the interruption pathway with a negative-binomial GLM (Prophet cross-check), the projected annual case count declines and the upper 95% bound falls below one case per year before 2030, a credible, if conditional, interruption pathway. A recurrence-free survival analysis gives a 21.3% probability of remaining case-free for a full 36 months nationally; eighteen states are already past the WHO 36-month certification threshold, while six (Taraba, Kogi, Adamawa, Borno, Gombe, Kano) remain inside the window and gate national certification, with Borno and Kano the live reservoir. Surveillance timeliness comfortably meets WHO standards for stool collection and laboratory turnaround, so a no-new-case signal reflects genuine absence rather than missed detection; investigation timeliness (73.5% within 7 days) is one step below the target.

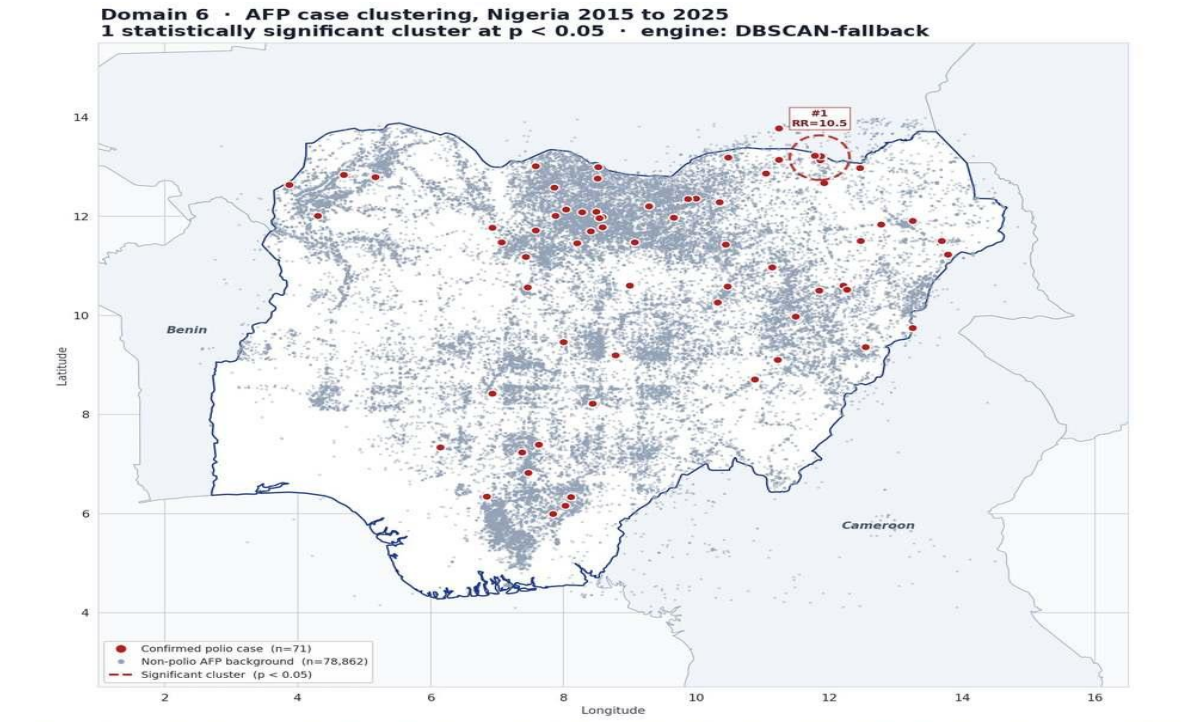


Figure 17. AFP case clustering, 2015–2025 (Bernoulli case-control spatial scan); one significant cluster on the north-eastern border.

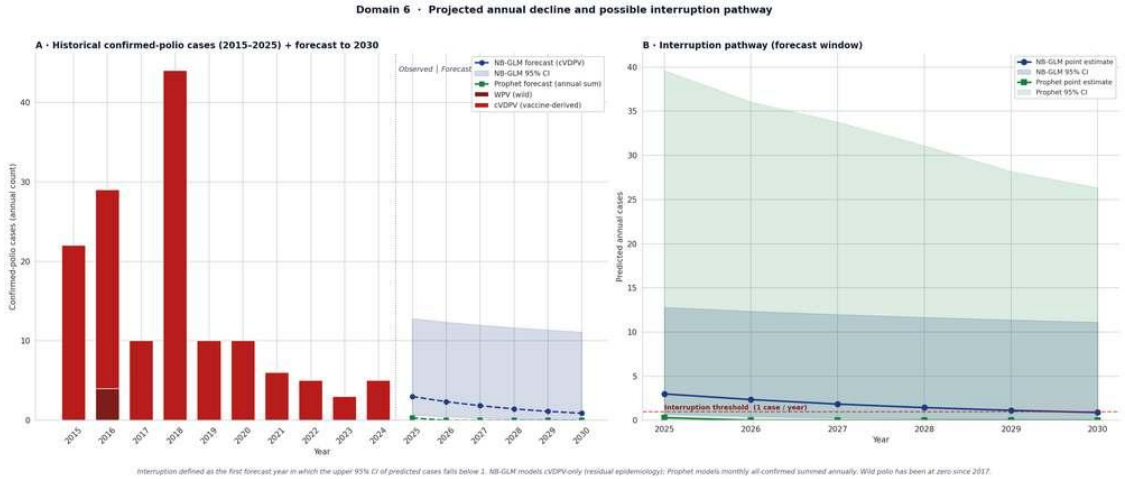


Figure 18. Historical confirmed-polio cases and forecast to 2030; interruption threshold below one case per year.

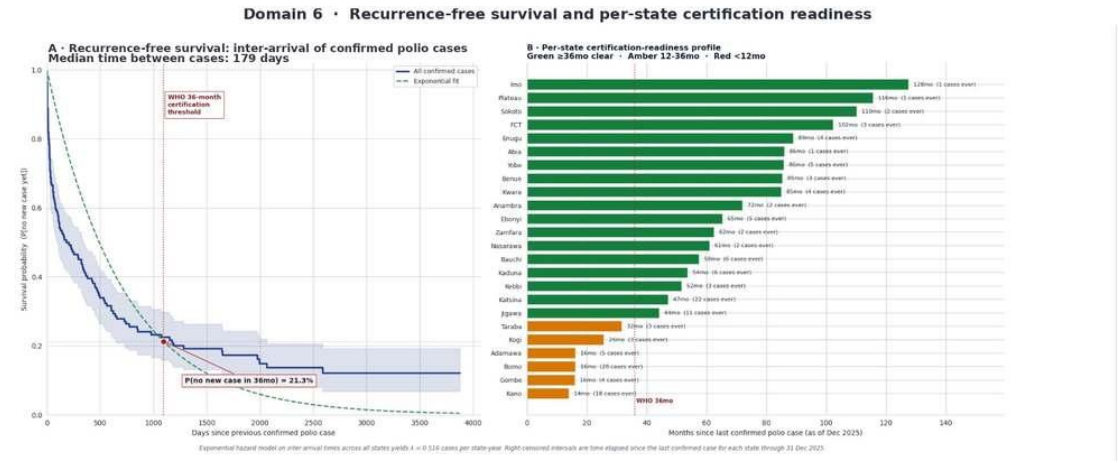


Figure 19. Recurrence-free survival of confirmed polio cases against the WHO 36-month certification threshold.

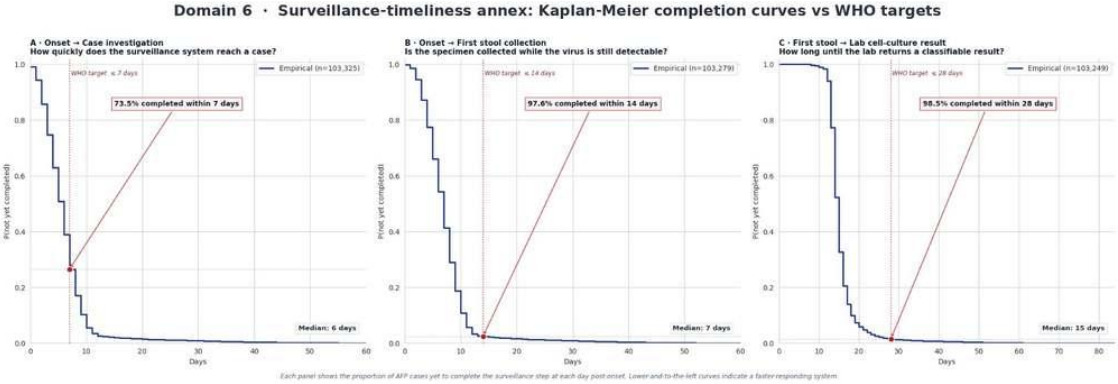


Figure 20. Surveillance-timeliness completion curves against WHO performance targets.

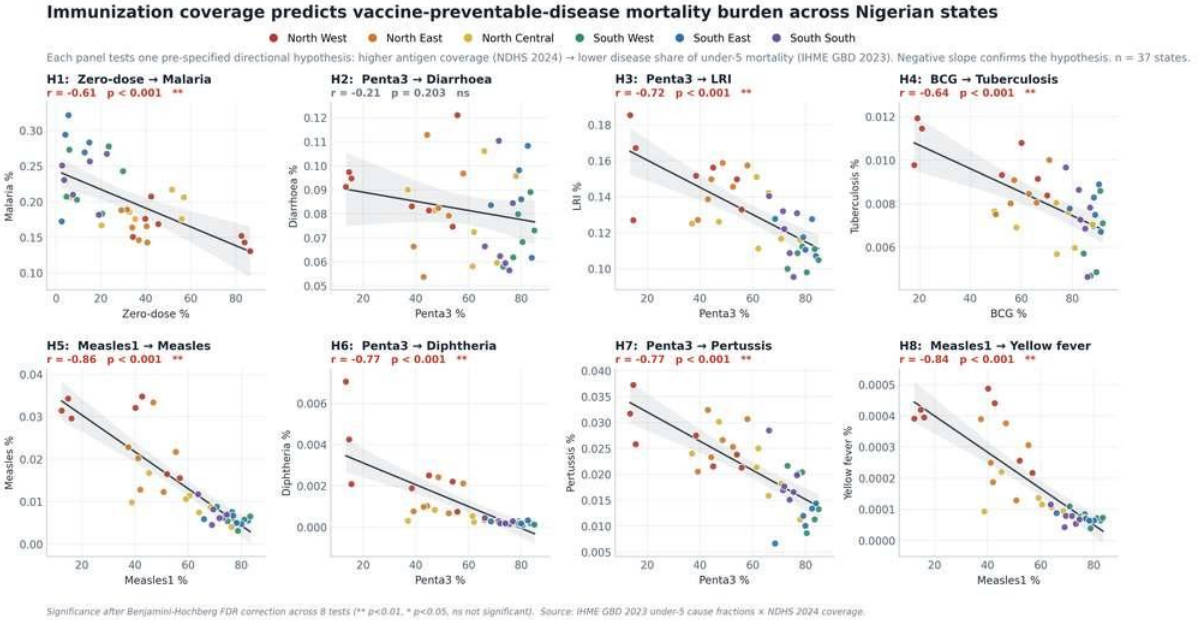
5.7 Domain 7 — Outbreak Risk and the Coverage-to-Mortality Link

Domain 7 tests the study's central hypothesis. Eight pre-specified directional tests with higher antigen coverage leading to a lower disease-specific share of under-five mortality were run against IHME GBD 2023 cause fractions and FDR-corrected. Seven of the eight are confirmed; the strongest are Measles1 to measles deaths (r = -0.86) and Measles1 to yellow-fever (r = -0.84). Because measles coverage tracks measles and yellow-fever mortality most tightly, raising the measles first dose and conducting timely follow-up campaigns in the Tier-1 North-West is the single highest-yield action for averting vaccine-preventable child deaths in the near term, and it supplies the cost-per-death-averted rationale GAVI investment requires.

The composite outbreak-risk surface ranks Zamfara, Sokoto, and Kebbi highest. A triangulation against the independent zero-dose forecast is decisive: the vaccine-targeted share of mortality and the zero-dose forecast correlate at r = 0.87, and six states fall in the convergent high-risk quadrant: Zamfara, Sokoto, Kebbi, Katsina, Niger, and Kaduna. Two independently built data products point to the same places.

Table 7. The six highest outbreak-risk states on the composite VPD risk surface.

Rank	*State*	*Overall risk*	*Gi* class 2026*	*Priority tier*
1	Zamfara	99.2	Hot spot (p<0.01)	Tier 1: Critical
2	Sokoto	96.1	Hot spot (p<0.01)	Tier 1: Critical
3	Kebbi	95.0	Hot spot (p<0.01)	Tier 1: Critical
4	Katsina	61.4	Hot spot (p<0.05)	Tier 2: High
5	Niger	57.8	Hot spot (p<0.01)	Tier 2: High
6	Kaduna	55.7	Hot spot (p<0.05)	Tier 2: High



Figures 21–22. Coverage-to-mortality hypothesis grid and the composite VPD outbreak-risk surface.

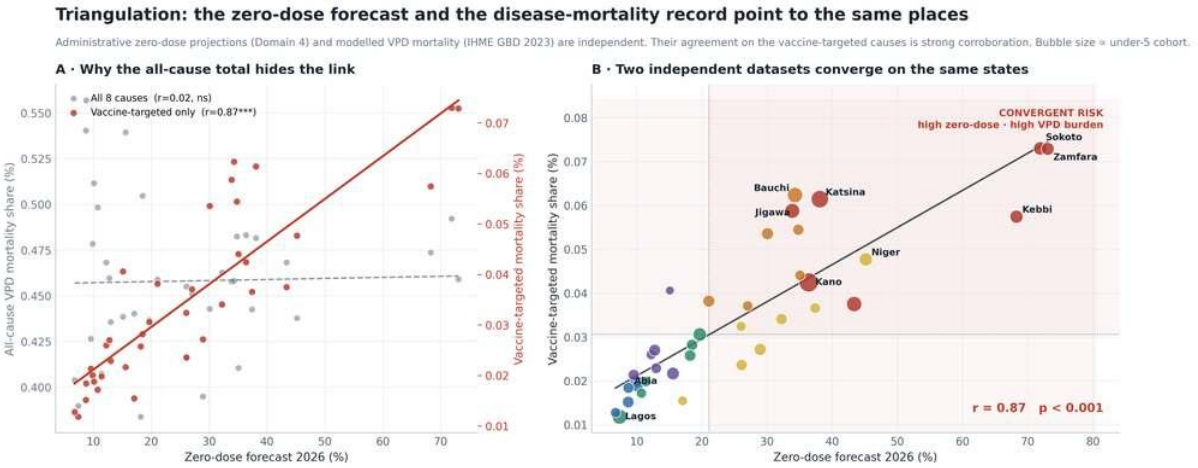


Figure 23. Triangulation: the vaccine-targeted mortality share tracks the zero-dose forecast (r = 0.87); the North-West forms the convergent-risk cluster.

5.8 The Predictive Modelling Platform

The analysis is delivered not as a static set of results but as a live, no-code web application that the NPHCDA Digital Innovation Hub can operate directly. The platform is deployed on Hugging Face under Docker, with secure login and usage tracking, and uploaded data remains in-session.

- **Live re-runs on new data.** Refreshed DHIS2 or survey files are uploaded through the browser; the application validates their structure and recomputes only what has changed. No code is written and no analyst is required.
- **Reproduces this report.** The platform regenerates the national total to within 0.01% of the figures reported here, and has been back-tested and wave-validated, so the report and the live tool cannot silently diverge.
- **The LGA Priority and Archetypes screen.** Filter by state or LGA to see antigen risks, estimated missed children, rank and recommended response package. Results export to Excel for microplanning.
- **Two coverage lenses, held apart.** Estimated coverage of the eligible cohort and the 2024 delivery index are displayed separately and triangulated against NDHS, so the distinction set out in Section 5.1.2 is preserved in the interface itself.
- **ZARA, a grounded analyst.** The Zero-dose Analytics and Risk Assistant answers plain-language questions in English, French, Hausa, Yoruba, Igbo, Pidgin, Swahili and other languages. It reads only the live outputs and uploaded documents, cites the page it drew from, and is instructed never to generate a figure that is not in the results.
- **Provenance on every view.** Each result carries its data date, reporting completeness and an indicator of whether the value is a model estimate, so that figures are interpreted with their limitations attached.
- **One-click documents.** A GAVI-style factsheet and a Word policy brief regenerate from the same live outputs, so report, factsheet, brief and slide deck remain a single source of truth.
- **Governance built in.** Role-based secure login, usage tracking, and a load–validate–run–review–approve–release sign-off cycle designed for DIH operation.

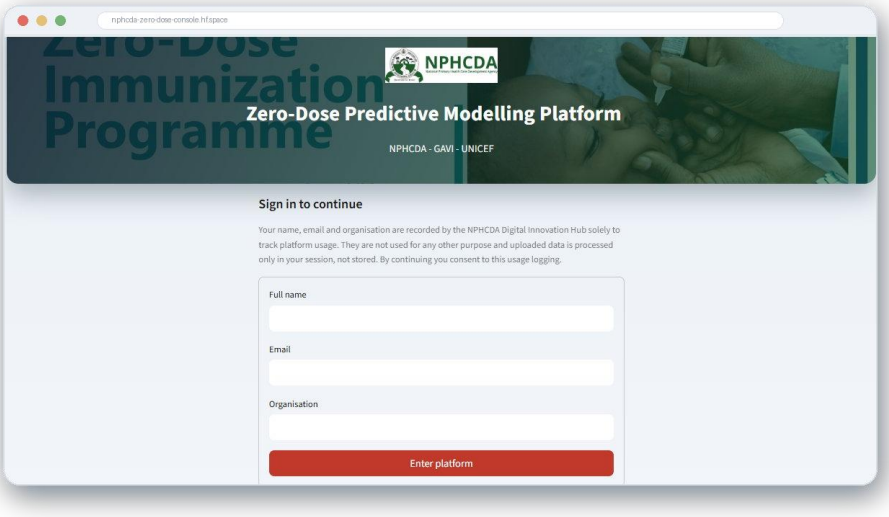


Figure 24. The predictive modelling platform: secure sign-in to the live Hugging Face deployment operated by the Digital Innovation Hub.

The platform converts the analysis from a set of consortium notebooks into a decision-support function the agency operates directly, and it is the mechanism through which the recommendations in Section 9.2 are to be sustained.

5.9 External Validation Against Independent Data

The forecasts were tested against two datasets that played no part in building them.

State level, against a survey that did not exist. The Nigeria mini Demographic and Health Survey (NmDHS) 2025-26 was released after this modelling concluded, which makes it a genuine out-of-sample test rather than a retrospective fit. Method 1 and the new survey rank the 37 states in almost the same order (Spearman ρ = 0.89); Method 2 gives ρ = 0.87. On levels, the North-West forecast of 48% sits within one point of the survey's 49%. Ordering and magnitude both hold against data the models never saw.

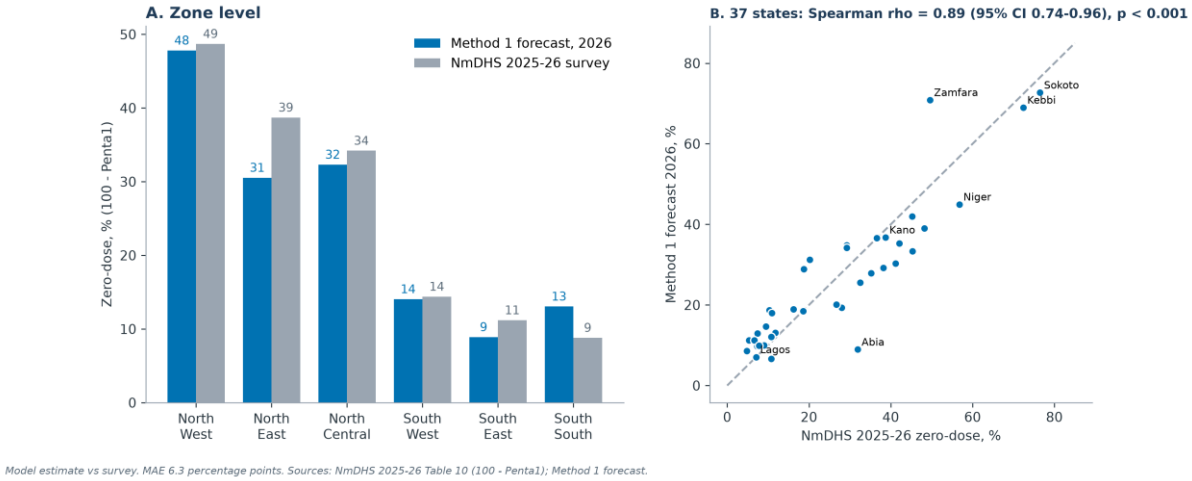
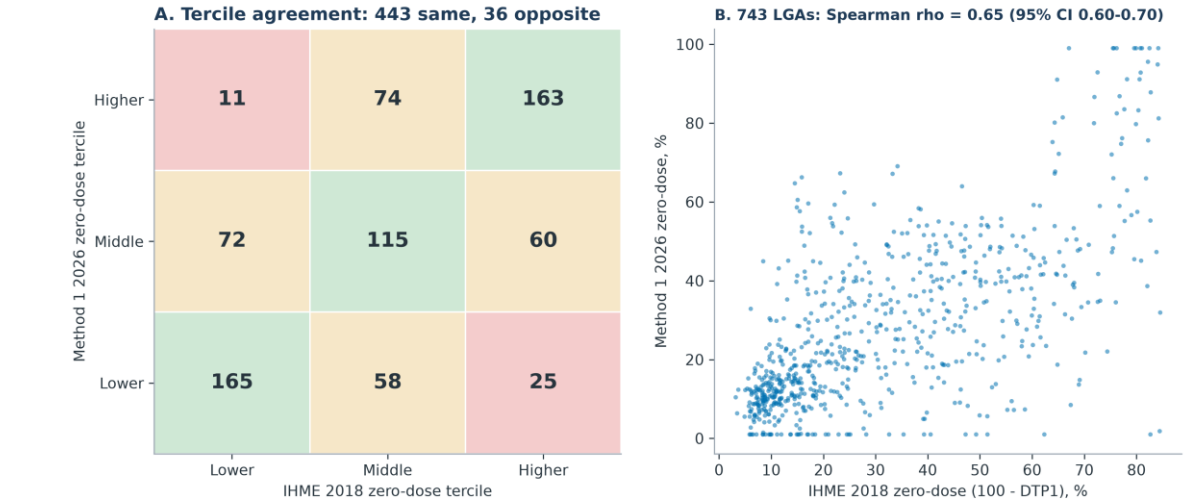


Figure 25. Prospective validation: the 2026 Method 1 state forecasts against the NmDHS 2025-26 survey, 37 states (ρ = 0.89).

LGA level, against an independently constructed surface. The LGA ranking was compared with the IHME Local Burden of Disease DTP1 admin-2 surface, built by a different team using different methods and inputs. Across 743 matched LGAs the Method 1 rank correlation is ρ = 0.65, and most LGAs fall in the same or an adjacent priority third; only 36 are placed in opposite thirds and are flagged for review. Method 2 agrees more closely (ρ = 0.89), including within states (ρ = 0.54 against 0.08 for Method 1). Absolute levels differ, as expected: the IHME surface ends in 2018 and these estimates are for 2026, and eight years shift the levels even where the geography agrees.



Cross-year comparison (2026 model vs 2018 IHME). Sources: IHME LBD DTP1 admin-2 2000-2018; Method 1.

Figure 26. Convergent validity: the 2026 Method 1 LGA estimates against the IHME Local Burden of Disease DTP1 surface (ρ = 0.65).

Table 7b. National comparison and agreement with independent data, both Domain 5 methods.

Measure	*Method 1*	*Method 2 (SAE)*
LGAs estimated	773	774
Zero-dose children, 2026 (95% interval)	2.10 million (1.90-2.31)	2.15 million (1.86-2.46)
National zero-dose rate	29.9%	30.6%
NmDHS 2025-26, 37 states: Spearman ρ (95% CI)	0.89 (0.74 to 0.96)	0.87 (0.71 to 0.95)
NmDHS 2025-26: mean absolute error / bias (points)	6.3 / -0.7	5.9 / +0.1
IHME DTP1 2018, 743 LGAs: Spearman ρ (95% CI)	0.65 (0.60 to 0.70)	0.89 (0.87 to 0.90)
IHME DTP1 2018, within states: Spearman ρ (95% CI)	0.08 (0.00 to 0.15)	0.54 (0.48 to 0.59)
States with positive within-state agreement	22 of 37	36 of 37
LGAs in the top 155 under both methods	107	107

ρ = Spearman rank correlation; 95% CI from 2,000 bootstrap resamples. All p < 0.001 except Method 1 within states (p = 0.03). Within-state agreement uses ranks computed within each state and pooled.

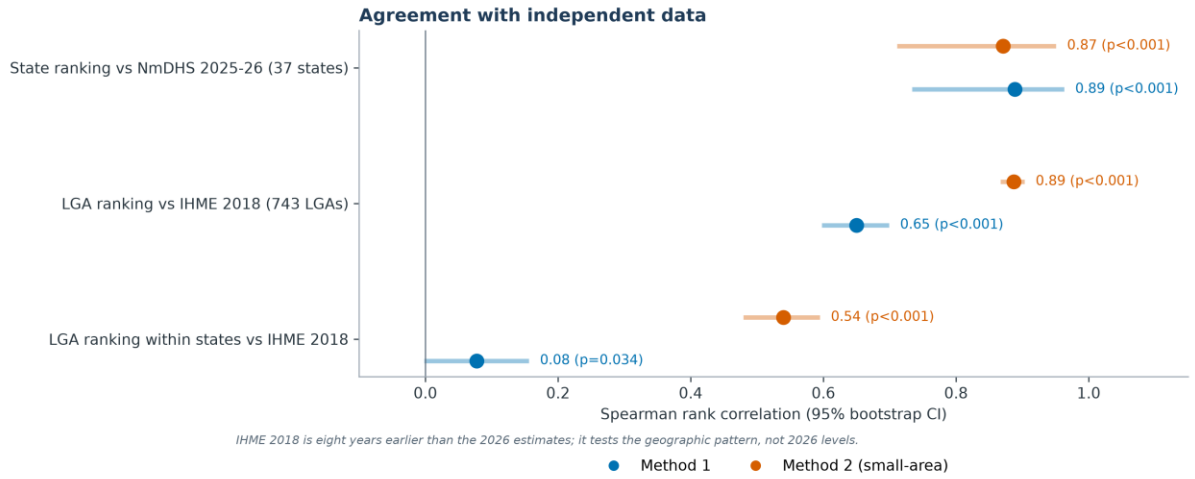


Figure 26b. Spearman rank correlation of each Domain 5 method with independent data: NmDHS 2025-26 (states) and IHME 2018 (LGAs, overall and within state).

A third, independent line of evidence points the same way. The pattern also matches the IHME Global Burden of Disease 2023 mortality data, which is the triangulation reported in Section 5.7 at r = 0.87.

Taken together, a survey published after the work concluded, a coverage surface constructed independently by another team, and an independent mortality series place the same states and the same LGAs in the same order. This convergence across independently produced data sources supports the use of these outputs for Gavi 6.0 targeting.

6. Readiness Assessment: Baseline and Endline

The modelling above is only as useful as the system's capacity to own and operate it. The baseline and endline assessments evaluated that capacity across twelve administrative units using a common instrument, covering data availability, analytics capacity, data-quality governance, systems integration, infrastructure adequacy, and the embedding of predictive modelling into routine governance. Following the agreed structure, baseline and endline results are presented separately below; they are then interpreted together in the Discussion (Section 7).

6.1 Baseline Findings

The baseline assessment combined a two-part quantitative readiness checklist (Part A on the presence of key data and decision-use functions; Part B on capacity, systems, and infrastructure) with key-informant interviews to contextualize the scores. Quantitative results are summarized first, then the qualitative themes.

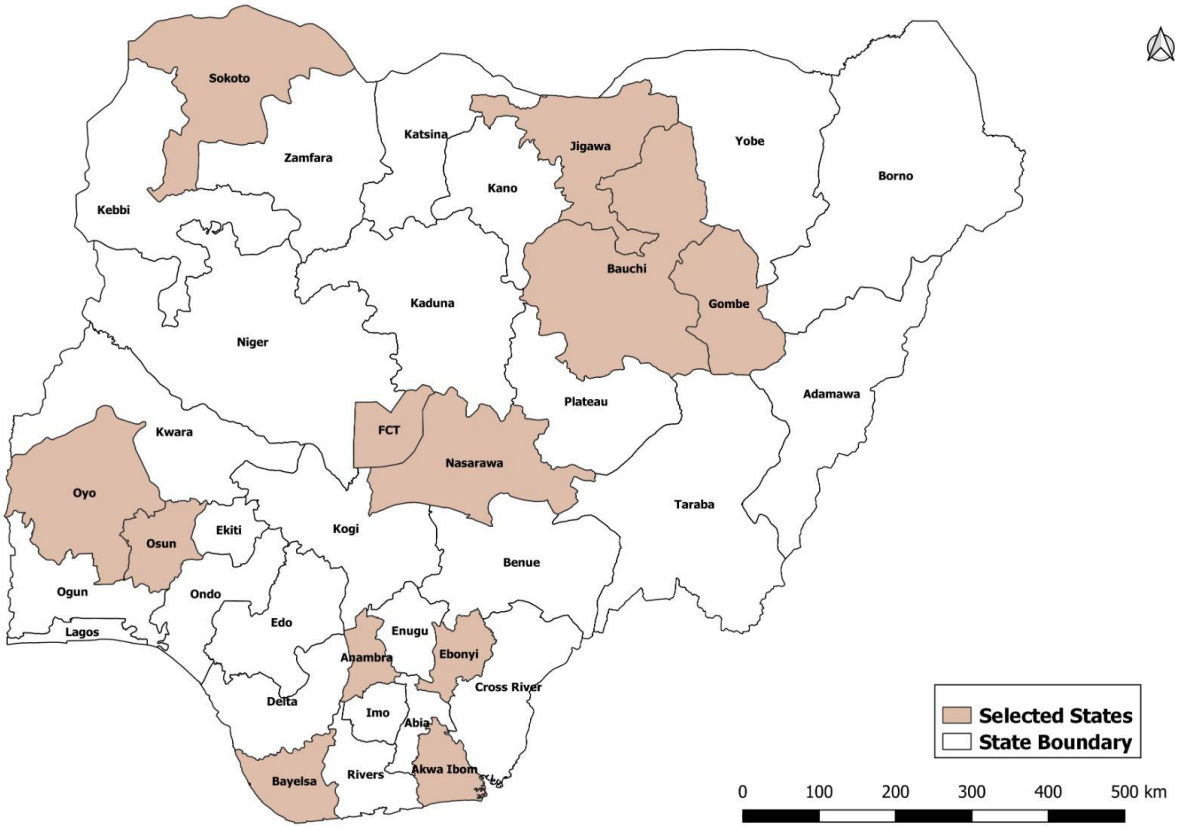


Figure 27: Study States

6.1.1 Quantitative readiness (baseline)

At baseline, overall Part A checklist scores ranged from 59.5% to 97.6%, with Nasarawa highest (97.6%) and Bayelsa lowest (59.5%). Performance was strongest in data availability, data use for planning and data analysis (most units at or near 100%), while the readiness-for-predictive-modelling sub-domain and governance-related domains showed the widest variation. Bayelsa was the clear outlier, recording low scores in data management and quality (20%), performance monitoring (33.3%), and data use for planning (20%) despite full data-availability compliance.

On Part B, capacity scores clustered at the upper end of the 1–5 scale: ten units recorded an overall capacity of 4, and three (Anambra, Bauchi, Ebonyi) recorded 3, with none below 3. Data-quality process maturity was uniformly high (4 for all units except Oyo at 5). On the overall modelling-readiness assessment, most units were classified as High readiness, and Anambra, Bauchi, and Ebonyi Moderate; systems readiness was Strong for Bayelsa, Nasarawa, and Osun, and Weak for Akwa Ibom, Anambra, and Oyo (reflecting weaker infrastructure). Estimated time-to-readiness ranged from 3–6 months for several units to 12+ months for Bauchi and Ebonyi. The baseline Part A scores by unit are shown below; Part B and the overall classification follow the same structure as the endline tables.

Table 8. Baseline Part A readiness scores by administrative unit and section

Unit	*Overall %*	*S1*	*S2*	*S3*	*S4*	*S5*	*S6*	*S7*
Akwa Ibom	81.0	80.0	80.0	100	44.4	85.7	100	100

Unit	*Overall %*	*S1*	*S2*	*S3*	*S4*	*S5*	*S6*	*S7*
Anambra	78.6	100	80.0	83.3	22.2	100	100	100
Bauchi	92.9	100	100	100	88.9	71.4	100	100
Bayelsa	59.5	100	20.0	33.3	77.8	71.4	20.0	80.0
Ebonyi	78.6	100	100	100	44.4	42.9	100	100
FCT Abuja	89.7	93.3	100	94.4	59.3	100	100	100
Gombe	92.9	100	100	100	88.9	71.4	100	100
Jigawa	95.2	100	100	100	100	71.4	100	100
Nasarawa	97.6	100	100	100	88.9	100	100	100
National	95.2	100	100	100	88.9	85.7	100	100
Osun	92.9	100	100	100	88.9	71.4	100	100
Oyo	92.9	100	100	100	88.9	71.4	100	100
Sokoto	88.1	100	100	83.3	55.6	100	100	100

6.1.2 Qualitative findings (baseline)

The baseline key-informant interviews, organized under the same themes as the endline, established the starting picture. On data systems and governance, DHIS2 and HMIS were the central platforms, with supplemental campaign and surveillance data; encryption and audit trails were cited as safeguards, but respondents noted that data security and governance policies were not uniformly enforced. On staff capacity, a significant skills gap existed at national and subnational levels, with foundational skills (data management, statistics, visualisation) present and working knowledge of MySQL and R among some staff, but Python was concentrated in a single individual, a clear single point of failure. Data-quality challenges were persistent and multi-level, rooted in irregular supportive supervision, insufficient logistics (laptops, power, connectivity), human error, and weak policy enforcement.

On data use, indicators such as antigen coverage and Penta dropout were reviewed descriptively through dashboards and Excel, not through predictive models. On predictive-modelling awareness, respondents understood the concept and were enthusiastic, citing the RTS, malaria vaccine state selection as prior modelling experience, and identifying clear perceived benefits (outbreak prediction, vaccine allocation, target population estimation, budgeting). The barriers identified were structural and consistent: continuous loss of trained staff to better-resourced partners, weak governance enforcement, a subnational infrastructure deficit, narrow programming capacity, and a reactive analytical culture focused on describing the past rather than forecasting. At the state level, capability was real but typically concentrated in one individual (for example, the Akwa Ibom State M&E Officer), with the absence of R and Python skills and a non-functional DIH limiting broader readiness, a pattern that recurs across the assessed states.

Several states illustrated the baseline picture concretely. Nasarawa had the most advanced setup, with an M&E officer running a near-real-time monitoring system built on an SMS gateway, a custom Excel dashboard, QGIS settlement mapping, and a red-yellow-green risk framework, producing the fastest data-to-action cycle in the assessment (emergency

declarations within 24 hours of AFP surveillance data) but resting entirely on one individual. Oyo employed a dedicated analyst using Power BI, Power Query, and SPSS with consumption-based vaccine forecasting already in practice, yet its entire seven-person data team shared a single computer. Jigawa presented a stronger modelling baseline than most, with two to three staff having some Python knowledge and 281 computers deployed to facilities, though training was limited to data entry. Osun operated a three-copy immunisation-card system and community-mobiliser networks, but staff reported using personal funds for DHIS2 uploads. In Sokoto, awareness of predictive modelling was very limited, and the Digital Innovation Hub was unknown to all respondents. Across the lowest-scoring southern states (Akwa Ibom, Anambra, Bayelsa, Ebonyi), respondents could articulate the value of predictive tools but lacked the enabling conditions (stable power, functional computers, connectivity, and trained staff) to support routine analytics.

6.2 Endline Findings

The endline assessment revisited the same thirteen administrative units using the same instruments and analytical framework to document changes in readiness and the extent to which predictive modelling can be embedded in routine immunization planning and decision-making.

6.2.1 Quantitative Readiness (Endline)

The endline quantitative checklist assessed institutional preparedness for predictive-modelling integration across thirteen administrative units: the NPHCDA (National), the FCT, and eleven participating states. It comprises Part A, a readiness checklist covering data availability and decision-use functions, and Part B, an assessment of capacity, systems, and infrastructure on a 1–5 scale.

Part A — Readiness checklist.

Overall Part A scores ranged from 78.6% to 97.6%, with Bauchi highest (97.6%) and Anambra lowest (78.6%); eight of the thirteen units scored at or above 90%, indicating that basic data-use structures are broadly established at national and subnational levels. Performance was strongest and most uniform in data availability and access, data use for planning (100% across all units), and data analysis and interpretation, where most units recorded full confirmation. The clearest differentiator was readiness for predictive modelling and advanced analytics, which ranged from 55.6% (Akwa Ibom, Anambra, Oyo, Sokoto) to 100% (Bauchi, Bayelsa, Jigawa), confirming that advanced analytics capacity, rather than routine data handling, is where targeted, differentiated support is still needed. System constraints and opportunities showed the National unit lowest at 42.9%.

Operational indicators varied across units: reporting lag time ranged from quarterly (Sokoto) through monthly (most units) to real-time/same-day (Bauchi, Gombe, Osun); data-linkage capacity ranged from lowest-observed at National to manual or support-dependent elsewhere; and sex-disaggregation practice ranged from rare (Bayelsa, Sokoto) to systematic (Akwa Ibom, Bauchi, Gombe, Jigawa, National, Osun).

Table 9. Part A readiness scores by administrative unit and section

Unit	*Overall %*	*S1*	*S2*	*S3*	*S4*	*S5*	*S6*	*S7*
Akwa Ibom	85.7	100	60.0	100	55.6	100	100	100

Unit	*Overall %*	*S1*	*S2*	*S3*	*S4*	*S5*	*S6*	*S7*
Anambra	78.6	80.0	80.0	83.3	55.6	71.4	100	100
Bauchi	97.6	100	100	100	100	85.7	100	100
Bayelsa	90.5	100	80.0	83.3	100	71.4	100	100
Ebonyi	90.5	100	100	100	66.7	85.7	100	100
FCT Abuja	95.2	100	100	83.3	88.9	100	100	100
Gombe	95.2	100	100	100	88.9	85.7	100	100
Jigawa	92.9	100	100	100	100	57.1	100	100
Nasarawa	92.9	100	100	100	88.9	71.4	100	100
National	85.7	100	100	100	88.9	42.9	100	80.0
Osun	92.9	100	100	100	88.9	71.4	100	100
Oyo	85.7	100	100	100	55.6	71.4	100	100
Sokoto	88.1	100	100	100	55.6	100	100	80.0

(S1 Data availability; S2 Data management & quality; S3 Performance monitoring & feedback; S4 Readiness for predictive modelling; S5 System constraints & opportunities; S6 Data use for planning; S7 Data analysis & interpretation).

Part B — Capacity, systems, and infrastructure.

Part B assessed staff competency, data-quality processes, data use and decision impact, systems integration, data standards, and infrastructure adequacy across the thirteen administrative units. Exact overall capacity scores ranged from 3.1 in Anambra to 4.7 in Jigawa. Eight units; Akwa Ibom, Bauchi, Bayelsa, FCT Abuja, Jigawa, Nassarawa, Oyo, and Sokoto were classified as having high capacity readiness, while Anambra, Ebonyi, Gombe, National, and Osun were classified as moderate; no unit fell within the low-capacity category. Staff competency showed the widest variation, ranging from 2.0 in Anambra and Osun to 5.0 in Jigawa and Sokoto. Data-quality process scores were comparatively stable, ranging from 3.3 to 5.0, while data use and decision impact ranged from 3.0 to 5.0.

Table 10. Part B Capacity, Systems, and Infrastructure Assessment Scores by Administrative Unit

Unit	*Overall Capacity*	*Staff Competency*	*Data Quality Process*	*Data Use / Decision Impact*	*Systems Integration Level*	*Data Standards Strength*	*Infrastructure Adequacy*
	Capacity Score			**Systems Integration**			
Akwa Ibom	4	3	4	5	No integration	Moderate	No
Anambra	3	2	3	4	Manual	Weak	No
Bauchi	4	4	4	3	Semi-automated	Moderate	Yes
Bayelsa	4	3	4	4	Manual	Moderate	Partially
Ebonyi	3	3	4	3	Automated	Moderate	Partially

Unit	*Overall Capacity*	*Staff Competency*	*Data Quality Process*	*Data Use / Decision Impact*	*Systems Integration Level*	*Data Standards Strength*	*Infrastructure Adequacy*
FCT Abuja	4	5	4	4	Automated	Moderate	Yes
Gombe	3	3	4	3	Semi-automated	Moderate	Yes
Jigawa	5	5	4	5	Fully interoperable	Moderate	Yes
Nasarawa	4	4	4	4	Manual	Strong	Yes
National	3	3	4	4	Semi-automated	Moderate	Partially
Osun	3	2	4	4	Manual	Moderate	Partially
Oyo	4	4	4	4	Manual	Moderate	Partially
Sokoto	5	5	5	4	Manual	Moderate	Yes

Capacity scores are reported on a 1–5 scale, with higher values indicating stronger capacity. Systems integration categories are: No integration, Manual, Semi-automated, Automated, and Fully interoperable. Data standards categories are Strong, Moderate, and Weak. Infrastructure adequacy categories are Yes, Partially, and No.

Overall Modelling Readiness

Overall system readiness was predominantly moderate: ten units were classified as moderate, Nassarawa was classified as strong, and Akwa Ibom and Anambra were classified as weak. Systems integration ranged from no integration in Akwa Ibom to fully interoperable in Jigawa. Data standards were moderate in eleven units, strong in Nassarawa, and weak in Anambra. Infrastructure was assessed as adequate in six units (Bauchi, FCT Abuja, Gombe, Jigawa, Nassarawa, and Sokoto), partially adequate in five (Bayelsa, Ebonyi, National, Osun, and Oyo), and inadequate in two (Akwa Ibom and Anambra). Estimated time to modelling readiness was 3–6 months in five units, 6–9 months in three, 9–12 months in three, and more than 12 months in two units (Ebonyi and Jigawa).

Table 11. Overall Capacity, Systems Readiness, And Estimated Time to Modelling Readiness by Unit.

Unit	*Capacity*	*Readiness level*	*Systems readiness*	*Data std.*	*Infra.*	*Time to readiness*
Akwa Ibom	4	High	Weak	Moderate	No	3–6 months
Anambra	3	Moderate	Weak	Weak	No	3–6 months
Bauchi	4	High	Moderate	Moderate	Yes	9–12 months
Bayelsa	4	High	Moderate	Moderate	Partially	3–6 months
Ebonyi	3	Moderate	Moderate	Moderate	Partially	12+ months
FCT Abuja	4	High	Moderate	Moderate	Yes	3–6 months

Unit	*Capacity*	*Readiness level*	*Systems readiness*	*Data std.*	*Infra.*	*Time to readiness*
Gombe	3	Moderate	Moderate	Moderate	Yes	9–12 months
Jigawa	5	High	Moderate	Moderate	Yes	12+ months
Nasarawa	4	High	Strong	Strong	Yes	6–9 months
National	3	Moderate	Moderate	Moderate	Partially	6–9 months
Osun	3	Moderate	Moderate	Moderate	Partially	6–9 months
Oyo	4	High	Moderate	Moderate	Partially	9–12 months
Sokoto	5	High	Moderate	Moderate	Yes	3–6 months

High (>3.5), Moderate (2.5–3.5), Low (<2.5).

6.2.2 Qualitative Findings (Endline)

The endline qualitative findings draw on key-informant interviews at the national level (NPHCDA Planning, Research & Statistics) and across twelve administrative units, organized under the same themes used at baseline, so the two rounds read side by side. National findings are presented in full because they carry the most technically grounded account and the clearest new endline evidence; state findings are then synthesized thematically, with a state-by-state summary table.

6.2.2.1 National-level findings

Data Systems, Infrastructure, and Governance.

DHIS2 remains the central national data system, with the established facility-to-LGA-to-national reporting pathway unchanged since baseline. What has changed is the growing use of parallel streams alongside routine reporting: campaign tools, SMS same-day reporting in some states, and integrated supportive-supervision checklists, which are increasingly triangulated rather than treated as a single source of truth. Governance has advanced meaningfully: an emerging formal architecture under the Digital Innovation Hub (DIH) 'blueprint' now includes a data-sharing agreement, credential-based access, and multi-level authorisation, with active liaison with NITDA and the Nigeria Data Protection Commission, a shift from internal agency policy toward alignment with national data-protection law.

Staff Capacity In Data Analytics

The respondent rated overall advanced analytics capability at about 6 out of 10, distinguishing widespread general ICT skills from the scarcer ability to apply them in an epidemiological context. Around 12–15 staff are proficient in visualisation tools (Power BI, geospatial, Excel, ODK), narrowing to roughly five with modelling-related capability within the routine-immunisation unit, supported by partners. A new endline signal is the spontaneous emergence of AI tools as self-directed staff learning, both an opportunity (latent motivation to upskill) and a risk (unstructured self-learning without epidemiological grounding).

> *"Most people might know how to use Python, but applying it in the context of epidemiology and public health might be the issue."* — M&E Officer, NPHCDA PRS (National)

Data Quality

The dominant issue remains denominator instability, now described with greater analytical clarity: several states have repeatedly been flagged for implausible coverage above 100% (reaching 125–130%), sometimes alongside disease outbreaks, raising data-integrity concerns. In response, the national team tracks absolute numbers of children immunised month-on-month where population estimates are unreliable, rather than relying on coverage percentages. Two further developments are new at endline: an Identify-Enumerate-Vaccinate (IEV) case-based physical enumeration across 14 states as a denominator-correction mechanism, and DHIS2 validation rules embedded directly in the system that flag quality issues at the point of entry, a more mature, less supervision-dependent quality architecture than at baseline.

Use of Data and Modelling-Informed Planning

A structured monthly analytical cycle tracks antigen-specific coverage, session-frequency compliance, dropout, stockouts, supervision, and zero-dose burden. The most concrete evidence of an iterative, modelling-informed planning cycle is the active reprioritisation of the 100 zero-dose LGAs: an eight-month intervention plan was rolled out against the original modelling-identified list, an evaluation has been conducted, and the agency is now determining whether the list should be revised against updated data.

Predictive Modelling Awareness, Readiness, and Confirmed Use

The national respondent gave the most technically precise account in the assessment, defining predictive modelling accurately as time-bound forecasting from historical data. Critically, and unlike every state-level respondent, he confirmed that modelling has already been used operationally for target-population estimation across recent campaign rounds and, most significantly, as the main analytical tool in the original identification of the 100 zero-dose LGAs. This capability still depends heavily on external partners rather than internal staff, and engagement is largely driven by individual interest rather than institutional requirement. He also gave the most detailed DIH account: ECHIS for community data, a facility EMR to replace paper registers, and infrastructure modernisation, with modelling and analytics embedded as a core DIH function rather than a separate activity.

> *"For zero-dose, for the first time, modelling was the only tool used to do that. But the modelling capacity within the agency is limited, so most of the time it has been outsourced."* — M&E Officer, NPHCDA PRS (National)

Barriers To Sustaining Modelling

The barrier analysis distinguished three levels with unusual precision. First, a technical-to-public-health translation gap: proficiency in Python or geospatial tools does not guarantee the epidemiological understanding needed to apply them correctly. Second, training depth: short two-to-three-week trainings produce only surface understanding, with at least two months of intensive, disease-specific training recommended. Third, and most distinctive, organisational

tasking: even well-trained staff lose skills unless leadership assigns specific modelling responsibilities and follows up. A further new barrier is data sufficiency; routinely collected data is not always structured or granular enough to support modelling, making readiness a data-architecture question the DIH is intended to address.

> *"Somebody might have the modelling skills, but nobody is giving him any task that will kick off his modelling capability."* — M&E Officer, NPHCDA PRS (National)

Governance And Recommendations (National)

The respondent's recommendations were the most structured in the assessment, organised around four conditions: select a small group of genuinely interested individuals rather than broad, shallow training; use simple, well-scoped data templates and a jointly developed worked example; provide sustained post-training mentorship with regular follow-up; and ensure leadership-level supervision of modelling work as the accountability mechanism that turns training into sustained practice.

6.2.2.2 State-Level Findings: Thematic Synthesis

Across the twelve administrative units, the state interviews echo the national picture and converge on a consistent set of themes:

- **Performance and coverage.** Several states report strong routine coverage (Jigawa above 80–100% on key antigens; Gombe's Penta3 rising from ~33% in 2018 to ~57% in 2023/24). The recurring threat to data validity is the private and tertiary facility reporting gap, especially in urban LGAs where completeness often falls below 80%, systematically understating true activity.
- **Data flow and management.** Most states run DHIS2 for monthly reporting alongside a weekly SMS gateway (Jigawa, Nasarawa across 817 facilities), with month-end reconciliation between the two acting as a built-in cross-validation check.
- **Data quality.** Common challenges are denominator instability (Nasarawa coverage exceeding 120% from population underestimation), private-facility non-reporting, and tally-to-summary transcription errors, often where one worker both vaccinates and records. Missing monthly summary forms were flagged in FCT as a critical structural gap.
- **Analysis and decision-making.** The analytical toolkit is uniformly Excel, Power BI, and the DHIS2 visualiser; no Python, R, or Stata use was identified at the state level. Gombe and Nasarawa show the most structured data-to-action cycles, though a one-to-two-week lag between analysis and implementation is common, driven by the separation of analysts from decision-makers and by logistics constraints.
- **Zero-dose identification.** Approaches combine settlement-level register tracking, community agent networks (TBAs, CHIPS), and administrative analysis; the strongest validation came from house-to-house surveys (Gombe's Zero-Dose Learning Hub) and surveillance triangulation (Nasarawa, where AFP findings showed 12 of 13 LGAs with unimmunised children, prompting a gubernatorial emergency declaration).

- **Modelling readiness and sustainability.** No predictive tools are in use at the state level. Readiness rests on three consistently named requirements: continuous capacity-building, formal integration of modelling into routine decision structures, and a clear progression pathway from Excel toward advanced platforms. The binding constraint named repeatedly is that skills fade without regular hands-on use: sustainability depends on continuous engagement, not one-off training. Gombe's 2025 move to fully domestic immunisation funding makes it a strong early-pilot candidate.

The state-by-state summary below captures each unit's distinctive features against the recurring themes.

Table 12. State-By-State Summary of Qualitative Findings Against the Recurring Themes

Unit	*Notable strengths / features*	*Key constraints*
FCT (AMAC)	Advanced LGA zero-dose tool (2023/24 mobile app, real-time settlement data); strong community engagement	Missing monthly summary forms; high private-facility volume; Excel-only; staffing gaps
Gombe	Penta3 ~33%→57% (2018→2023/24); domestic immunisation funding (2025); house-to-house ZD validation	Private/tertiary reporting gap; no analytic capacity below state; Excel/Power BI only
Jigawa	Coverage 80–100% on key antigens; dual SMS+DHIS2 with reconciliation; birth-tracking via TBAs/CHIPS	Seasonal HTR access; private-facility under-reporting; no Python/R; credentials lost without use
Nasarawa	Three parallel data systems; last-mile (>3km) strategy; surveillance-driven emergency response	Denominator instability (coverage >120%); analyst–decision separation
Akwa Ibom	Established routine reporting	Lower data-management score; advanced-analytics readiness limited
Anambra	Functional core data structures	Lowest overall readiness; weak systems integration and infrastructure
Bauchi	Highest Part A score; real-time reporting capacity	Outreach data-tool gaps; advanced-analytics depth limited
Bayelsa	Marked readiness improvement from baseline	Sex-disaggregation rare; moderate systems readiness
Ebonyi	Robust data-quality processes	Weak systems readiness and infrastructure adequacy
Osun	Large reported staff base; full data-use score	Weak data-standards strength; longer time-to-readiness
Oyo	Strong routine data handling	Predictive-modelling readiness low; Excel-only

Unit	*Notable strengths / features*	*Key constraints*
Sokoto	High capacity and staff competency; strong systems readiness	Quarterly (slower) reporting lag; advanced-analytics readiness mixed

Strengths and constraints are drawn from the endline key-informant interviews and, where noted, the Part A/B readiness scores.

7. Stakeholder Engagement, Digital Gap Remediation and Close-out Activities

This section documents the stakeholder engagement, digital remediation planning and close-out activities undertaken after completion of the modelling, readiness assessment and platform-development work. It complements the modelling-domain findings and the baseline–endline readiness assessment by showing how the results were presented to decision-makers, how the Digital Innovation Hub (DIH) gaps were translated into a remedy plan, and how stakeholder feedback shaped the next phase of implementation.

7.1 Stakeholders Engaged

The consultancy engaged stakeholders across policy leadership, programme management, data systems, partner coordination and technical implementation. The key stakeholder groups are summarized below.

- National Primary Health Care Development Agency (NPHCDA) leadership, including the Executive Director, NPHCDA, directors and senior programme staff who reviewed the modelling outputs and provided strategic direction for national adoption.
- NPHCDA technical units, including the Digital Innovation Hub, the Routine Immunization team, Planning, Research and Statistics, data and analytics personnel, and staff responsible for immunization monitoring, reporting and programme review.
- Routine Immunization Technical Working Group members, who reviewed the outputs from the perspective of routine programme planning, microplanning, campaign targeting, LGA prioritization and practical use by national and subnational immunization teams.
- GAVI stakeholders, including the GAVI in-country team and technical reviewers engaged during virtual and in-person close-out discussions on zero-dose targeting, Gavi 6.0 prioritization, modelling outputs and platform use.
- UNICEF Nigeria, which supported the broader programme objective of integrating predictive modelling into routine immunization decision-making and strengthening national and state-level capacity for data-driven programme adaptation.
- State Primary Health Care Development Agencies and designated national/state analytics focal persons, who are expected to participate in dashboard use, data submission, model-output interpretation and routine feedback once DIH workflows are institutionalized.
- The CIDRE–Quantium Insights Consortium, which implemented the baseline and endline assessments, facilitated the modelling fellowship, developed the modelling outputs, prepared documentation and supported handover of the platform and decision-support products.

7.2 Preparatory Engagement, State Entry and Participant Selection

Prior to commencement of field activities, the consortium carried out a structured preparatory process to ensure that the baseline and endline assessments were properly introduced, coordinated and implemented across the participating administrative units. Formal letters of

introduction and notification were sent to the selected states and relevant institutional stakeholders to inform them of the purpose of the predictive modelling readiness assessment, the expected scope of engagement, the categories of respondents required and the proposed timelines for data collection. These letters served as the official entry point for securing cooperation from State Primary Health Care Development Agencies, Routine Immunization teams, Planning, Research and Statistics units, Monitoring and Evaluation officers, LGA-level officers and facility-level respondents. Following the dispatch of official letters, state entry and community entry processes were conducted in collaboration with state and local immunization structures. The consortium engaged state-level focal persons and programme officers to confirm appropriate respondents, schedule interviews, identify relevant offices and facilities, and ensure that field teams were properly introduced before data collection. At the LGA and facility levels, entry processes included briefing relevant officers on the purpose of the assessment, confirming their roles in routine immunization data generation and reporting, and obtaining their cooperation for administration of the readiness checklist and key informant interviews. This approach helped reduce disruption to routine work, strengthened ownership of the assessment and ensured that responses were obtained from personnel directly involved in immunization data management, reporting, supervision, analysis and decision-making.

The data collection process was implemented using a role-based selection approach. Respondents were purposively selected for their direct involvement in routine immunization data systems, surveillance-linked reporting, data quality assurance, programme monitoring, performance review, planning, or service delivery. At national and state levels, the assessment prioritized officers responsible for immunization coordination, Planning, Research and Statistics, Monitoring and Evaluation, data management and Digital Innovation Hub-related functions. At LGA level, respondents included officers involved in PHC coordination, immunization implementation, disease surveillance, monitoring and data validation. At facility level, respondents included PHC in-charges and routine immunization service providers responsible for recording, summarizing and submitting immunization service data. Participant selection for the fellowship was also undertaken to ensure representation across national and sub-national levels and to support the institutionalization of predictive modelling capacity within the immunization system. A total of 20 national-level participants were selected, comprising 18 participants from NPHCDA and 2 participants from NCDC. These participants represented central-level technical and programme functions relevant to immunization data use, surveillance, modelling, analytics, programme planning and decision-making. In addition, 37 sub-national participants were selected to represent state-level routine immunization and data-use structures. The inclusion of sub-national participants was important as predictive modelling outputs are only useful when they can be interpreted and applied by the officers responsible for planning, supervision, outreach prioritization and corrective action at state and lower levels.

Overall, the preparatory engagement, state entry, community entry and participant-selection processes ensured that the assessment and fellowship were embedded within existing NPHCDA, state, LGA and facility structures. This strengthened access to relevant respondents, improved the quality of the data collected, supported ownership among participating states and

created a practical foundation for translating the modelling outputs into routine immunization planning and decision-making.

7.3 Remedy Plan to Address Digital and DIH Gaps

The digital gap analysis was used as a guide for interpreting the gaps that affect predictive modelling deployment and DIH functionality. The key finding was that Nigeria has a functional routine immunization data foundation, with DHIS2/HMIS and routine review structures already in use, but the system still requires targeted remediation before predictive modelling can be sustained as a routine government-owned workflow.

7.3.1 DIH Governance, Ownership and Workflow Activation

The DIH should serve as the institutional home for routine predictive modelling. This requires defined roles for data custody, model operation, output review, approval and release. The workflow should include data submission, validation, model execution, quality assurance, interpretation, sign-off and dissemination of final outputs to NPHCDA leadership, programme teams and partners. Standard operating procedures, model assumptions, data dictionaries, user guides, approval workflows and model-version records should be maintained by the DIH so that outputs remain traceable and auditable.

7.3.2 Hardware, Power and Connectivity Remediation

The remedy plan should prioritize the equipment required for national and state participation in the modelling workflow. This includes high-specification laptops for national analytics users and one focal analytics user in each state and the FCT, dedicated analytics servers, secure backup storage, uninterrupted power protection, a national DIH solar/inverter system, state/FCT power-backup arrangements, reliable business connectivity for the DIH and data bundles for designated national and state focal persons. These items are necessary because unstable devices, power and internet directly affect data submission, dashboard access, model refresh and analytics continuity.

7.3.3 Data Integration, Hosting and Digital Workflow

The DIH should move from manual data assembly to an integrated digital workflow. Priority actions include DHIS2-surveillance integration, an API exchange layer for DHIS2, LMIS, campaign, surveillance and external data sources, ETL/data warehouse development, cloud hosting, automated DHIS2 refresh, historical-data access and a multi-source data linkage platform. These actions will reduce manual data handling, improve reproducibility and allow modelling outputs to be refreshed when routine monthly data are updated.

7.3.4 Data Quality, Denominator Validation and Reporting Completeness

Data-quality improvement should proceed alongside modelling adoption. Priority actions include automated QA scripts, anomaly detection, outlier checks, completeness monitoring, denominator validation, source-record reconciliation, monthly Penta1 completeness checks, and routine review of private and tertiary facility reporting. These actions are essential because denominator instability, missing Penta1 data, incomplete private/tertiary reporting, and delayed facility submissions can distort zero-dose, dropout, and coverage estimates if they are not flagged and corrected.

7.3.5 Capacity Development and Routine Use

Capacity development should shift from broad exposure to sustained applied competence. The next phase should establish a small DIH/RI analytics core, supported by state focal persons, with clear terms of reference and protected time for analysis. Training should be cohort-based, disease-specific and anchored in real state datasets. At least six months of post-training mentorship, recurring model-refresh assignments and review of outputs during RI performance meetings should be included so that modelling becomes a routine programme function rather than a one-off training output.

7.3.6 Platform Expansion Based on Stakeholder Feedback

Feedback from NPHCDA leadership and partners should guide platform expansion. Immediate enhancements should include OPV and IPV alongside the current tracer antigens; LGA ranking from worst to best for prioritization; and LGA-specific profiles showing the main problem, likely driver and proposed intervention. These improvements will make the platform more useful for outbreak preparedness, campaign planning, microplanning and routine supervision.

7.4 Close-out Meetings and Dissemination Activities

The close-out phase was used to present the final modelling outputs, demonstrate the platform, validate the practical interpretation of results and agree on priority next steps for DIH adoption and Gavi 6.0 targeting.

7.4.1 GAVI Engagements

The close-out process included both virtual and in-person engagements with GAVI stakeholders. The documented in-country engagement took place on Tuesday, 7 July 2026. During this engagement, the team presented the zero-dose modelling findings, the LGA priority list, the community archetypes, the platform outputs and the proposed use of the results for Gavi 6.0 targeting. Discussions emphasized the need for clear, actionable outputs that can support investment decisions, phased prioritization and microplanning at LGA level.

7.4.2 NPHCDA Leadership and Technical Staff Meeting

Following the GAVI in-country engagement, the consortium held a second presentation on Tuesday, 7 July 2026 with the Director of Disease Control and Immunization NPHCDA, other Directors and key NPHCDA staff, including the Head of DIH and the Head of Routine Immunization. This meeting presented the consolidated modelling outputs, platform functionality, DIH handover requirements and the practical implications of the ranked LGA list.

7.4.3 Meeting with the Executive Director

On Wednesday, 8 July 2026, the team presented the modelling work and platform outputs to the Executive Director of NPHCDA. The Executive Director provided specific inputs for strengthening the next phase of the work. These included expanding the antigen scope to include OPV and IPV; ranking LGAs from the worst-performing to the best-performing for prioritization; and providing LGA-specific problems, issues and proposed interventions or solutions. These inputs are important because they move the platform from general risk identification toward direct programme use for outbreak preparedness, targeted interventions and LGA-specific corrective action.

7.4.4 DIH Handover and Operationalization Meeting

The DIH close-out engagement focused on operational ownership of the platform and modelling workflow. The handover discussion covered platform access, routine data upload, model refresh, data-quality checks, output review, documentation, version control, approval processes, user administration, SOPs, data dictionaries and maintenance responsibilities. The key agreement was that the DIH should become the operational hub for routine modelling, while continued platform maintenance, enhancement and expansion should remain collaborative between NPHCDA and the CIDRE–Quantium Insights Consortium during the transition period.

7.4.5 Routine Immunization Technical Working Group Meeting

The modelling outputs were also presented to the Routine Immunization Technical Working Group on Friday, 17 July 2026. The presentation focused on practical programme use: how the ranked LGA list can support microplanning, how community archetypes can guide intervention packages, how antigen-specific alerts can inform supervision and outreach, and how the platform can be used for routine model refresh and Gavi 6.0 targeting.

8. Discussion

Read together, the seven modelling domains tell one coherent story. A stable, broadly on-target national average masks a deep north-western deficit, and four independent analytical routes to the equity-deprivation index, the Bayesian hotspot surface, the outbreak-risk surface, and the GBD triangulation, converge on the same small set of states: Sokoto, Kebbi, and Zamfara, with Niger, Katsina, Kaduna, and Jigawa close behind. That this convergence arises from administrative, survey, and mortality data that were modelled separately is the strongest form of corroboration available short of a trial.

Two findings reshape programme design. First, the coverage-to-mortality link is real and quantified, so closing the zero-dose gap in these states is expected to translate into avoided child deaths, the argument GAVI investment rests on. Second, demand and supply problems do not share a map: hesitancy is concentrated differently from zero-dose burden, so demand-generation messaging and supply-and-access expansion must be targeted independently rather than bundled into one national campaign.

The readiness assessment frames whether these models can be owned and sustained in-country. The endline quantitative results are encouraging: across the thirteen assessed units, basic data-use structures are broadly established (eight of thirteen scored at or above 90% on Part A), no unit fell below a moderate overall capacity level, and most units are estimated to reach modelling readiness within three to nine months. The consistent differentiator is advanced analytics capacity rather than routine data handling: the predictive modelling readiness sub-domain showed the widest spread (55.6% to 100%), and systems integration varied from fully automated to none. This points to where differentiated, unit-specific support should concentrate. Comparing rounds (per the agreed approach, comparison is drawn here rather than in the results), some units show marked movement from baseline; Bayelsa, for example, rose from 59.5% at baseline to 90.5% at endline on the Part A checklist. The two rounds are interpreted together below, so that improvements, steady states, and any declines are each read on their merits rather than as simple before-and-after change.

Read across the two rounds, the readiness picture has strengthened in several respects. The lowest baseline performers improved markedly; Bayelsa rose from 59.5% to 90.5% on the Part A checklist, and the overall distribution tightened, with the endline range (78.6% to 97.6%) sitting higher and narrower than at baseline (59.5% to 97.6%). On the qualitative side, governance moved from informal, unevenly enforced policy toward a formal DIH architecture aligned with national data-protection law, and modelling shifted from an understood-but-aspirational concept to confirmed operational use in the identification and active reprioritisation of the 100 zero-dose LGAs.

The stakeholder engagement and close-out activities further strengthened the interpretation and uptake of the findings. The presentation to the GAVI in-country team provided an opportunity to align the modelling outputs with Gavi 6.0 targeting and financing discussions. Further presentations to the Deputy Director, NPHCDA, other directors and key NPHCDA staff, including the Head of DIH and Head of RI, created a management-level review platform within NPHCDA. The Executive Director’s review further sharpened the operational direction of the work by recommending the inclusion of OPV and IPV, ranking of LGAs from worst to best,

and production of LGA-specific problem statements with corresponding interventions. The subsequent presentation to the Routine Immunization Technical Working helped position the outputs within the technical forum responsible for routine immunization planning and follow-up.

These engagement activities are important as predictive modelling can only influence immunization outcomes if decision-makers understand, accept and routinely use the outputs. The meetings with GAVI, NPHCDA leadership, the Executive Director, the DIH team and the RI Technical Working Group therefore served as validation, interpretation and institutionalization moments.

The digital gap analysis reinforces the same message. Routine immunization data are available, and DHIS2/HMIS provides an important foundation, but the digital structure is not yet strong enough to sustain routine predictive modelling without targeted remediation. The system must be able to extract clean data, link multiple sources, run reproducible models, refresh dashboards, govern access, document assumptions, protect data, and translate outputs into decisions. Current gaps in hardware, power, connectivity, hosting, systems integration, private and tertiary reporting, data-quality automation, access control and DIH workflow design directly affect whether the modelling platform can become a routine function rather than a project output.

Systems integration and infrastructure adequacy still vary widely, with Anambra and Ebonyi weakest on both at endline; the private- and tertiary-facility reporting gap persists as a structural source of undercount; and the sustainability of skills depends on continuous hands-on use, echoing the baseline concern about staff attrition and the reactive analytical culture. Crucially for adoption, the highest-burden modelling states identified in Section 5 Sokoto, Jigawa, Bauchi and others in the North-West score comparatively well on endline readiness (Sokoto and Jigawa at the top of the capacity scale, most in the 3–6 month readiness band), which is favourable: the places where the modelling matters most are among the better positioned to operate it, provided the named capacity, governance and infrastructure conditions are met.

9. Conclusions and Recommendations

9.1 Conclusions

Nigeria's zero-dose burden is a concentrated, entrenched, and convergent one, centred on a small group of North-Western states where low coverage, structural deprivation, outbreak risk, and vaccine-preventable mortality coincide. The modelling localizes this burden to the LGA level, explains its drivers, distinguishes demand from supply geographies, and links coverage gaps to child deaths, turning a national average into an actionable, prioritized map. Polio interruption is within credible reach, conditional on sustaining campaign quality and surveillance in a small number of gating states.

Whether these gains are realized depends on institutional readiness. The readiness assessment identifies the conditions, namely analytics capacity, data-quality governance, systems integration, and infrastructure, that determine whether predictive modelling becomes embedded in national practice or remains a project output.

The close-out engagements confirmed strong stakeholder interest and provided clear direction for the next phase. GAVI engagement positioned the outputs for GAVI 6.0 targeting. NPHCDA leadership engagement supported internal ownership. The Executive Director’s inputs sharpened the operational agenda by calling for inclusion of OPV and IPV, worst-to-best LGA prioritization, and LGA-specific problem-and-solution profiles. The Routine Immunization Technical Working Group presentation created an important technical platform for further uptake.

The DIH remains the most appropriate institutional home for sustaining the modelling workflow. However, it will require targeted digital remediation to function as a routine predictive-modelling hub. Without stronger data pipelines, model governance, user roles, approval workflows, power, connectivity, hosting, documentation, access control and data-quality automation, the modelling platform risks remaining a project output rather than becoming an embedded decision-support system.

9.2 Recommendations

- **Target the convergent Tier-1 North-West first.** Concentrate the largest, most sustained package on Sokoto, Kebbi, and Zamfara, with Niger, Katsina, Kaduna, and Jigawa close behind, using the ranked LGA list for microplanning rather than whole-state campaigns. Sequence the investment in two waves: an initial wave across about 151 LGAs reaches half of missed children and is fundable as a single package, with expansion to about 342 LGAs required to reach 80%. The 107 LGAs ranked in the top 155 by both Domain 5 methods should form the core of the first wave. Budget planning should provide for the second wave from the outset rather than treating it as contingent.
- **Prioritize the measles first dose and timely follow-up campaigns.** This is the single highest-yield action for averting vaccine-preventable child deaths in the near term, per the coverage-to-mortality link.
- **Differentiate delivery by measured barriers.** Start from a state package and refine it with each LGA's flagged barriers (Table 6l), using the contextual profiles as a planning summary; one national package agrees least with LGA barrier profiles. Validate

components locally through microplanning, and treat demand and supply as separate maps.

- **Close the second-measles-dose gap in the high-dropout states of the South-East and South-South and in Oyo, and audit any negative dropout as a denominator or reporting issue.**
- **Protect the polio interruption pathway** by concentrating campaign quality and surveillance in the six gating states, especially the north-eastern reservoir, and strengthening investigation timeliness.
- **Institutionalize the platform in the Digital Innovation Hub** and sustain analytics capacity against attrition, so the models are refreshed and used in routine planning rather than as a one-off exercise.
- **Implement the digital and DIH remediation plan.** Priority remediation should include DIH reactivation, cloud/data warehouse setup, DHIS2 refresh automation, integration with surveillance and other data systems, data linkage, automated QA scripts, anomaly detection, private/tertiary reporting bridge, role-based access control, model-output approval workflow, secure backup, power support, connectivity and state-level access.
- **Concentrate capacity-building on a small, committed analytics core** rather than broad training, with disease-specific depth, sustained mentorship, and a jointly built working model, the conditions the national assessment identified as decisive.
- **Assign explicit organisational responsibility for modelling** with leadership-level supervision and follow-up, since trained skills fade without assigned tasks and accountability.
- **Prioritize systems integration and infrastructure in the weakest units** and close the private- and tertiary-facility reporting gap, which systematically undercounts urban activity and distorts zero-dose targeting.
- **Pilot local modelling adoption in well-positioned high-burden states** such as Sokoto and Jigawa, where readiness scores are strong, and in Gombe, which now funds immunisation domestically, converting the favourable readiness-burden overlap into early operational use.
- **Build modelling-ready data architecture through the DIH** so that routinely collected data is structured and granular enough to support modelling, addressing the data-sufficiency barrier directly.

9.3 Limitations and Next Steps

- Domain 7 currently uses IHME GBD 2023 modelled mortality as a proxy because the IDSR/NCDC line-lists are pending; the risk surface is a structural-susceptibility measure, not a dated outbreak forecast, and will be re-estimated once surveillance data arrive.
- Administrative DHIS2 counts are subject to reporting completeness and denominator error; survey anchoring and Bayesian shrinkage mitigate but do not eliminate this.
- Hesitancy is measured from ACSM field-activity coding, which captures concerns that were voiced and recorded; absence of recorded concern is not proof of confidence.

- Domain 8 (counterfactual campaign design, cost-effectiveness frontiers, and budget-constrained LGA optimization) is the planned next step/phase, awaiting programme and expenditure data from the NPHCDA.
  - Of the three areas for strengthening identified during NPHCDA leadership review on 8 July 2026, two are addressed in this report: the antigen panel now extends to nine further antigens including OPV3, IPV1 and IPV2 (Section 5.1.1), and all modelled LGAs are ranked worst-to-best by burden (Section 5.5.2). The third, LGA-specific problem-and-intervention profiles, is partially addressed through the archetype-matched packages in Tables 6 and 6c and the LGA barrier-to-intervention mapping in Table 6l, and will be completed at LGA level once a GRID3 or National Health Facility Registry facility dataset is available.
  - Method 1 does not estimate Guzamala (Borno), which reported no Penta1 doses in 2021-2024, and its LGA allocation follows reported Penta1 doses, so LGAs whose reported doses are out of line with their population carry larger error. Method 2, which does not use routine coverage, provides the cross-check (Section 5.5.5); it relies on modelled covariate surfaces (2016-2024) and survey anchors, and its LGA estimates are model-based. Priority LGAs should be confirmed locally before large investments.
- The readiness assessment covers thirteen administrative units (National, FCT, and eleven states); findings are indicative of these units and not a full national census of subnational readiness.
- Qualitative findings reflect the views of the specific key informants interviewed; in several states, capability was concentrated in one individual, so unit-level readiness can be sensitive to staff movement.
- Denominator instability (administrative coverage exceeding 100% nationally for all four tracer antigens and in most LGAs) affects the routine data that underpins both readiness judgements and zero-dose targeting, and is mitigated but not eliminated by absolute-number tracking and enumeration exercises.

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